

EG QUESTION BANK

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23IT2T05, 23AD2T05, 23AM2T05 ENGINEERING GRAPHICS						

Course Objectives:

- To enable the students with various concepts like dimensioning, conventions and standards related to Engineering Drawing
- To impart knowledge on the projection of points, lines and plane surfaces
- To improve the visualization skills for better understanding of projection of solids
- To develop the imaginative skills of the students required to understand Section of solids and Developments of surfaces.
- To make the students understand the viewing perception of a solid object in Isometric and Perspective projections.

Course Outcomes: At the end of the course, student will be able to

CO1: Understand the principles of engineering drawing, including engineering curves, scales, orthographic and isometric projections.

CO2: Draw and interpret orthographic projections of points, lines, planes and solids in front, top and side views

CO3: Understand and draw projection of solids in various positions in first quadrant

CO4: Explain principles behind development of surfaces.

CO5: Prepare isometric and perspective sections of simple solids.

COURSE ARTICULATION MATRIX (Correlation between COs, POs & PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	1	1	1	-	-	-	-	-	-	-	1	3	3
CO2	2	2	1	1	-	-	-	-	-	-	-	1	3	3
CO3	3	2	2	2	-	-	-	-	-	-	-	1	3	3
CO4	3	2	2	2	-	-	-	-	-	-	-	1	3	3
CO5	3	2	2	2	-	-	-	-	-	-	-	1	3	3

Note: Enter Correlation Levels **1** or **2** or **3**. If there is no correlation, put **‘-’** **1**- Slight (Low), **2** - Moderate (Medium), **3** - Substantial (High).

SYLLABUS

UNIT-I:

Introduction: Lines, Lettering and Dimensioning, Geometrical Constructions and Constructing regular polygons by general methods.

Curves: construction of ellipse, parabola and hyperbola by general, Cycloids, Involute, Normal and tangent to Curves.

Scales: Plain scales, diagonal scales and vernier scales.

UNIT-II:

Orthographic Projections: Reference plane, importance of reference lines or Plane, Projections of a point situated in any one of the four quadrants.

Projections of Straight Lines: Projections of straight lines parallel to both reference planes, perpendicular to one reference plane and parallel to other reference plane, inclined to one reference plane and parallel to the other reference plane. Projections of Straight Line Inclined to both the reference planes

Projections of Planes: regular planes Perpendicular to both reference planes, parallel to one reference plane and inclined to the other reference plane; plane inclined to both the reference planes.

UNIT-III:

Projections of Solids: Types of solids: Polyhedra and Solids of revolution. Projections of solids in simple positions: Axis perpendicular to horizontal plane, Axis perpendicular to vertical plane and Axis parallel to both the reference planes, Projection of Solids with axis inclined to one reference plane and parallel to another plane.

UNIT-IV:

Sections of Solids: Perpendicular and inclined section planes, Sectional views and True shape of section, Sections of solids in simple position only.

Development of Surfaces: Methods of Development: Parallel line development and radial line development. Development of a cube, prism, cylinder, pyramid and cone.

UNIT-V:

Conversion of Views: Conversion of isometric views to orthographic views; Conversion of orthographic views to isometric views.

Computer graphics: Creating 2D&3D drawings of objects including PCB and Transformations using Auto CAD (*Not for end examination*).

Vocabulary: Technical Jargons

Text Books

1. N. D. Bhatt, Engineering Drawing, Charotar Publishing House, 2016

Reference Books

1. Engineering Drawing, K.L. Narayana and P. Kannaiah, Tata McGraw Hill, 2013.
2. Engineering Drawing, M.B.Shah and B.C. Rana, Pearson Education Inc,2009.
3. Engineering Drawing with an Introduction to AutoCAD, Dhananjay Jolhe, Tata McGraw Hill, 2017.

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Introduction to Engineering Drawing

1.1 Introduction to Lines used in drawing:

Pencil Drawings: For drawings finalized with pencil, the lines can be divided into two line-groups as medium and thin lines. It is important to note that in the finished drawing, all lines except construction lines should be dense, clean and uniform. Construction lines should be thin and faint and should be hardly visible in the finished drawing.

Drawing pencils are graded according to increase in relative hardness. They are marked with the labeled as HB,H,2H,3H,4H,5H,6H etc. students are advised to use pencils as recommended in the table below.

Lines	Pencil
Initial work and construction lines, dimension lines, Extension lines Centre lines, section lines	2H (thin)
Outlines, arrow heads, dotted lines	H (medium)
Numbering (numerals), Lettering (alphabets)	HB (thick)

1.2 Types of Lines:

Outlines: Lines drawn to represent visible edges and surface boundaries of objects are called outlines or principal lines. They are continuous thick lines

Margin Lines: They are continuous thick lines along which the points are trimmed

Dimension Lines: These lines are continuous thin lines. They are terminated at the outer ends by pointed arrow heads touching the outlines, extension lines or centre lines.

Extension or Projection Lines: These lines also are continuous thin lines. They extend by about 3mm beyond the dimension lines.

Construction Lines: These lines are drawn for constructing figures. They are shown in geometrical drawings only. They are continuous thin light lines.

Hatching or Section Lines: these lines are made to make the section evident. They are continuous thin lines and are drawn generally at an angle of 45° to the main outline of the section. They are uniformly spaced about 1mm to 2mm apart.

Leader or pointer lines: Leader line is drawn to connect a note with the feature to which it applies. It is a continuous thin line.

Border Lines: Perfectly rectangular working space is determined by drawing the border lines. They are continuous thin lines.

Short-break lines: These lines are continuous, thin and wavy. They are drawn freehand and are used to show a short break or irregular boundaries.

Long-break lines: These lines are thin ruled lines with short zigzags within them. They are drawn to show long breaks.

Hidden or dotted lines: Interior or hidden edges are shown by hidden lines. They are also called dashed lines or dotted lines. They are of medium thickness and made up of short dashes of approximately equal lengths of about 2mm spaced at equal distances of about 1mm. when a hidden line meets or intersects another hidden line or an outline, their point of intersection or meeting should be clearly shown

Centre Lines: Centre lines are drawn to indicate the axes of cylindrical, conical or spherical objects or details, and also to show the centres of circles and arcs. They are thin, long chains composed of alternately long and short dashes spaced approximately 1mm apart. The longer dashes are about 6 to 8 times the short dashes which are about 1.5mm long.

Cutting-plane lines: The location of cutting plane is shown by this line. It is a long, thin, chain line, thick at ends only.

S. No.	Line Type	Representation	Width*	Applications
1	Continuous NARROW		THIN (0.13 mm)	A. Construction lines, B. Projection lines, C. Dimension lines, D. Extension lines, E. Leader lines, F. Section (Hatching) lines, G. Outlines of revolved sections, H. Imaginary lines of intersection
2	Continuous WIDE		MEDIUM (0.25 mm)	A. Visible outline of sectioned surface, B. Reference lines in projection, C. Ground Lines
3	Continuous EXTRA-WIDE		THICK (0.5 mm)	A. Visible outlines, B. Lines of special importance
4	Dashed NARROW		THIN	Hidden lines or edges
5	Dashed WIDE		MEDIUM	Hidden lines or edges
6	Long Dashed Dotted NARROW (Chain NARROW)		THIN	A. Centerlines, B. Lines of symmetry, C. Cutting planes
7	Long Dashed Dotted WIDE (Chain WIDE)		MEDIUM	Cutting planes (at ends & change of direction)
8	Long Dashed Double Dotted NARROW (Phantom NARROW)		THIN	A. Locus lines, B. Alternative and extreme positions of movable parts, C. Outlines of adjacent parts
9	Continuous Freehand NARROW		THIN	Short break line
10	Continuous Zigzag NARROW		THIN	A. Long break line, B. Limits of partial or interrupted views

1.3 LETTERING:

Writing of titles, dimensions, notes and other important particulars on a drawing is called *lettering*. It is an important part of a drawing. However accurate and neat a drawing may be drawn, its appearance gets spoiled and sometimes, its usefulness is impaired by poor lettering. Lettering should therefore be done properly in clear, legible and uniform style. It should be in plain and simple style so that it could be done *freehand* and *speedily*.

Note: Use of drawing instruments in lettering takes considerable time and hence, it should be avoided. Efficiency in the art of lettering can be achieved by careful and continuous practice.

I. Single stroke letters: The BIS recommends single stroke lettering for use in engineering drawing. These are the simplest forms of letters and are usually employed in most of the engineering drawings.

The word single-stroke should not be taken to mean that the letter should be made in one stroke without lifting the pencil.

Single Stroke letters are of two types:

(i) Vertical and (ii) inclined

Inclined letters lean to the right, the slope being 75° with the horizontal. The size of the letter is described by its height.

Recommended letter height for vertical lettering:

The main titles	6 to 8mm	6mm is recommended
Subtitles	3-6 mm	4mm is recommended
Notes, dimension figures	3- 5mm	4mm is recommended
The drawing number in the title block	10- 12mm	10mm is recommended

H or HB grade of pencil is recommended for this purpose. The spacing between two letters should not necessarily be equal. The letters should be so spaced that they do not appear too close or too much apart.

Judging by the eye, the back ground areas between the letters should be kept approximately equal. The distance between the words must be uniform and at least equal to the height of the letters.

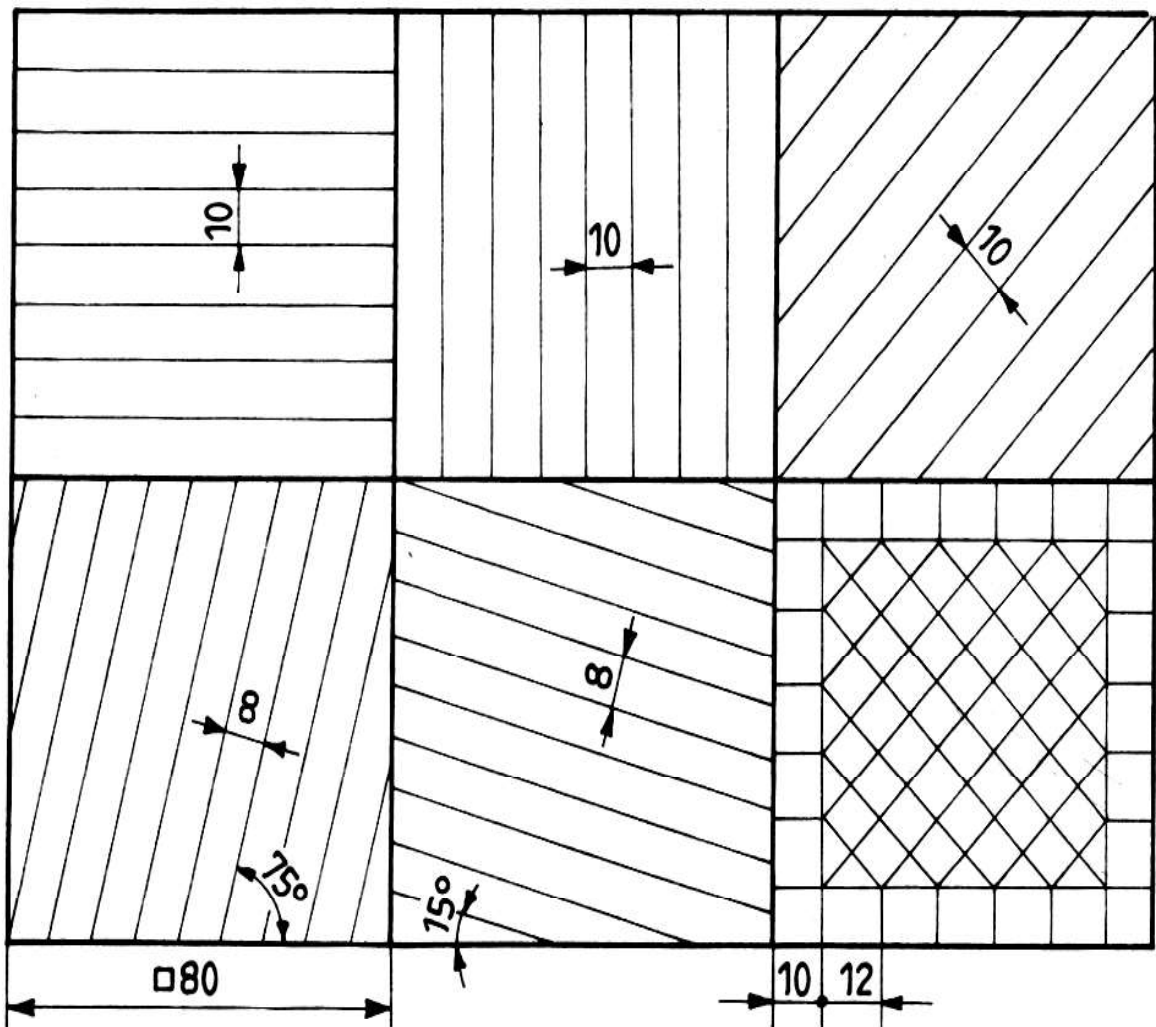
1.4 MEASUREMENTS

10 millimeters(mm) = 1 centimeter(cm)
10 centimeters(cm) = 1 decimeter(dm)
10 decimeters(dm) = 1 meter(m)
10 meters(m) = 1 deca meter(dam)
10 deca meters(dam) = 1 hecto meter(hm)
10 hecto meters(hm) = 1 kilo meter(km)
1 inch = 2.54 centimeter
1 foot = 12 inches.
3 feet = 1 yard.
22 yards = 1 chain.
10 chains = 1 furlong.
8 furlongs = 1 mile.
1 mile = 1.6 kilo meter(km)

SHEET No. 1 (UNIT-1)

DRAFTER USES

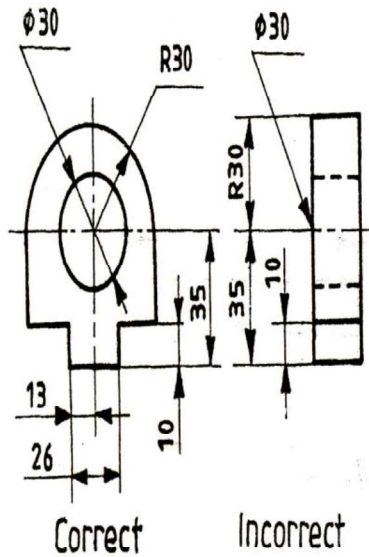
1. Using mini drafter set 15° , 30° , 45° , 60° , 75° & 120° angles with a horizontal line.
2. Using mini drafter, draw triangles with the following base angles on 50mm long line as the base. (i) 15° & 45° (ii) 30° & 60° (iii) 45° & 75° (iv) 60° & 85° (v) 120° & 15°
3. Draw two lines parallel to each other, separated by 15mm and making an angle of 30° with horizontal.
4. Draw a squares of sides 30mm and draw a rectangle of sides 50mm x 30mm
5. Draw a circle of 50mm diameter and without using a protractor divide it into 8,12 equal parts.
6. Reproduce the figure shown below.



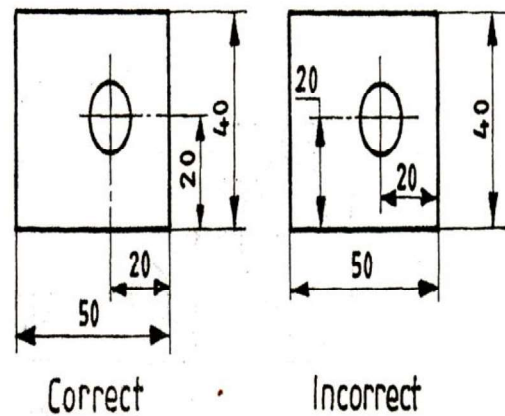
SHEET No. 1 (UNIT-1)

LETTERING & DIMENSIONING

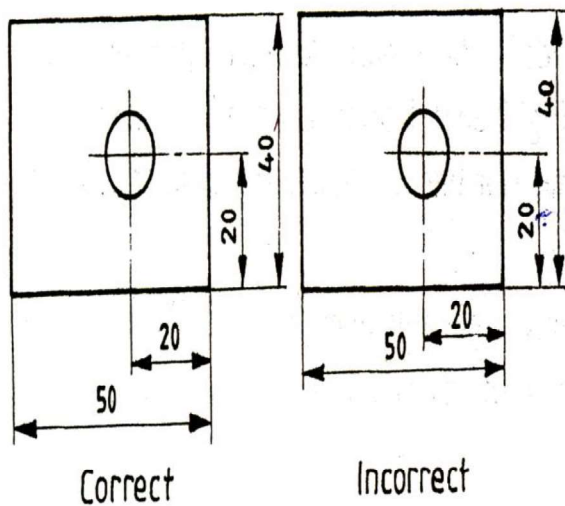
1. Alphabets A to Z, College Title, Name, Branch (full), No.s 1 to 10, height- 6mm



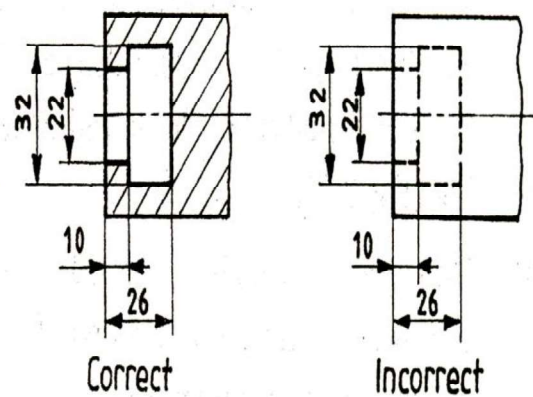
Dimensioning where the shape
is best shown



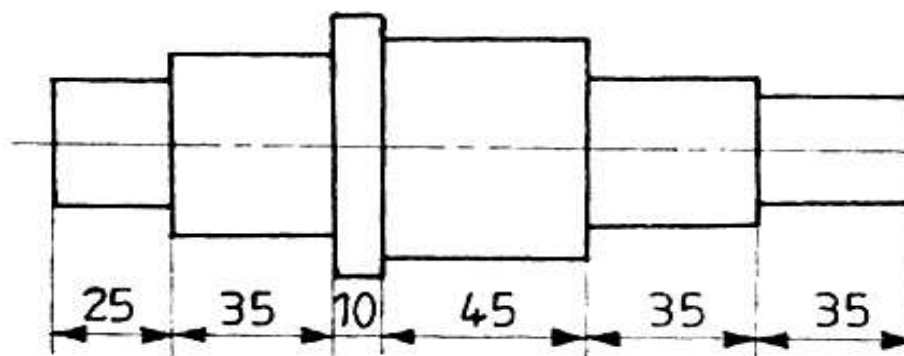
Placing dimensions outside the view



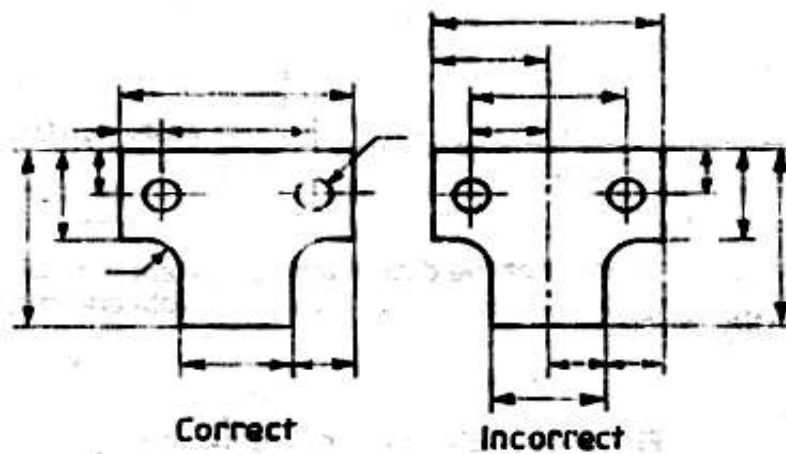
Spacing of dimension lines



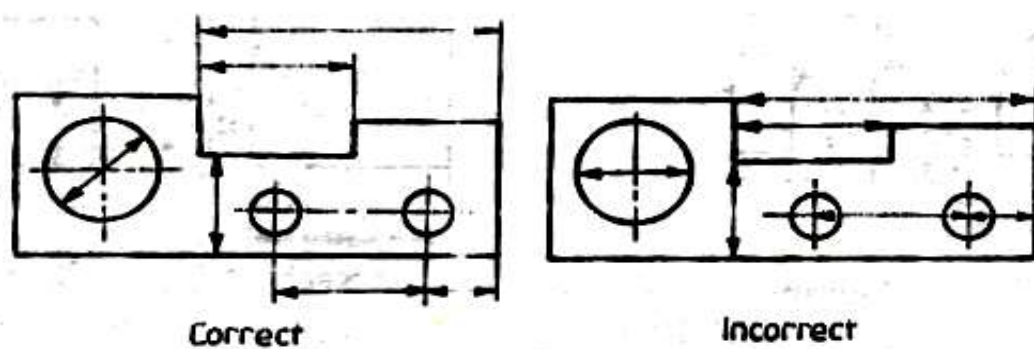
Placing dimensions from the visible
Outlines



Chain dimensioning



Avoiding centre line for dimensioning



Avoiding contour lines for dimensioning

SHEET No. 2 (UNIT-1) GEOMETRICAL CONSTRUCTION

- 1(a) Bisect a line of 85mm
- 1 (b) Bisect an angle of 75°
- 1(c) Divide the line 108 into 8 number of equal parts

SHEET No. 3 (UNIT-1) POLYGONS

- 1(d) Construct a regular hexagon of side 32mm when one side is vertical.
- 1(e) Inscribe a regular polygon of sides, say 5, in a circle of 64mm diameter when one side is horizontal.
- 1 (f) Construct a polygon, say hexagon of length of side 35mm using External Angle method.
- 1(g) Construct a polygon (say pentagon) of 30mm side using base angle of side.
- 1(h) Circumscribe an equilateral triangle about a circle of 30mm diameter.
- 1(i) Circumscribe / Describe / Superscribe of a regular polygon about a circle. (say 4, 5, 6, 7, 8)
- 2. Construct a square, pentagon, hexagon, heptagon and octagon of 40mm side using general construction method

SHEET No. 4 (UNIT-1) CONIC SECTIONS

- 3) The major axis of an ellipse is 100mm and the minor axis is 55mm. Find the foci and construct the ellipse by Intersecting Arcs Method.
- 4. The major & minor axes of an ellipse are 120mm and 80mm respectively. Draw an ellipse by using Oblong Method (Rectangle Method)
- 5. Draw an ellipse when a distance of its focus from directrix is 50mm & eccentricity is $\frac{2}{3}$. Draw a tangent & normal at a point 70mm from the directrix. (Eccentricity Method)
- 6. Draw a parabola when a distance between its focus and directrix is 50mm . Draw a tangent & normal at a point 70mm from the directrix. (Eccentricity Method)
- 7. Draw a hyperbola when a distance of its focus from directrix is 50mm & eccentricity is $\frac{3}{2}$. Also draw a tangent & normal to the hyperbola at a point 25mm from the directrix. (Eccentricity Method)

SHEET No. 5 (UNIT-1) CYCLOIDS AND INVOLUTES

- 8. A coin of 40mm diameter rolls over a horizontal table without slipping. A point on the circumference of the coin is in contact with the table surface in the beginning and after one complete revolution. Draw the path traced by the point.

9. Draw a cycloid of a circle of diameter 50mm for one revolution. Also draw a tangent & normal to the curve at a point 35mm above the base line.

10. Draw an involute of a pentagon of side 25mm.

SHEET No. 6 (UNIT-1) SCALES

11.

Construct a plain scale to show meters when 1 centimeter represents 4 meters and long enough to measure up to 50 meters. Find the R.F. and mark on your scale a distance of 36 meters.

12

Construct a plain scale of R.F. 1 : 50,000 to show kilometers and hectometers and long enough to measure up to 7 kilometers. Measure a distance of 54 hectometers on your scale.

13

Construct a diagonal scale of R.F. = 1 : 32,00,000 to show kilometers and long enough to measure upto 400 km. Show distances of 257 km and 333 km on your scale.

14

The Marina Beach at Chennai, is 2.5 km long. On inspection of the road map, its equivalent distance measures 5 cm. Devise a diagonal scale to read 50 m minimum. Mark on the scale a distance of 6350 m.

15

*Construct a vernier scale to read meters, decimeters and centimeters and long enough to measure up to 4 m. R.F. of the scale is 1/20. Mark on your scale a distance of 2.28 m. (**Backward / Retrograde Vernier**)*

16

*Construct a vernier scale to show readings of 1/10th of a meter when 3 cm represents 10 m. Construct the scale to read up to 60 m and mark the distances of 35.3 m and 47.3 m on your scale. (**Backward Vernier**)*

SHEET No. 7 (UNIT-II)

PROJECTIONS OF POINTS:

1. Draw the projections of the following points, keeping the distance between the projectors as 25 on the same reference line
A-25 above HP and 45 in front of VP
B-35 above HP and 50 behind VP
C-40 below HP and 30 behind VP
D-30 below HP and 40 in front of VP
E-50 above HP and on VP
F-45 below HP and on VP
G-On HP and 35 in front of VP
H-On HP and 25 behind VP
I-On both HP and VP
2. A point at 25 above the reference line XY is the front view of two points A & B. The top view of A is 49 behind VP and the top view of B is 50 in front of VP. Draw the projections of the points and state their positions relative to the planes of projection and the quadrants in which they lie.
3. A point A is 40 above HP and 25 in front of VP. Another point B is 20 behind VP and 30 below HP. The horizontal distance between the points is 100. Draw the three projections of the points A & B and join their front views, top views and left side views.
4. A point A is on HP and 40 in front of VP. Another point B is on VP and below HP. The line joining their front views makes an angle of 45° with XY, while the line joining their top views makes an angle of 30° . Find the distance of the point B from HP.

SHEET No. 8 (UNIT-II)

PROJECTIONS OF LINES:

1. Draw the projection of a straight line AB, of 60 long in the following positions:
 - (a) Parallel to both HP and VP and 20 from each
 - (b) Parallel to and 20 above HP and on VP.
 - (c) Parallel to and 30 in front of VP and on HP.
 - (d) Perpendicular to HP, 30 in front of VP and one end on HP.
 - (e) Perpendicular to HP, 30 in front of VP and one end 25 above HP.
 - (f) Perpendicular to and 20 in front of VP and on HP.
 - (g) Inclined at 30^0 to VP, in HP and one end on VP.
 - (h) Inclined at 45^0 to HP, one end 20 above HP and parallel to and 30 in front of VP
 - (i) Inclined at 60^0 to VP, one end 20 in front of VP and parallel to and 25 above HP
2. A line of 100 long is parallel to and 30 above HP. Its two ends are 25 and 50 in front of VP respectively. Find its inclination with VP.
3. A line of 90 long is parallel to and 20 in front of VP. Its one end is in HP While the other end is 40 above HP. Draw the projections of the line and find its inclination with HP.
4. The top view of a line, which is 75 long, measures 50. The line, is in VP, its one end being 25 above HP. Draw its projections.
5. The front view of a line, which is inclined at 30^0 to VP, is 45 long. Draw the projections of the line when it is parallel to and 30 above HP, its one end being 30 in front of VP. Determine the true length of the line.

SHEET No. 9 (UNIT-II)

LINE INCLINED TO BOTH THE PLANES:

Determinations of true length, angle of inclination and traces

1. The line CD= 80mm long, is inclined at 30^0 to H.P. and 45^0 V.P. its one end C is 20mm above H.P. and 30mm in front of V.P Draw its projections and locate its traces.
2. A line AB 65mm long has its end A 20mm above H.P and 25mm in front of V.P. the end B is 40mm above H.P. and 65mm in front of V.P. draw its projections and find its inclinations with the HP and VP. Mention the values of apparent angles also and locate its traces.
3. The mid point M of a straight line of AB is 40mm from H.P. and 50mm from V.P. the line AB is 70mm long and makes 40^0 with H.P and 30^0 with V.P. draw the projection of the line and locate its traces.
4. The length of the top view of a straight line is 45mm and the length of the front view is 60mm. The top view 'ab' is inclined at 30^0 to XY. Draw the projections of the line AB assuming the point A to be in the H.P and 30mm in front of V.P. Determine the length and inclination of the line with H.P and V.P and locate its traces.
5. The projections of a line measure 80mm in the top view and 70mm in the front view. The mid point of the line is 45mm in front of V.P. and 35mm above H.P. one end is 10mm in front of VP and nearer to it. The other end is nearer to H.P Draw the projections of the line; find its true length and true inclinations and locate its traces.

SHEET No. 10 (UNIT-II)

PROJECTIONS OF PLANES

Parallel/ Perpendicular to one plane and inclined to other plane

1. A rectangle ABCD of size 40x60 has a corner on H.P and 20 away from V.P. all the sides of the rectangle are equally inclined to H.P and parallel to V.P. draw its projections.
2. A square of 30 side has one side on H.P. its plane is inclined at 60^0 to H.P and perpendicular to V.P. draw its projections
3. A pentagon of 30 side has one corner on V.P. its plane is inclined at 60^0 to H.P and perpendicular to V.P. draw its projections.
4. A thin rectangular plate 60x40 size has its shorter edge on H.P and inclined at 30^0 to V.P. draw the projections of the plate, when its top view is a square of 40 side.
5. A thin 30^0 - 60^0 set square has its longest edge (diagonal) on H.P and inclined at 30^0 to V.P. its surface makes an angle 45^0 with H.P. draw the projections, choosing suitable size of the set square.

SHEET No. 11 (UNIT-II)

PROJECTIONS OF PLANES INCLINED TO BOTH THE PLANES

1. A regular pentagon of 30 side is resting on one of its edges on H.P, which is inclined at 45^0 to V.P. its surface is inclined at 30^0 to H.P. draw its projections.
2. A rhombus has its diagonals 100 and 60 long. Draw the projections of rhombus, when it is so placed that its top view appears to be a square of diagonal 60 long and the vertical plane through the longer diagonal makes 30^0 with V.P.
3. A semi – circular plate of 80 diameters, has its straight edge on V.P and inclined at 30^0 to H.P, its surface is inclined at 45^0 to VP. Draw the projections of the plate.
4. A regular hexagonal plane of 45 side has a corner on H.P and its surface is inclined at 45^0 to H.P. draw the projections when the diagonal through the corner, which is on H.P makes 30^0 with V.P.
5. A circle of 40 diameter is resting on H.P on a point, with its surface inclined at 30^0 to H.P. Draw the projections of the circle, when (i) the top view of the diameter, through the resting point, makes an angle 45^0 with xy and (ii) the diameter passing through the resting point makes an angle 45^0 with V.P.

SHEET No. 12 (UNIT III)

PROJECTIONS OF SOLIDS (PRISMS , CYLINDERS,PYRAMIDS & CONES)

1. Draw the projections of the following solids, situated in their respective positions, taking the side of the base equal to 40 and the length of axis as 70.
 - a) A square prism with base on H.P and a side of the base inclined at 45^0
 - b) A pentagonal prism with a rectangular face on H.P, axis perpendicular V.P and one base on V.P.
 - c) A cube of edge 40, with a base on H.P and a face perpendicular to V.P
2. Draw the projections of the following solids, situated in their respective positions, taking the diameter of the base equal to 50 and the length of the axis 70.
 - a) . A cylinder, with its axis perpendicular to V.P and 40 above H.P and one end at 20 from V.P.
 - b). A cylinder, resting with its base on H.P and axis 30 in front of V.P
3. Draw the projections of the following solids, situated in their respective positions, taking the side of the base equal to 40 and the length of axis as 70.
 - a. A hexagonal pyramid with base on H.P and a side of the base parallel to and 25 from V.P
 - b. A triangular pyramid with the base on H.P, an edge of the base inclined at 45^0 to V.P and the apex at 40 from V.P.
 - c. A square pyramid with its base on V.P and one edge of the base inclined at 30^0 to H.P. and a corner contained by that edge is on H.P.
 - d. A hexagonal pyramid with its base on V.P and an edge of the base inclined at 20^0 to H.P.
4. a. A cone with its apex on H.P and 40 from V.P., the axis being perpendicular to H.P.
 - b. Draw the projection s of a tetrahedron of side 50, when it is resting on H.P on one of its faces, with an edge of its face (i) inclined at 45^0 to V.P (ii) parallel to V.P and (iii) inclined at 60^0 to V.P.

SHEET No. 13 (UNIT- III)

PROJECTIONS OF SOLIDS (PRISMS , CYLINDERS,PYRAMIDS & CONES)

1. A hexagonal prism, side of base 25mm and axis 50mm long rests with one of its base corners on H.P. such that its base makes an angle of 60^0 to H.P. and its axis is parallel to V.P. Draw its Projections.
2. Draw the projections of a cylinder, base 30mm diameter and axis 40mm long, rests with a point of its base circle on H.P. such that the axis is making an angle of 30^0 with H.P.
3. A regular pentagonal prism of side of base 30 and length 70 lies with one of its rectangular faces on H.P and its axis is inclined to V.P at 45^0 . Draw the projections of the prism.
4. Draw the projections of a cylinder of diameter 50 and length 70, lying on H.P with its axis inclined at 30^0 to V.P
5. A hexagonal pyramid side of base 25mm and axis 50mm long rests with one of the edges of its base on H.P. and its axis is inclined at 30^0 to H.P and parallel to V.P. draw its projections.
6. A square pyramid of base 40 side and axis 50 long, is resting on one of its triangular faces on H.P, with the edge of the base containing that face inclined at 45^0 to V.P. draw the projections of the pyramid.
7. Draw the projection of a cone, base 30mm diameter and axis 50mm long, resting on H.P., on a point of base circle with the axis making an angle of 45^0 with H.P.
8. Draw the projections of a cone 70 height and 50 diameter, resting on one of its generators on H.P and its axis lies in a vertical plane, inclined at 30^0 to V.P.

SHEET No. 14 (UNIT- IV)
SECTIONS OF SOLIDS :

- 1 *A hexagonal prism, side of base 30 mm and axis 60 mm long, rests with its base on HP such that one of its rectangular faces is parallel to VP. A section plane perpendicular to HP and parallel to VP cuts the prism at a distance of 10 mm from its axis. Draw its top and sectional front views.*
- 2 *A cube of 45 mm side rests with a face on HP such that one of its vertical faces is inclined at 30° to VP. A section plane, parallel to VP cuts the cube at a distance of 15 mm from the vertical edge nearer to the observer. Draw its top and sectional front views.*
- 3 *A square prism, side of base 30 mm and axis 60 mm long, rests with its base on HP and one of its rectangular faces is inclined at 30° to VP. A section plane perpendicular to VP and inclined at 60° to HP cuts the axis of the prism at a point 20 mm from its top end. Draw sectional top view and true shape of section.*
- 4 *A pentagonal pyramid, side of base 35 mm and axis 60 mm long, rests with its base on HP such that one of the edges of the base is perpendicular to VP. A section plane perpendicular to HP and parallel to VP cuts the pyramid at a distance of 20 mm from the corner of the base nearer to the observer. Draw its top and sectional front views.*
- 5 *A pentagonal pyramid, side of base 35 mm and axis 60 mm long, rests with its base on HP such that one of the edges of the base is perpendicular to VP. A section plane perpendicular to HP and parallel to VP cuts the pyramid at a distance of 20 mm from the corner of the base nearer to the observer. Draw its top and sectional front views.*

6

A pentagonal pyramid, side of base 30 mm and axis 60 mm long, rests with its base on HP and one of the edges of its base is perpendicular to VP. It is cut by a section plane perpendicular to VP, parallel to HP and passing through the axis at a point 35 mm above the base. Draw the front and sectional top views.

7

A cylinder of 50 mm diameter and 60 mm height stands vertically with its base on HP. It is cut by a section plane perpendicular to HP, parallel to VP and 15 mm from the axis. Draw its top and sectional front views.

8

A cone, base 40 mm diameter and axis 60 mm long, rests with its base on HP. It is cut by a section plane perpendicular to VP, parallel to one of the end generators and passing through a point on the axis 25 mm from the apex. Draw the sectional top view and true shape of the section.

Section plane parallel to HP

1) A cube of 40 edge, is resting on H.P. on one of its edges, with a face parallel to VP. One of the faces containing the resting edge is inclined at 30^0 to HP. The solid is cut by a section plane, parallel to HP and 10 above the axis. Draw the projections of the remaining solid.

2) A cone with base 60 diameter and axis 75 long, is resting on its base on HP. It is cut by a section plane parallel to HP and passing through the midpoint of the axis. Draw the projections of the cut solid.

Section plane parallel to VP

3) A cylinder of 40 diameter and axis 55 long, stands vertically with its base on HP. It is cut by a section plane, parallel to VP and passes a distance of 10 from the axis. Draw the projections of the retained solid.

4) A pentagonal pyramid of side of base 35 and axis 60 long, stands with its base on HP such that, one of the base edges is perpendicular to VP. A section plane parallel to VP, cuts the solid at a distance of 15 from the corner of the base which is nearer to the observer. Draw the top and sectional front views of the cut solid.

Section plane inclined to HP and perpendicular to VP

5) A hexagonal pyramid of side of base 30 and axis 60 long, is resting on its base on HP, with an edge of the base perpendicular to VP. It is cut by a section plane, inclined at 30^0 to HP and passing through the axis at 20 from the base. Draw the three views of the solid and obtain the true shape of the section.

6) A cylinder of 45 diameter and 70 long, is resting on one of its bases on HP. It is cut by a section plane, inclined at 60^0 with HP and passing through a point on the axis at 15 from one end. Draw the three views of the solid and also obtain the true shape of the section.

Section plane inclined to VP and perpendicular to HP

7) A hexagonal prism of side of base 25 and axis 60 long, is resting on its base on HP, such that an edge of the base is parallel to VP. It is cut by a section plane, inclined at 45° to VP and 10 away from the axis . Draw the projections of the solid. Also obtain an auxiliary front view, showing the true shape of the section.

8) A pentagonal pyramid with edge of base 25 and axis 65 long, is resting on H.P. on its base with an edge nearer to the observer, parallel to V.P. It is cut by a section plane, inclined at 60° to V.P. and at a distance of 6 from the axis. Draw the projections and obtain the true shape of the section.

SHEET No. 15 (UNIT-IV)

DEVELOPMENT OF SURFACES

1

Draw the development of the lateral surfaces of a right square prism of edge of base 30 mm and axis 50 mm long.

2

Draw the development of the outside case and tray of a match box of size 45 mm x 33 mm x 16 mm.

3

Draw the development of the complete surface of a G. cylindrical drum with lid. Diameter is 30 cm and height is 1.6 times the diameter.

4

A hexagonal prism, edge of base 20 mm and axis 50 mm long, rests with its base on HP such that one of its rectangular faces is parallel to VP. It is cut by a plane perpendicular to VP, inclined at 45° to HP and passing through the right corner of the top face of the prism. (i) Draw the sectional top view. (ii) Develop the lateral surfaces of the truncated prism.

5

Draw the development of the lateral surface of the lower portion of a cylinder of diameter 50 mm and axis 70 mm when sectioned by a plane inclined at 40° to HP and $\perp$ to VP and bisecting the axis.

6

Draw the development of the lateral surfaces of a square pyramid, side of base 25 mm and height 50 mm, resting with its base on HP and an edge of the base parallel to VP.

*Draw the
development of the lateral surface of a
cone of base diameter 48 mm and
altitude 55 mm.*

Parallel line development

- 1) A square prism of side of base 40 and axis 80 long, is resting on its base on H.P. such that, a rectangular face of it is parallel to V.P. Draw the development of the prism.

- 2) A hexagonal prism of side of base 30 and axis 75 long, is resting on its base on H.P. such that, a rectangular face is parallel to V.P. It is cut by a section plane, perpendicular to V.P. and inclined at 30° to H.P. The section plane is passing through the top end of an extreme lateral edge of the prism. Draw the development of the lateral surface of the cut prism.

- 3) A cylinder of diameter of base 40 and axis 55 long, is resting on its base on H.P. It is cut by a section plane, perpendicular to V.P. and inclined at 45° to H.P. The section plane is passing through the top end of an extreme generator of the cylinder. Draw the development of the lateral surface of the cut cylinder.

Radial line development

- 4) A pentagonal pyramid of side of base 30 and axis 60 long, is resting on its base on H.P. with an edge of the base parallel to V.P. Draw the development of the lateral surface of the pyramid.

- 5) A cone of diameter of base 50 and axis 60 long, is resting on its base on H.P. Draw the projections of the cone and show on it, the shorter path traced by a point, starting from a point on the circumference of the base of the cone, moving around it and reaching the same point.

Other examples

- 6) A cylinder of base 80 diameter and axis 110 long, is resting on its base on H.P. It has a circular hole of 60 diameter, drilled through centrally such that, the axis of the hole is

perpendicular to V.P. and bisects the axis of the cylinder at right angles. Develop the lateral surface of the cylinder.

7) A hexagonal prism of 20 side of base and 50 height, rests on a base on H.P., with a vertical face parallel to V.P. A circular hole of 35 diameter, is drilled through the prism such that, the axis of the hole bisects the axis of the prism and is perpendicular to V.P. Draw the development of the prism.

SHEET No. 16 (UNIT-V)

CONVERSION OF ORTHOGRAPHIC VIEWS TO ISOMETRIC VIEWS

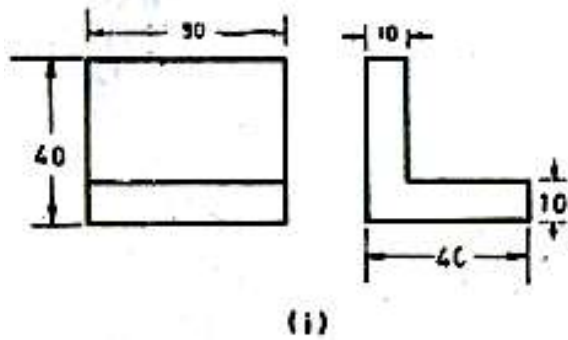
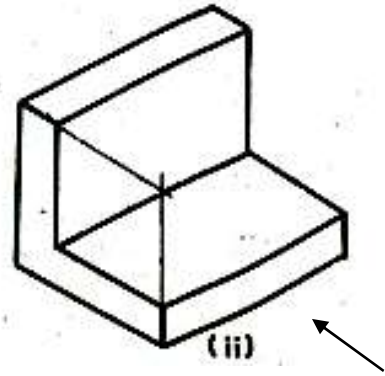


FIG 1



Answer

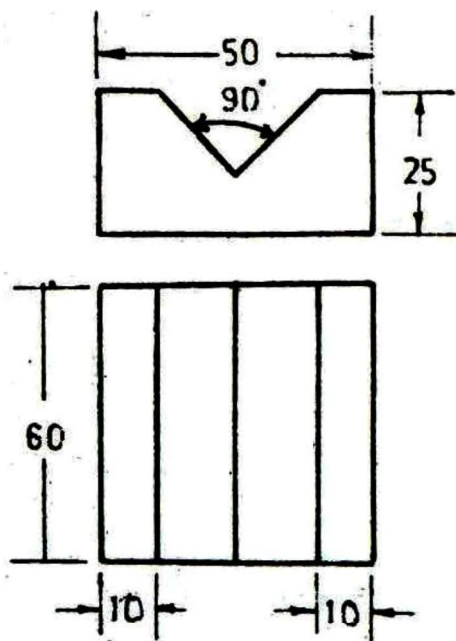


FIG 2

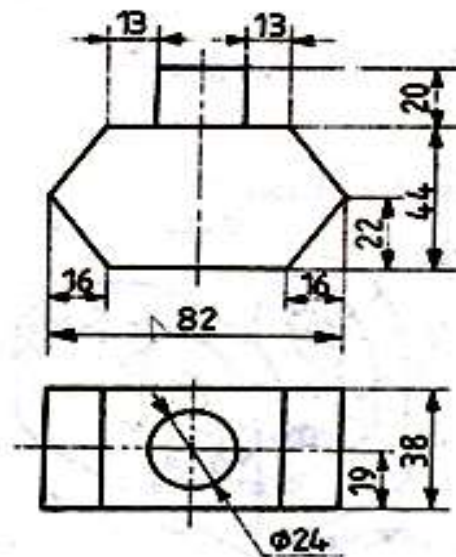


FIG 3

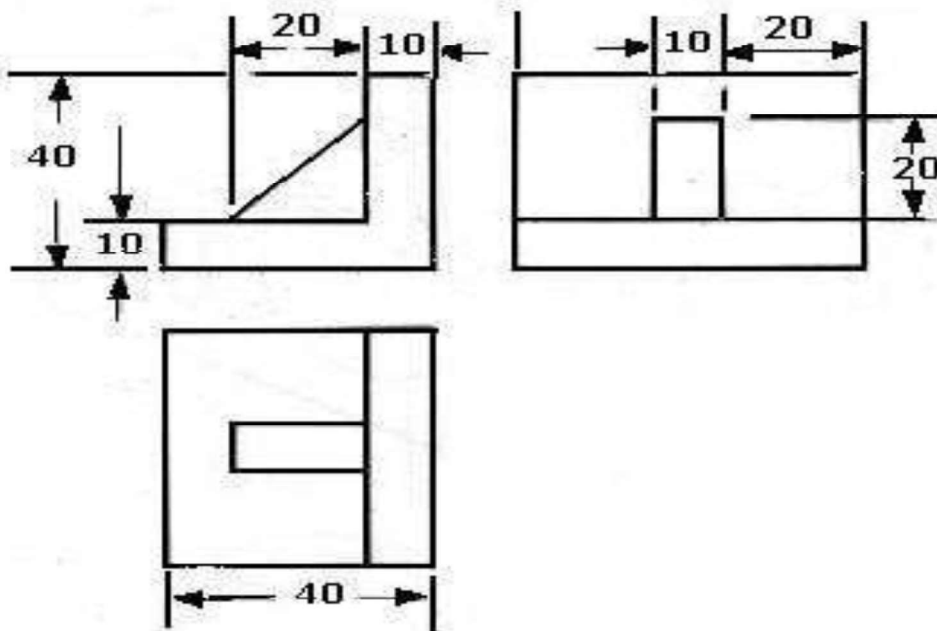


FIG 4

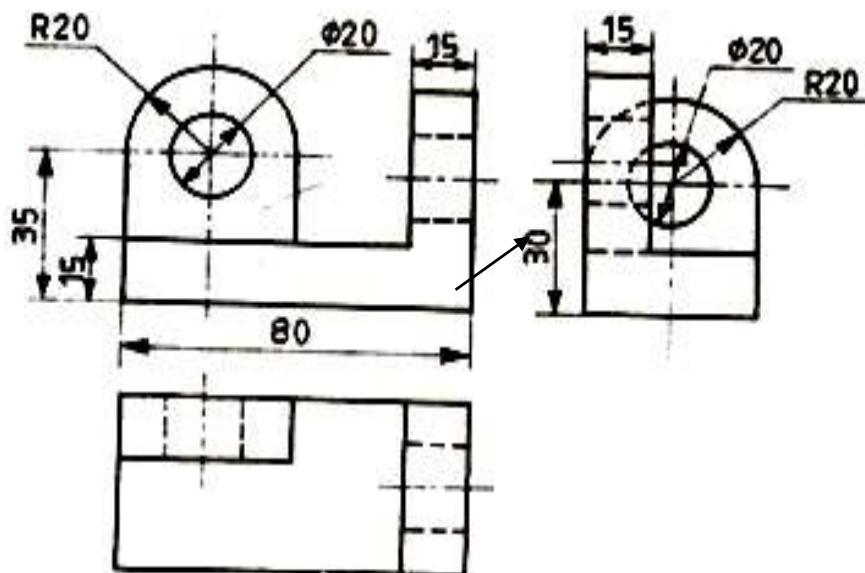


FIG 5

SHEET No.17 (UNIT-V)

CONVERSION OF ISOMETRIC VIEWS TO ORTHOGRAPHIC VIEWS

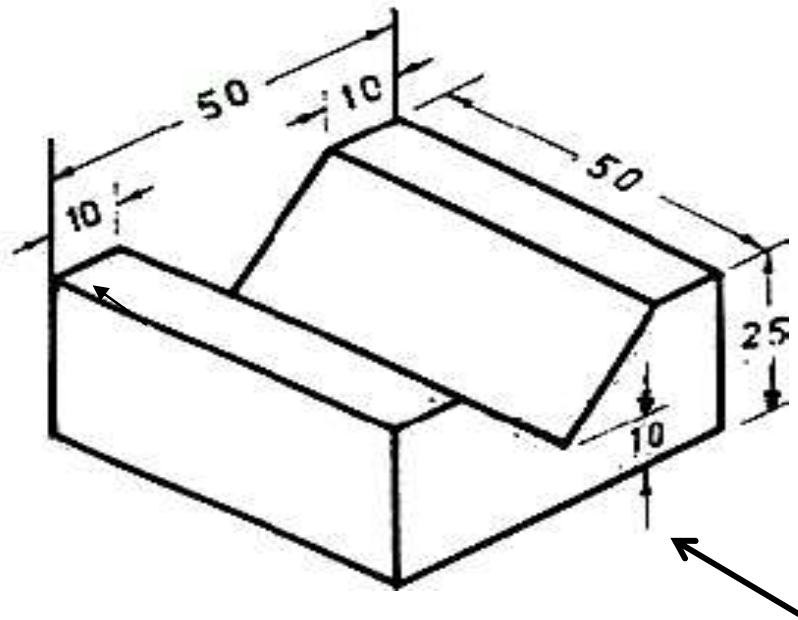


FIG 1

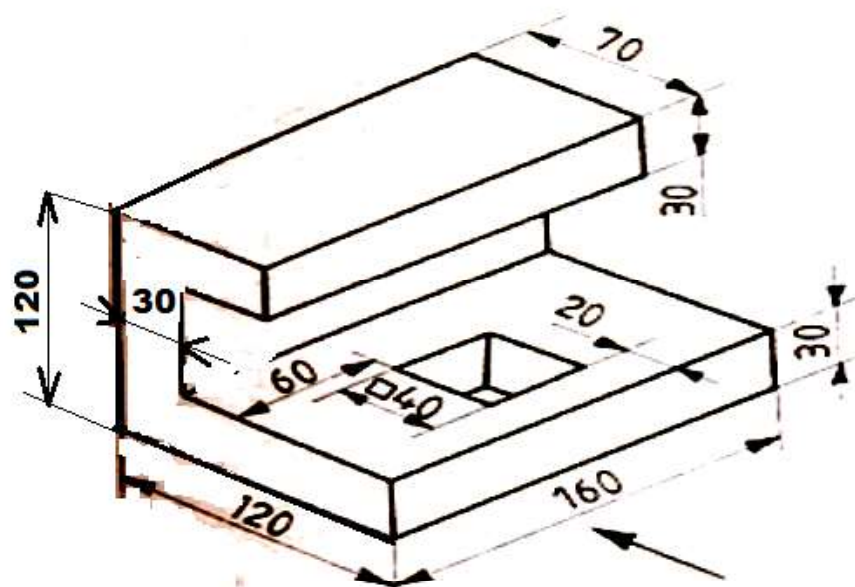


FIG 2

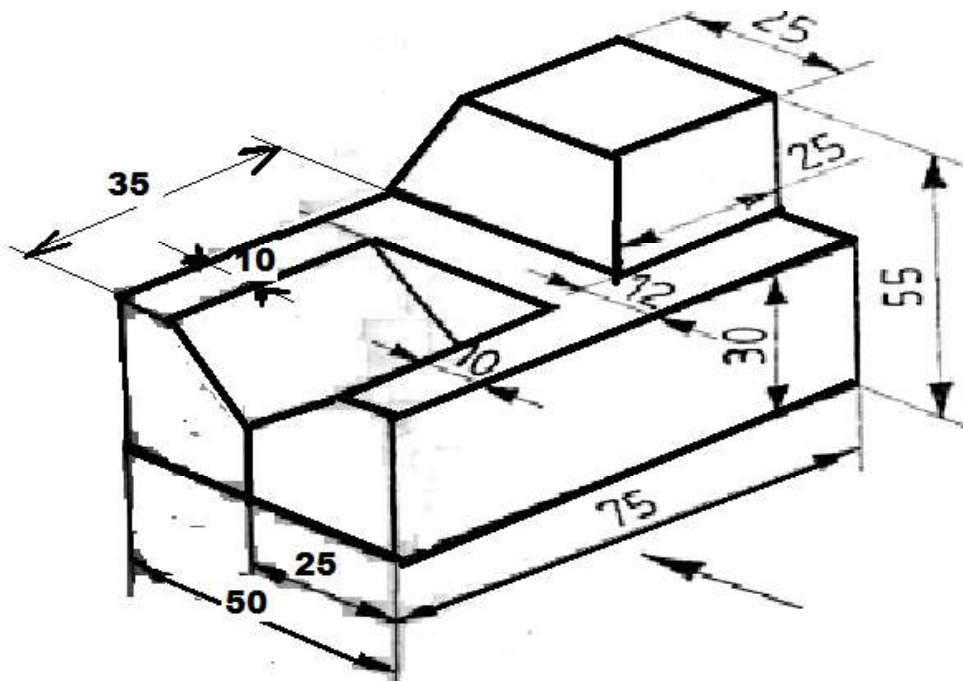


FIG 3

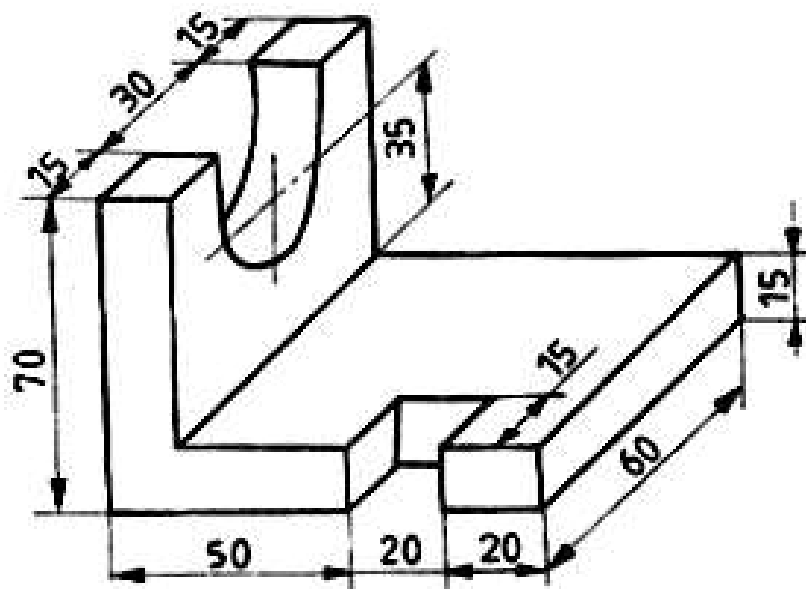


FIG 4

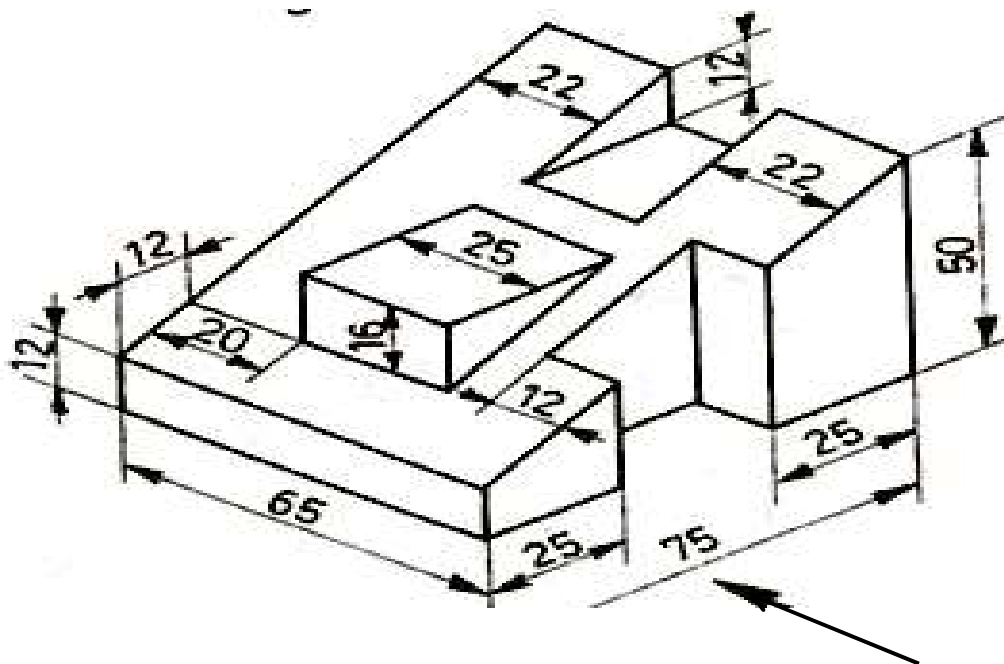


FIG 5

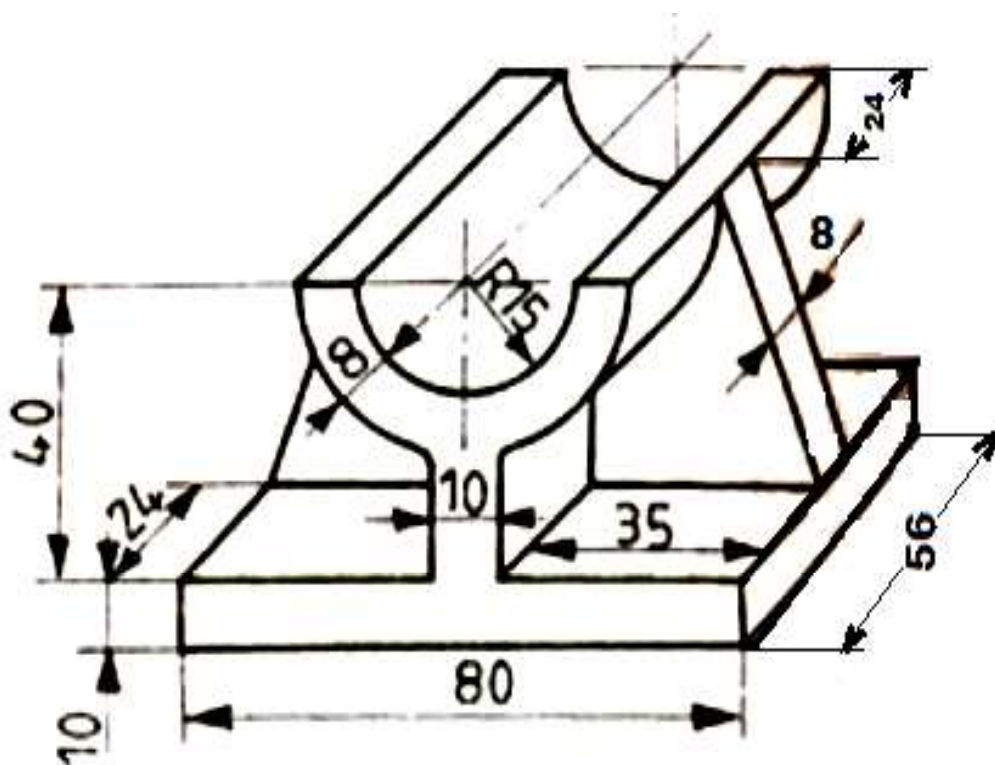


FIG 6

EG QUESTION BANK SOLUTIONS

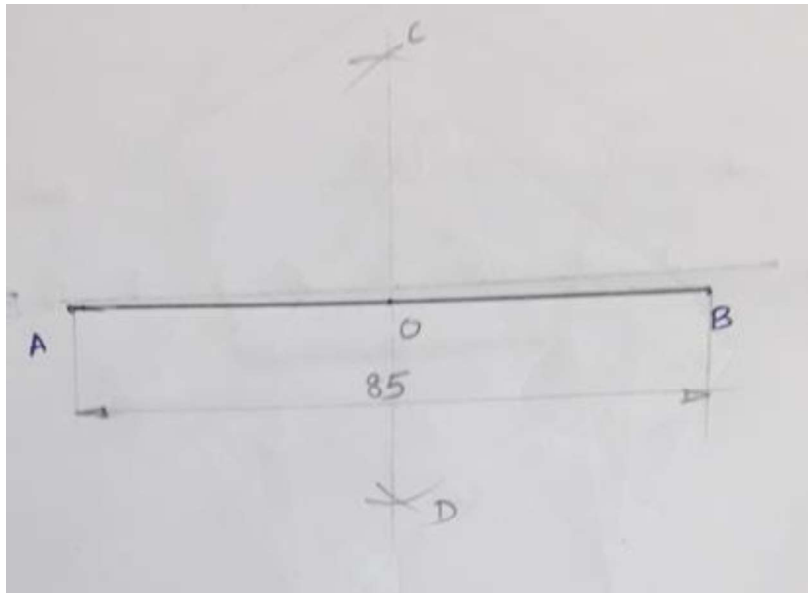
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SHEET No. 2 (UNIT-1) GEOMETRICAL CONSTRUCTION

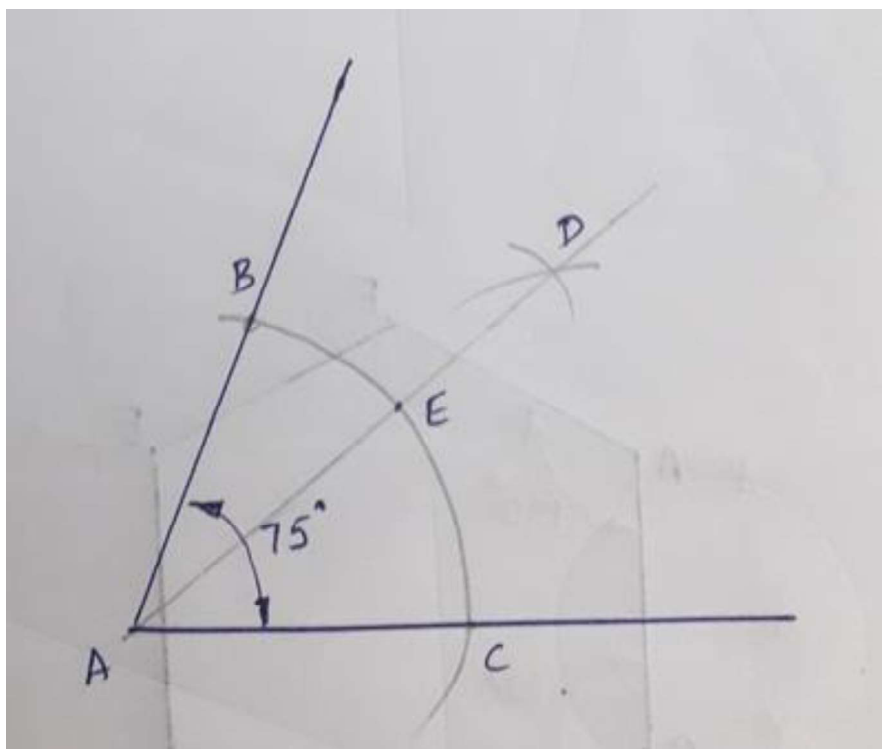
1(a) Bisect a line of 85mm

<https://www.youtube.com/watch?v=oKD-dTA3J8g>



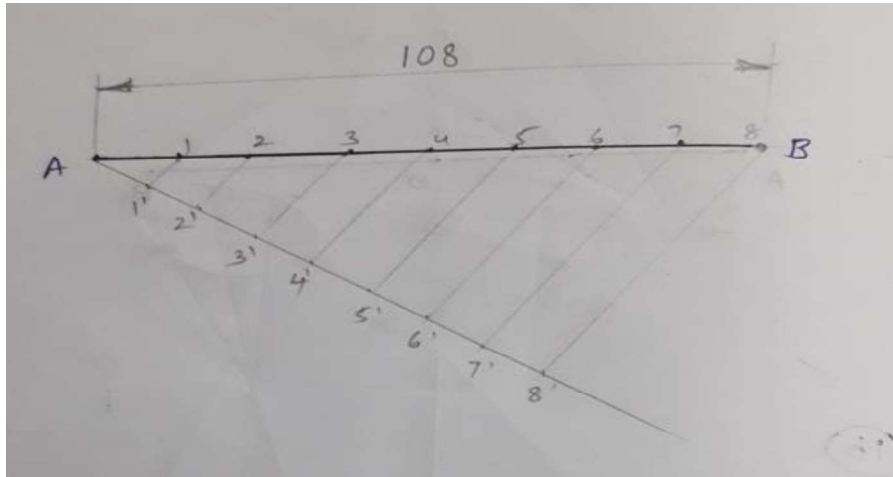
1 (b) Bisect an angle of 75°

<https://www.youtube.com/watch?v=oKD-dTA3J8g>



1(c) Divide the line 108 into 8 number of equal parts

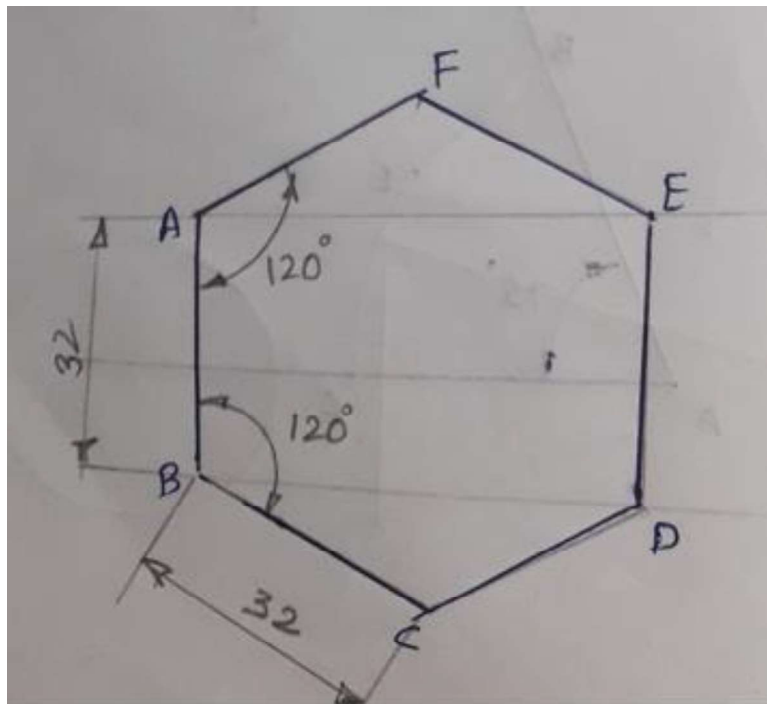
<https://www.youtube.com/watch?v=oKD-dTA3J8g>



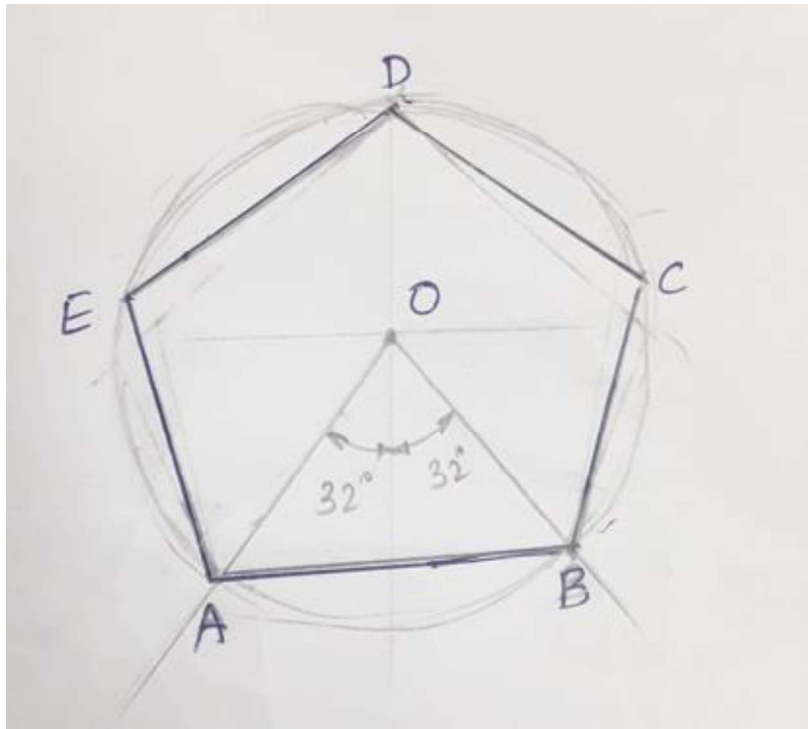
SHEET No. 3 (UNIT-1) POLYGONS

1(d) Construct a regular hexagon of side 32mm when one side is vertical.

<https://www.youtube.com/watch?v=hsglws8A2mg>



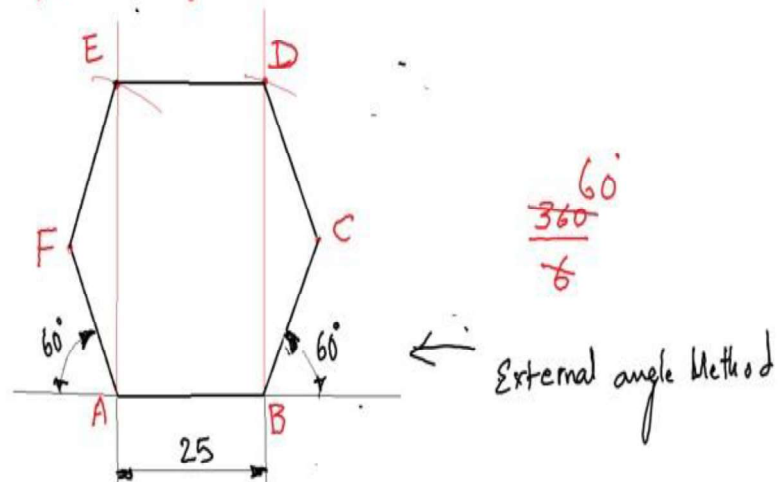
1(e) Inscribe a regular polygon of sides, say 5, in a circle of 64mm diameter when one side is horizontal.



1 (f) Construct a polygon, say hexagon of length of side 35mm using External Angle method.

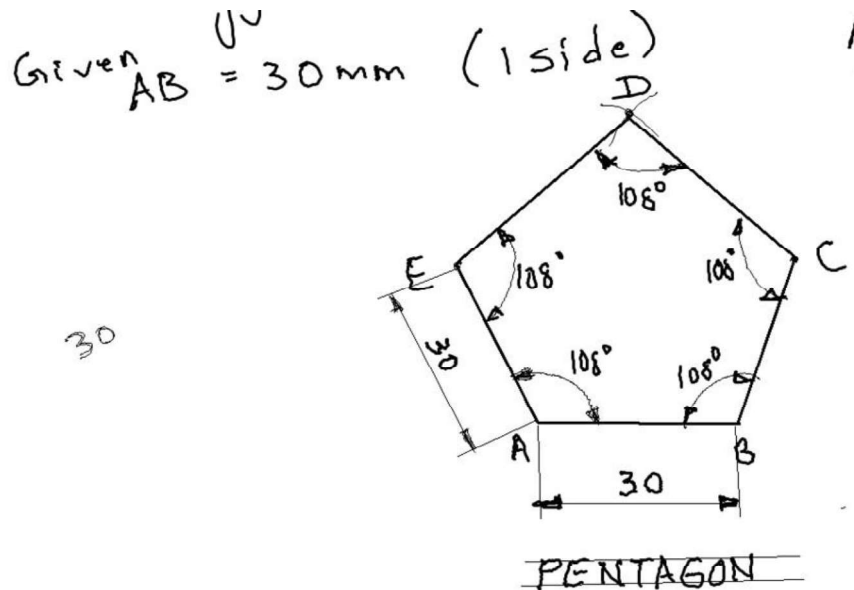
<https://www.youtube.com/watch?v=6Ig6PX0ee1M>

Construction 6: External Angle Method
Construct a polygon, say hexagon of length of side 25mm



1(g) Construct a polygon (say pentagon) of 30mm side using base angle of side.

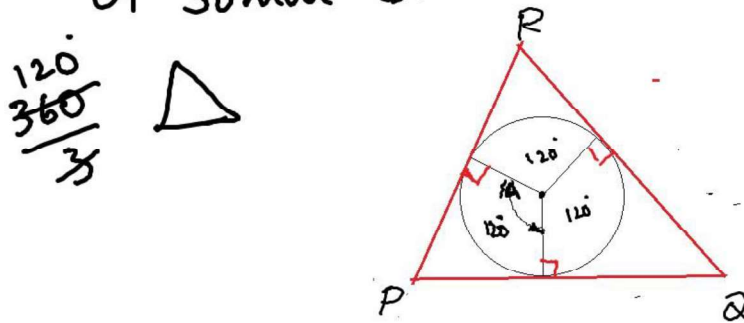
<https://www.youtube.com/watch?v=6Ig6PX0ee1M>



1(h) Circumscribe an equilateral triangle about a circle of 30mm diameter.

<https://www.youtube.com/watch?v=dKNW44tWWL4>

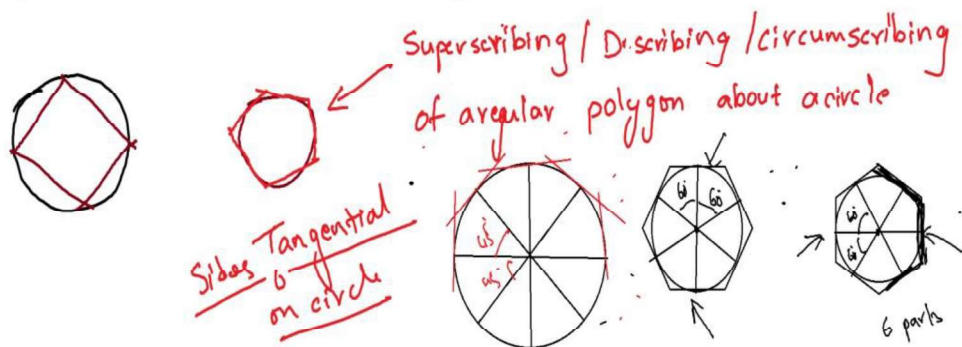
① Circumscribe an equilateral triangle about a circle of 30mm diameter.



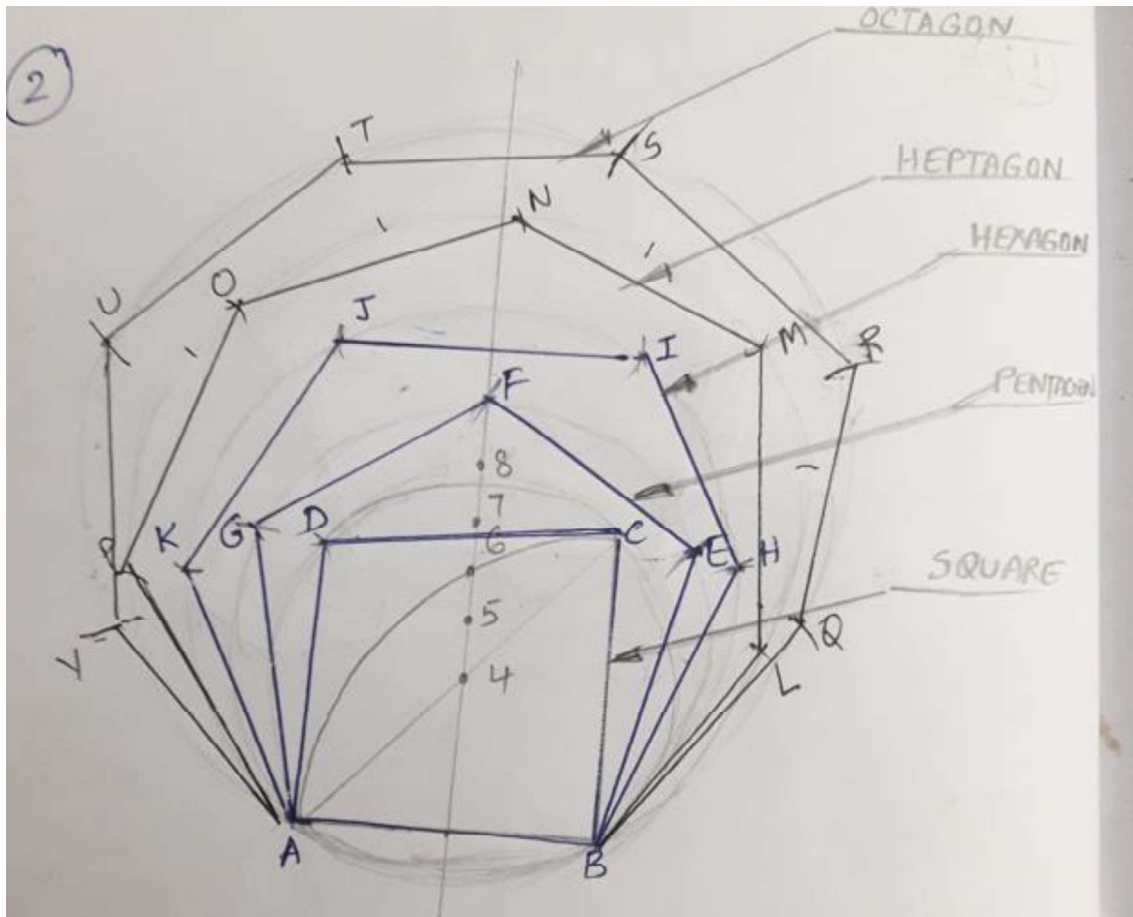
1(i) Circumscribe / Describe / Superscribe of a regular polygon about a circle. (say 4, 5, 6, 7, 8)

<https://www.youtube.com/watch?v=R2DXOgoT-po>

Superscription / Circumscription of Polygons



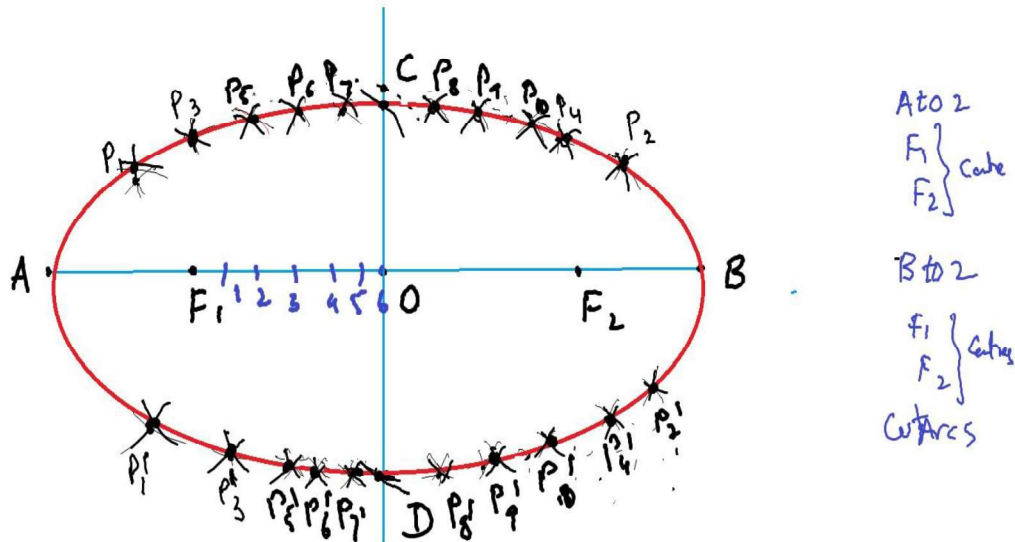
2. Construct a square, pentagon, hexagon, heptagon and octagon of 40mm side using general construction method. <https://www.youtube.com/watch?v=xHm7hB2iBgI>
<https://www.youtube.com/watch?v=NSDGN3EZr70>



SHEET No. 4 (UNIT-1) CONIC SECTIONS

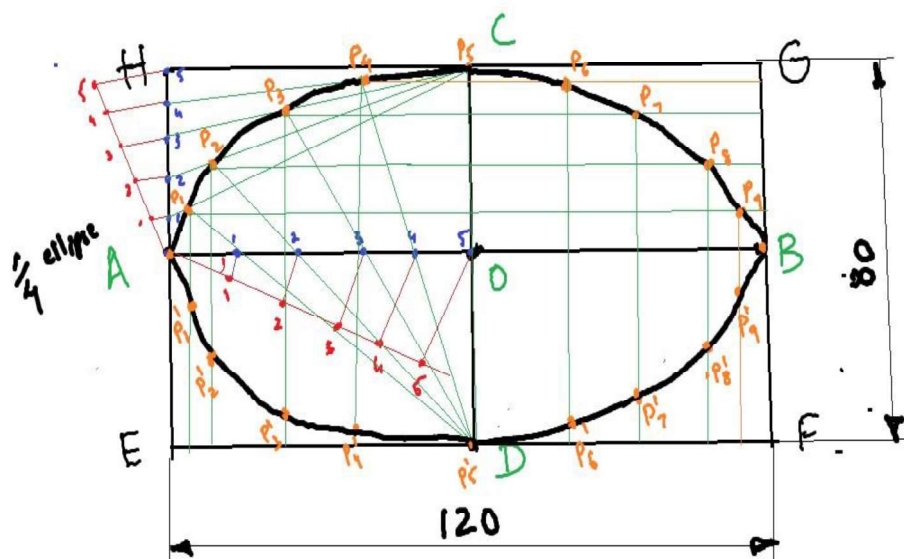
3) The major axis of an ellipse is 100mm and the minor axis is 55mm. Find the foci and construct the ellipse by Intersecting Arcs Method.

<https://www.youtube.com/watch?v=wIWrw1vhaI>



4. The major & minor axes of an ellipse are 120mm and 80mm respectively. Draw an ellipse by using Oblong Method (Rectangle Method) **Video Link**

<https://www.youtube.com/watch?v=yq65RZTqIPs>



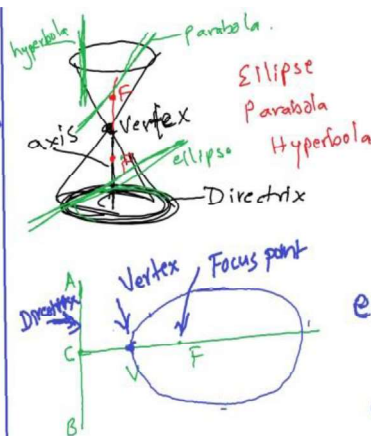
Draw a tangent & normal at a point 70mm from the directrix. (Eccentricity Method)

$e = \frac{VF}{VC} = \frac{2}{3}$
 VF with compass
 V as centre
 draw arc on V line
 Join CE
 For Normal
 $TT \perp$ Draw
 from

1 to 1' distance
 Fascentre
 Draw arcs
 on 1 1'
 line
 2-2'
 Fascentre
 2-2'
 line
 curves

Eccentricity method

- Draw a tangent & normal
at a point 70 mm from
the directrix.



$$e = \frac{VF}{VC} = \frac{\text{Vertex to Focus Distance}}{\text{Vertex to Directrix Distance}}$$

$$e = \frac{50}{70} < 1 \text{ ellipse}$$

$CF=50$ $e=1$ Parabola

$e > 1$ $\frac{30}{20} > 1$ Hyperbola

505 parts $2+3=5$

6. Draw a parabola when a distance between its focus and directrix is 50mm .

Draw a tangent & normal at a point 70mm from the directrix. (Eccentricity Method)

https://www.youtube.com/watch?v=KoPf_7N3Los

$e < 1$ ellipse
 $e = 1$ Parabola
 $e > 1$ hyperbola

$$e = \frac{VF}{VC} = \frac{\text{Vertex to Focus}}{\text{Vertex to Directrix}}$$

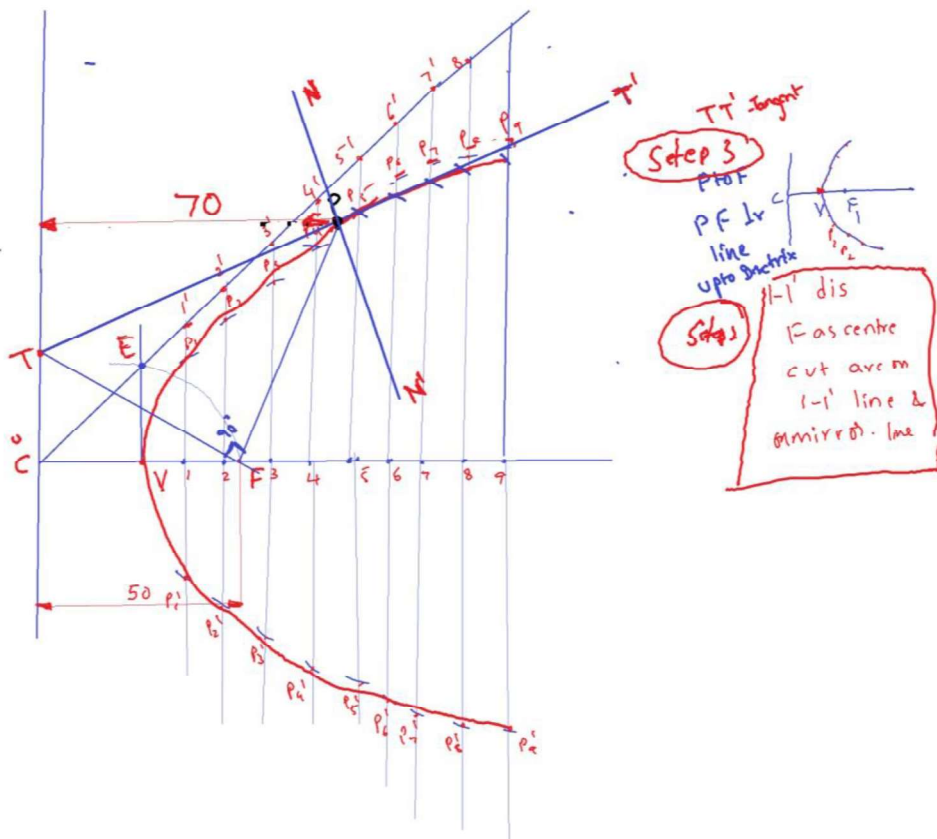
(2) Draw a parabola when the distance b/w its focus & directrix is 50 mm. Also draw a tangent & a normal at a point 70mm from the directrix.

Given : $FC = 50 \text{ mm}$
 $FC = 50 \text{ mm}$

$$e = 1 = \frac{VF}{VC} = 1$$

Step 1
 V as centre
 VF as radius
 Cut arc

ANSWER:-

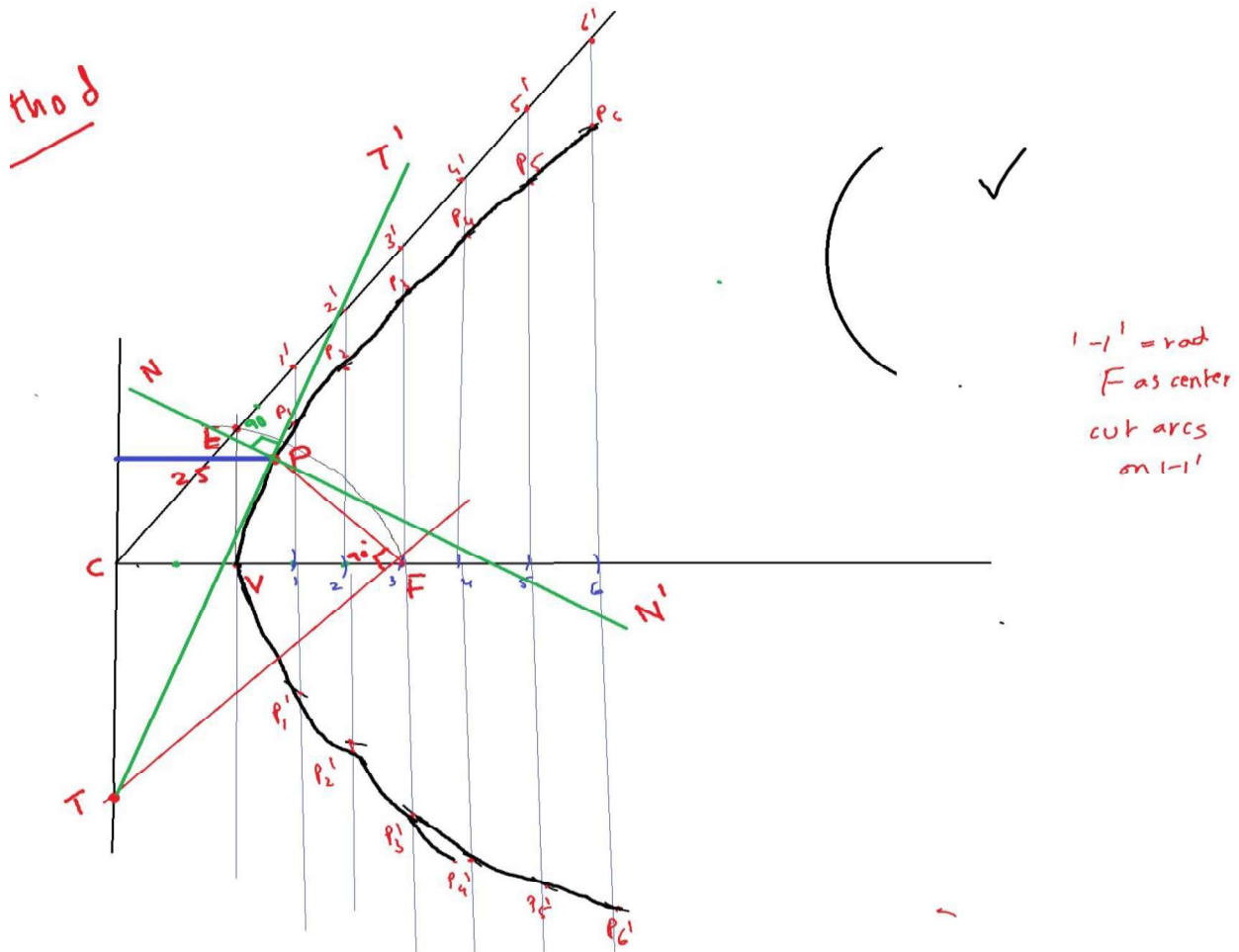


7. Draw a hyperbola when a distance of its focus from directrix is 50mm & eccentricity is $\frac{3}{2}$.

Also draw a tangent & normal to the hyperbola at a point 25mm from the directrix.

(Eccentricity Method)

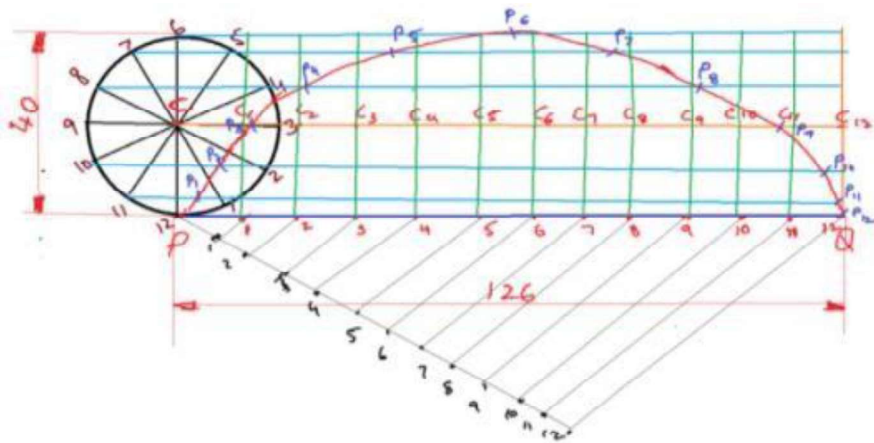
[Hyperbola By Eccentricity Method//Engineering Drawing - YouTube](#)



SHEET No. 5 (UNIT-1) CYCLOIDS AND INVOLUTES

8. A coin of 40mm diameter rolls over a horizontal table without slipping. A point on the circumference of the coin is in contact with the table surface in the beginning and after one complete revolution. Draw the path traced by the point.

<https://www.youtube.com/watch?v=Rr1UhFOU8sg>

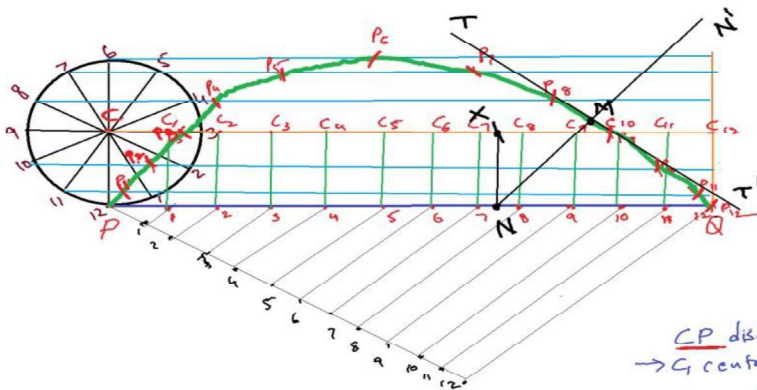


$\phi = 40\text{mm}$
 $r = 20\text{mm}$
 Divide
 circle
 into 12
 equal parts

$$PQ = 2\pi r = 2\pi(20) = 126$$

9. Draw a cycloid of a circle of diameter 50mm for one revolution. Also draw a tangent & normal to the curve at a point 35mm above the base line.

above the base line.



M as centre
 $MX \rightarrow 25\text{mm}$

Take circle radius
 & cut on centre
 line

$\phi = 50\text{mm}$
 $r = 25\text{mm}$

Divide
 circle
 into 12
 equal parts

$$PQ = 2\pi r$$

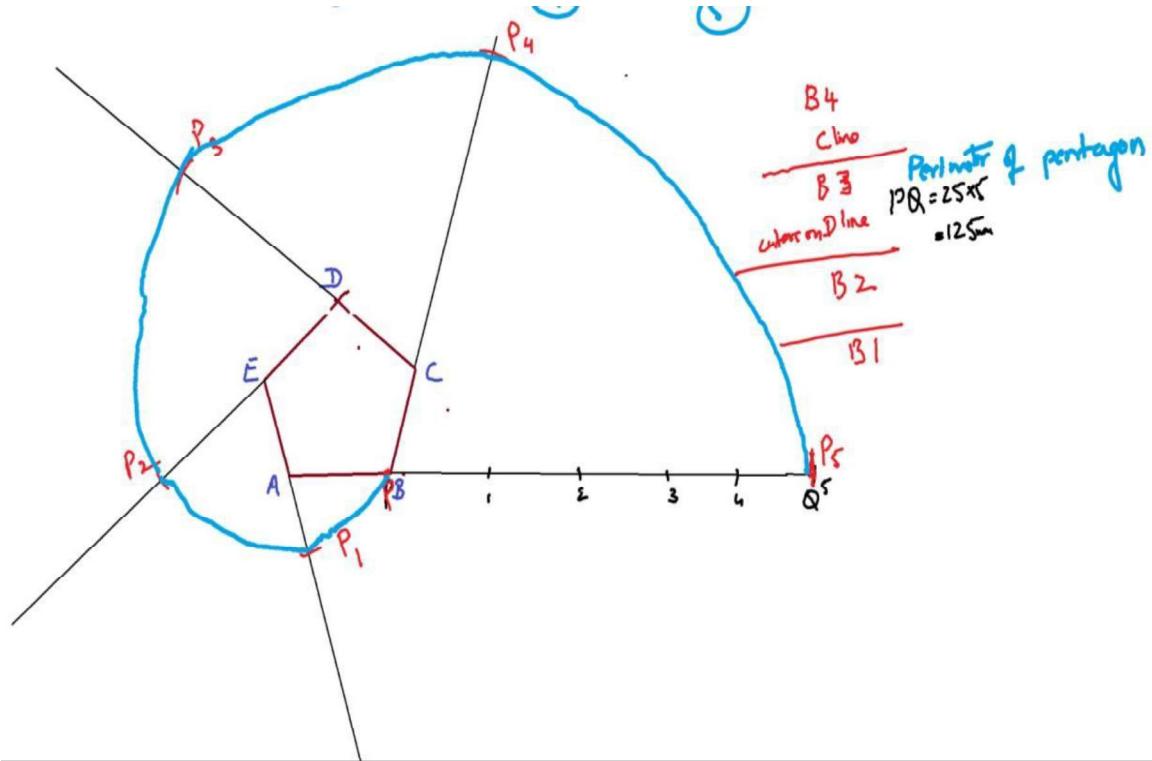
Take CP as radius
 C as centre
 arc on line 1

CP distance.
 $\rightarrow$ C centre draw
 arc on line 1

$$PQ = 2\pi r = 2\pi(25) = 157\text{mm} \\ = 15.7\text{cm}$$

10. Draw an involute of a pentagon of side 25mm.

<https://www.youtube.com/watch?v=vexwqVegQVw>

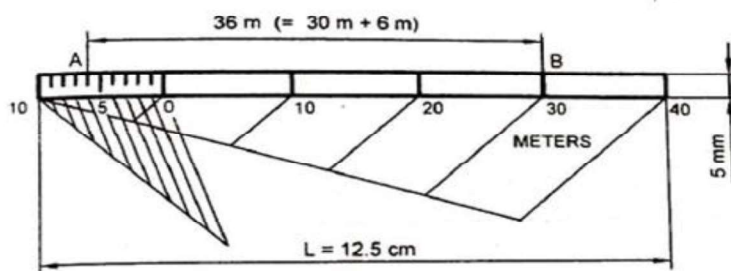


SHEET No. 6 (UNIT-1) SCALES

11.

Construct a plain scale to show meters when 1 centimeter represents 4 meters and long enough to measure up to 50 meters. Find the R.F. and mark on your scale a distance of 36 meters.

Answer



Note: Never rub-off the construction lines.

R.F. = 1/400

FIG. 1 PLAIN SCALE

Explanation

1. R.F. = $\frac{\text{Drawing size}}{\text{Actual size}}$ (in same units) = $\frac{1 \text{ cm}}{4 \times 100 \text{ cm}} = \frac{1}{400}$ [or Scale 1 : 400]
2. Length of scale = $L = \frac{\text{R.F.} \times \text{Maximum length to be measured}}{1} = \frac{1}{400} \times 50 \text{ m} = \frac{1}{400} \times 50 \times 100 \text{ cm} = 12.5 \text{ cm}$
3. Draw a horizontal line of length $L = 12.5 \text{ cm}$. Draw a rectangle of size $12.5 \text{ cm} \times 0.5 \text{ cm}$ on the horizontal line drawn above.

Note: Width of the scale is usually taken as 5 mm.

1. **Main Unit:** Total length to be measured is 50 m. Therefore divide the rectangle into 5 (= n) equal divisions, each division representing 10 m. [For dividing the length L into n number of equal parts, use the Geometrical Construction Method in Chapter 3.]

5. Mark 0 (zero) at the end of the first main division. From 0, number 10, 20, 30 and 40 at the end of subsequent main divisions towards **right** as shown.

Note: In general purpose scales, zero is marked at the beginning. But, in engineering scales, zero will be marked at the end of the first main division. This will enable sub-dividing the first main unit into sub-divisions.

6. **Sub-divisions of Main Unit:** By geometrical construction, divide the first main division into 10 sub-divisions to represent meters. Number the sub-divisions i.e., meters to the **left** of 0.
7. Write the names of main units and sub-units as METERS below the scale. Mark R.F. value.
8. Mark AB = 36 meters = 3 main divisions to the **right** side of 0 (zero) + 6 sub-divisions to **left** of 0 (zero) = 30 m + 6 m.

Construct a plain scale of R.F. 1 : 50,000 to show kilometers and hectometers and long enough to measure up to 7 kilometers. Measure a distance of 54 hectometers on your scale.

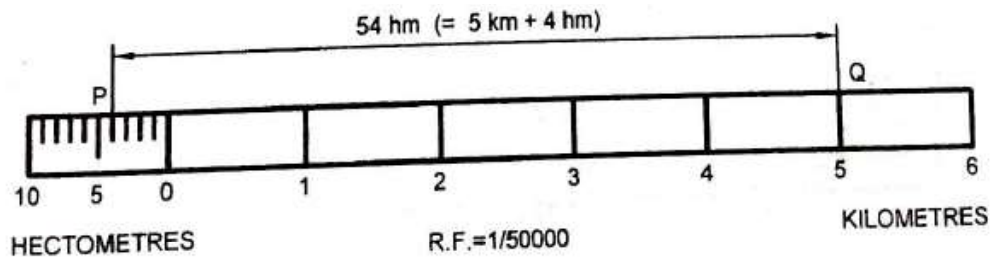


FIG. 2 PLAIN SCALE

1. Length of scale = $L = \frac{1}{50,000} \times 7 \text{ km} = \frac{1}{50,000} \times 7 \times 1000 \times 100 \text{ cm} = 14 \text{ cm}.$
2. Draw a rectangle of size 14 cm x 0.5 cm. Divide the rectangle into 7 equal divisions, each representing 1 km.
3. Mark 0 (zero) at the end of the first main division and 1, 2, 3, ..., 6 at the end of the subsequent main divisions towards right. Sub-divide the first division into 10 sub-divisions, each representing 1 hm. Number the sub-divisions to the left of 0 (zero).
4. Mark $PQ = 54 \text{ hm} = 5 \text{ km} + 4 \text{ hm}.$

Construct a diagonal scale of R.F. = 1 : 32,00,000 to show kilometers and long enough to measure upto 400 km. Show distances of 257 km and 333 km on your scale.

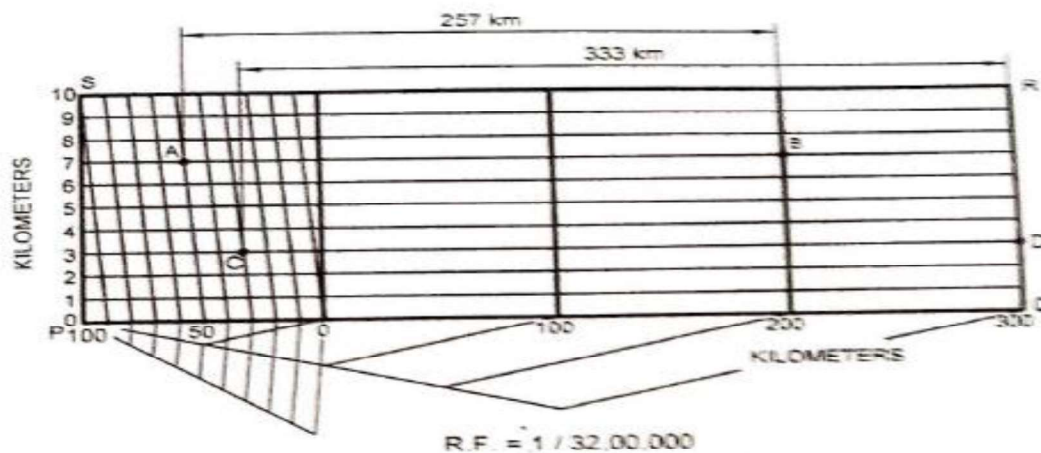


FIG. 7 DIAGONAL SCALE

1. Length of the scale = $L = R.F. \times \text{Maximum distance to be measured}$
 $= \frac{1}{32,00,000} \times 400 \text{ km} = \frac{1}{32,00,000} \times 400 \times 1000 \times 100 \text{ cm.} = 12.5 \text{ cm.}$
 2. Draw a line PQ of 12.5 cm long (L).
 3. Maximum length to be measured is 400 km. Minimum distance to be measured = 1 km (which is obtained from data 257 / 333 km).
 $\therefore \text{Max / Min} = 400 \text{ km} / 1 \text{ km} = 400$. This can be obtained in 3 steps as $4 \times 10 \times 10$.
 4. By geometrical construction, divide PQ into 4 main divisions, each representing 100 km. Mark 0 (zero) at the end of the first main division. Also mark 100, 200 and 300 towards the right of zero.
 5. Using geometric construction, sub-divide the first main division into 10 equal sub-divisions, each representing 10 km. [To avoid crowding of numbers mark only 50, 100 towards the left of zero.]
 6. Draw a line PS of 5 cm long perpendicular to PQ. Complete the rectangle PQRS.
- Note:** Height PS may conveniently be taken as 5 cm for $L = 12.5 \text{ cm}$.
7. Draw vertical lines from each main division on PQ.
 8. Divide PS into 10 equal divisions. Name the divisions as 0, 1, ..., 10 from P to S.
 9. Draw horizontal lines from each division on PS.
- To construct the Diagonal**
10. Join S to the first sub-division from P on the main scale PQ. Thus first diagonal line is drawn.
 11. Similarly draw remaining 9 diagonals parallel to the first diagonal. Thus each 10 km is divided into 10 equal parts.
 12. $AB = 257 \text{ km} = 200 \text{ km on Main division} + 50 \text{ km on Sub-division} + 7 \text{ km on Diagonal.}$
 13. $CD = 333 \text{ km} = 300 \text{ km on Main division} + 30 \text{ km on Sub-division} + 3 \text{ km on Diagonal.}$

The Marina Beach at Chennai, is 2.5 km long. On inspection of the road map, its equivalent distance measures 5 cm. Devise a diagonal scale to read 50 m minimum. Mark on the scale a distance of 6350 m.

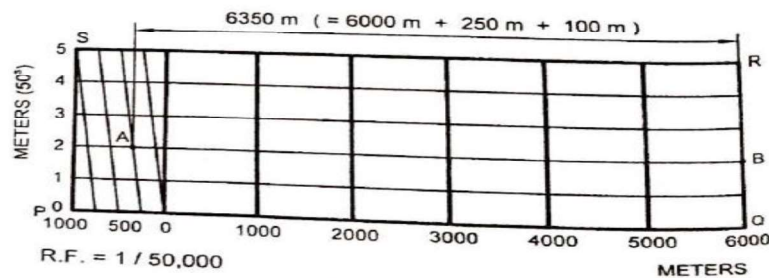


FIG. 9 DIAGONAL SCALE

1. $R.F. = \frac{5 \text{ cm}}{2.5 \text{ km}} = \frac{1}{50,000}$
2. $L = R.F. \times 7000 \times 100 \text{ cm} = 14 \text{ cm}$ (since a distance of 6350 m is to be marked).
3. Draw a rectangle PQRS of size 14 cm x 5 cm.
4. Maximum = 7000 m;
Minimum = 50 m;
Max/Min = 140 = 7 x 4 x 5.

Construct a vernier scale to read meters, decimeters and centimeters and long enough to measure up to 4 m. R.F. of the scale is 1/20. Mark on your scale a distance of 2.28 m. (Backward / Retrograde Vernier)

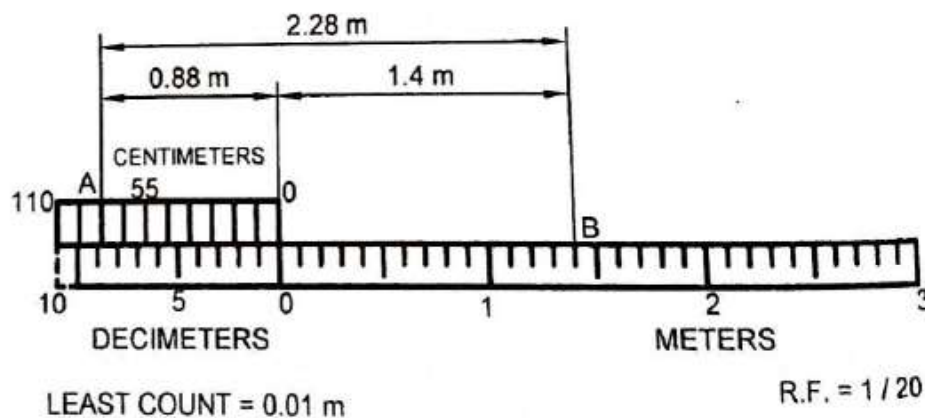


FIG. 16 BACKWARD / RETROGRADE VERNIER SCALE

1. Least Count = Smallest distance to be measured = 1 cm (given) = 0.01 m
2. $L = R.F. \times \text{Maximum distance to be measured} = (1/20) \times 4 \text{ m} = 20 \text{ cm}$

3. **Main Scale:** Draw a line of 20 cm length. Complete the rectangle of 20 cm x 0.5 cm.

Divide it into 4 equal parts each representing 1 meter. Sub-divide each

part into 10 main scale divisions. Hence 1 m.s.d. = $1 \text{ m} / 10 = 0.1 \text{ m} = 1 \text{ dm}$.

4. **Backward Vernier:** Take 11 divisions on main scale. Divide it into 10 equal parts on vernier scale. So, 1 v.s.d. = $11 \text{ m.s.d.} / 10 = 11 \times 1 \text{ dm} / 10 = 0.11 \text{ m} = 1.1 \text{ dm} = 11 \text{ cm}$.

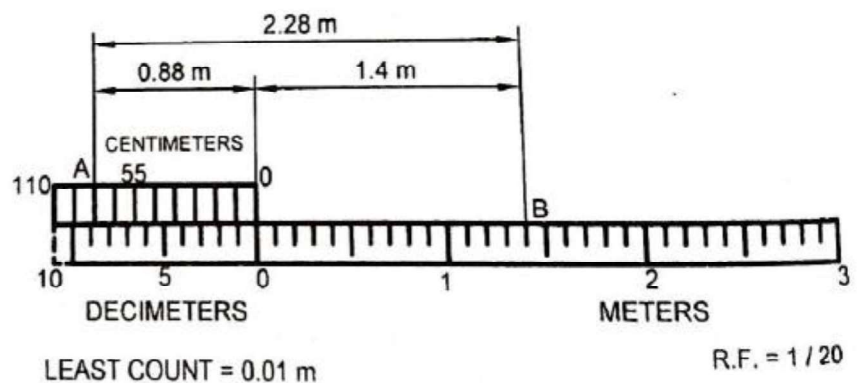


FIG. 16 BACKWARD / RETROGRADE VERNIER SCALE

Construct a vernier scale to show readings of $1/10^{\text{th}}$ of a meter when 3 cm represents 10 m. Construct the scale to read up to 60 m and mark the distances of 35.3 m and 47.3 m on your scale. (Backward Vernier)

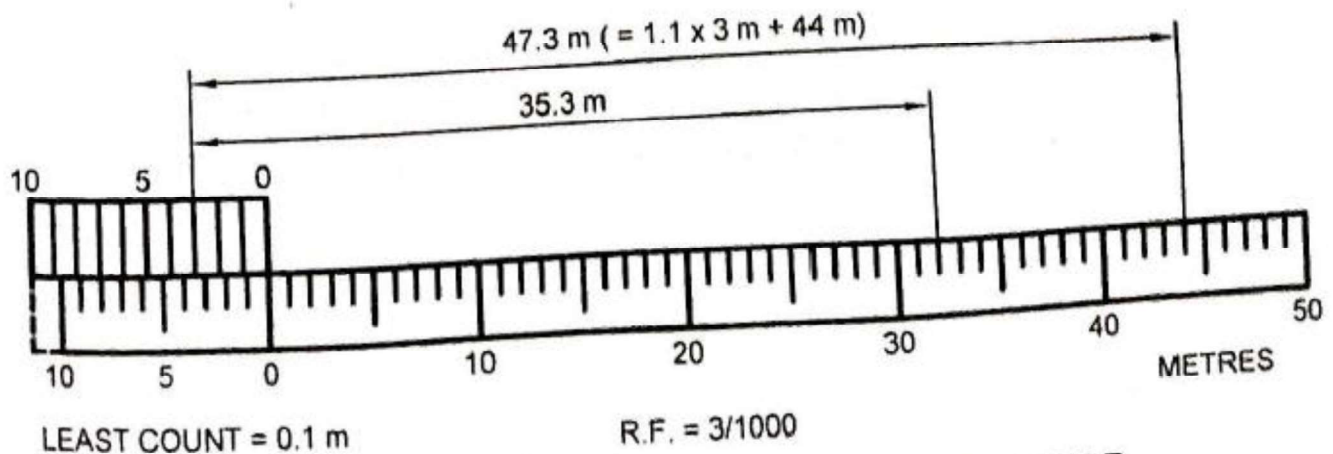


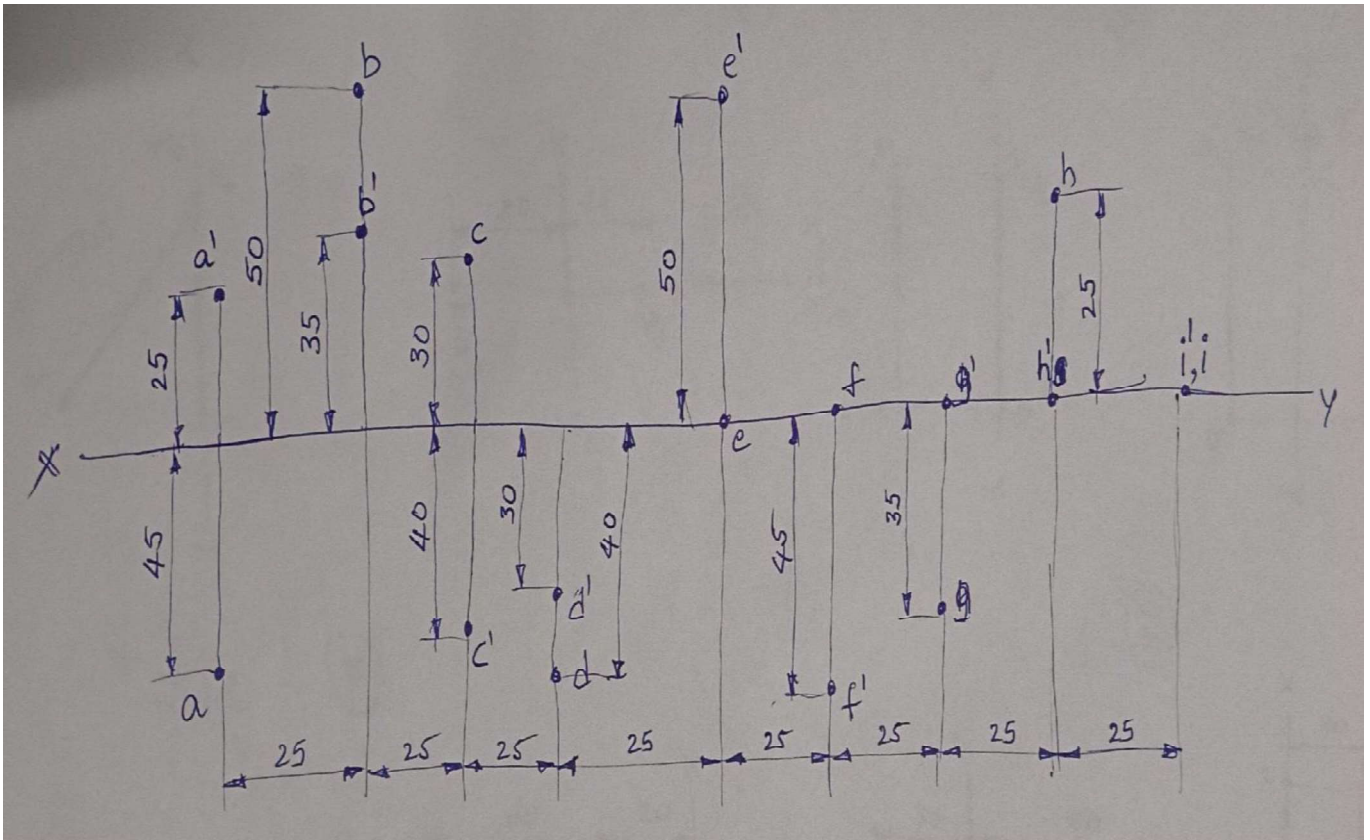
FIG. 18 BACKWARD / RETROGRADE VERNIER SCALE

1. L.C. = Smallest distance to be measured = $1 \text{ m} / 10 = 0.1 \text{ m}$.
2. Length of the Scale = $L = (3 \text{ cm} / 10 \text{ m}) \times 60 \text{ m} = 18 \text{ cm}$. Draw the main scale of 18 cm length. Divide it into 6 equal parts so that each part represents 10 m.
3. Each part is further divided into 10 main scale divisions. Therefore, 1 m.s.d. = 1 m.
4. To construct the vernier, take 11 divisions on the main scale. Divide it into 10 equal parts on the vernier.
5. 1 v.s.d. = $11 \times (1 \text{ m} / 10) = 1.1 \text{ m}$

SHEET No. 7 (UNIT-II)

PROJECTIONS OF POINTS:

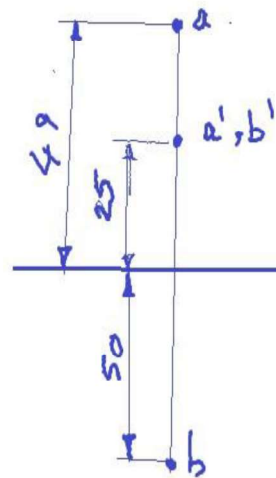
1. Draw the projections of the following points, keeping the distance between the projectors as 25 on the same reference line
 - A-25 above HP and 45 in front of VP
 - B-35 above HP and 50 behind VP
 - C-40 below HP and 30 behind VP
 - D-30 below HP and 40 in front of VP
 - E-50 above HP and on VP
 - F-45 below HP and on VP
 - G-On HP and 35 in front of VP
 - H-On HP and 25 behind VP
 - I-On both HP and VP



2. A point at 25 above the reference line XY is the front view of two points A & B. The top view of A is 49 behind VP and the top view of B is 50 in front of VP. Draw the projections of the points and state their positions relative to the planes of projection and the quadrants in which they lie.

A lies in Q_2 (2nd Quadrant)
 B lies in Q_1 (1st Quadrant)

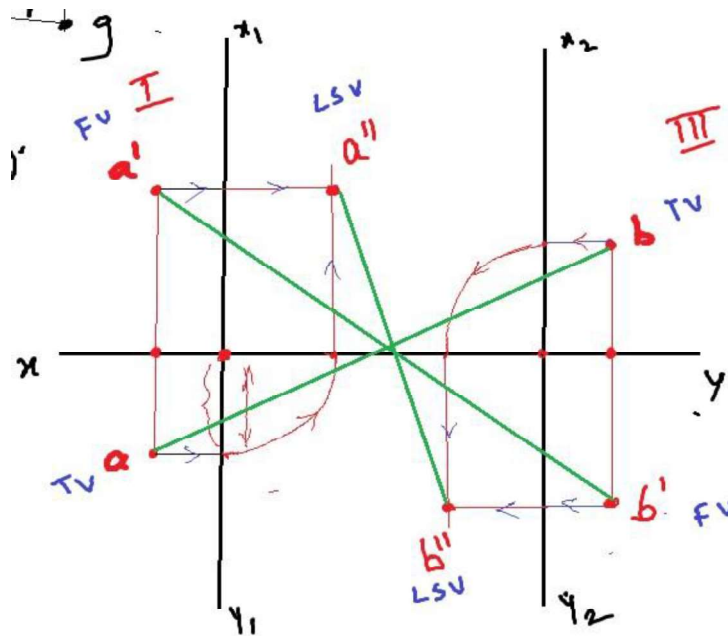
VP



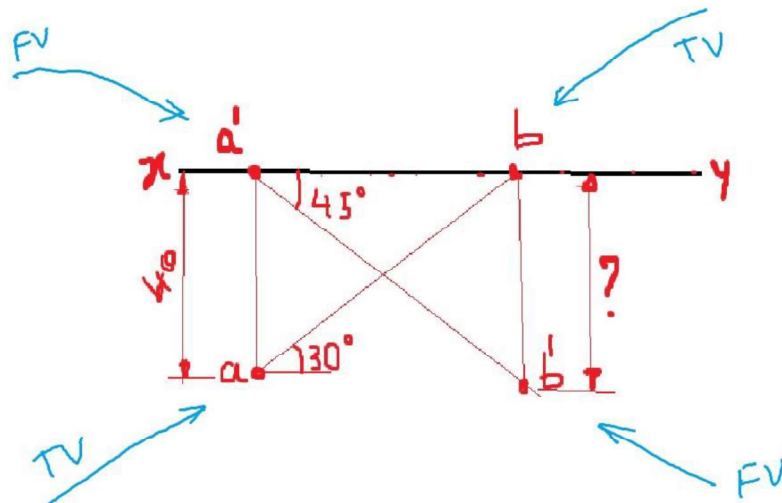
A 25 above HP
 & 49 behind VP

B 25 above HP
 & 50 in front VP

3. A point A is 40 above HP and 25 in front of VP. Another point B is 20 behind VP and 30 below HP. The horizontal distance between the points is 100. Draw the three projections of the points A&B and join their front views, top views and left side views.



4. A point A is on HP and 40 in front of VP. Another point B is on VP and below HP. The line joining their front views makes an angle of 45° with XY , while the line joining their top views makes an angle of 30° . Find the distance of the point B from HP.



SHEET No. 8 (UNIT-II)

PROJECTIONS OF LINES:

Line parallel to both the planes, Line parallel to one plane and inclined to other plane

1. Draw the projection of a straight line AB, of 60 long in the following positions:

(a) Parallel to both HP and VP and 20 from each

(b) Parallel to and 20 above HP and on VP.

(c) Parallel to and 30 in front of VP and on HP.

(d) Perpendicular to HP, 30 in front of VP and one end on HP.

(e) Perpendicular to HP, 30 in front of VP and one end 25 above HP.

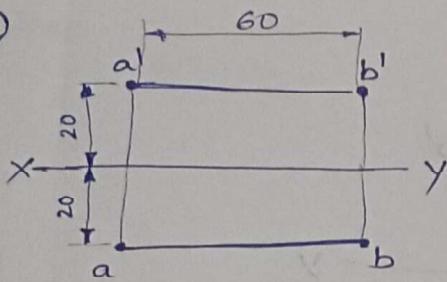
(f) Perpendicular to and 20 in front of VP and on HP.

(g) Inclined at 30^0 to VP, in HP and one end on VP.

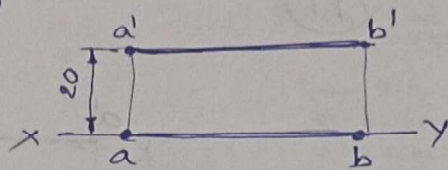
(h) Inclined at 45^0 to HP, one end 20 above HP and parallel to and 30 in front of VP

(i) Inclined at 60^0 to VP, one end 20 in front of VP and parallel to and 25 above HP

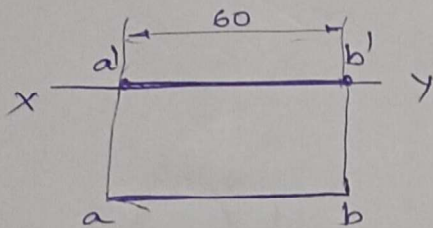
(1a)



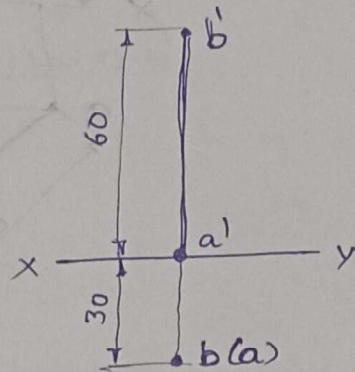
(1b)



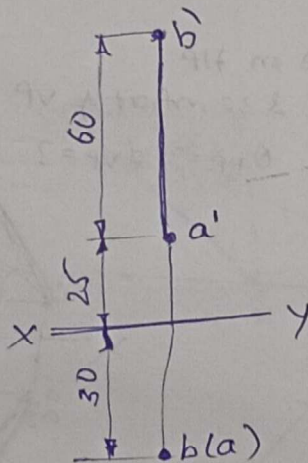
(1c)



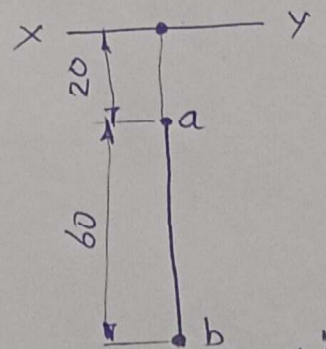
(1d)



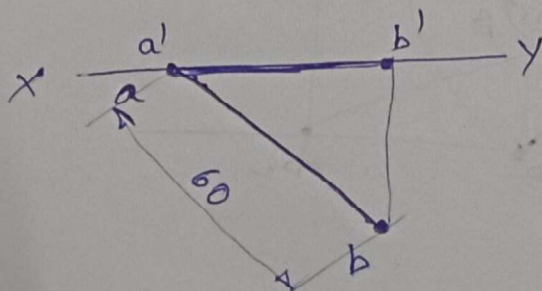
(1e)



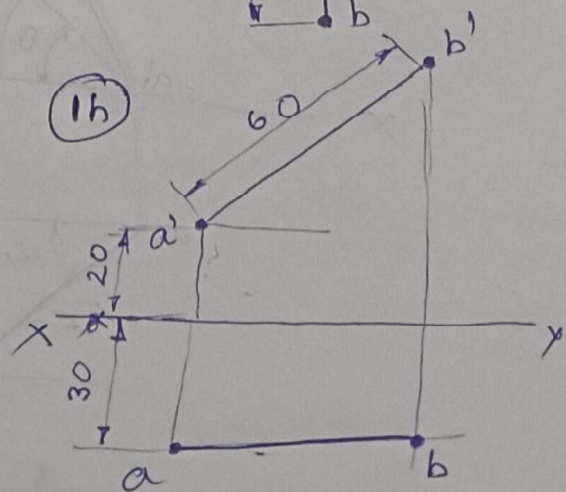
(1f)

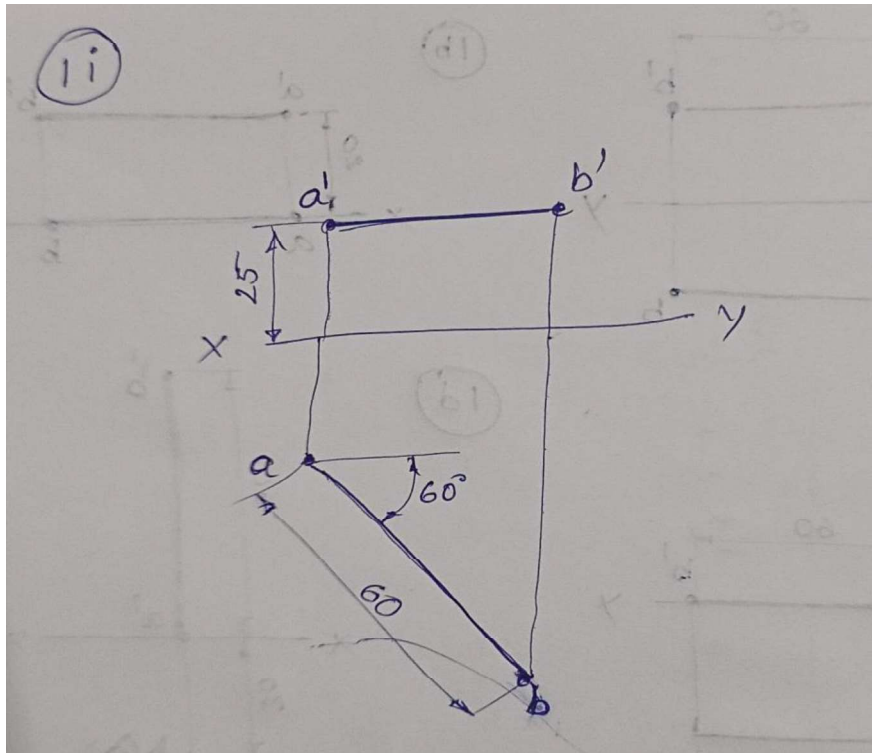


(1g)

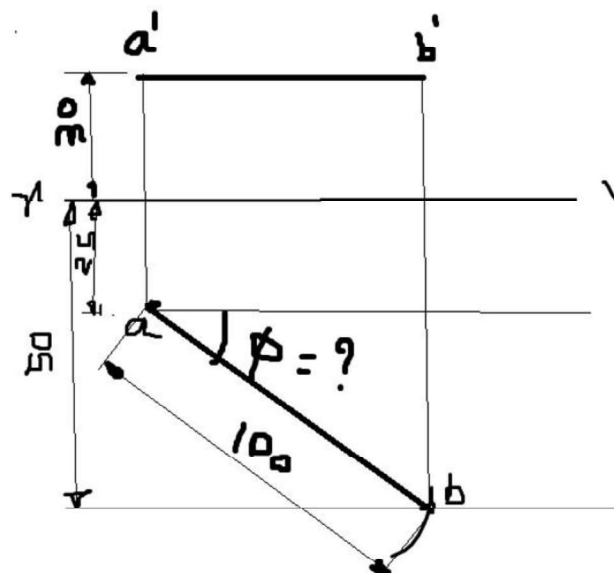


(1h)

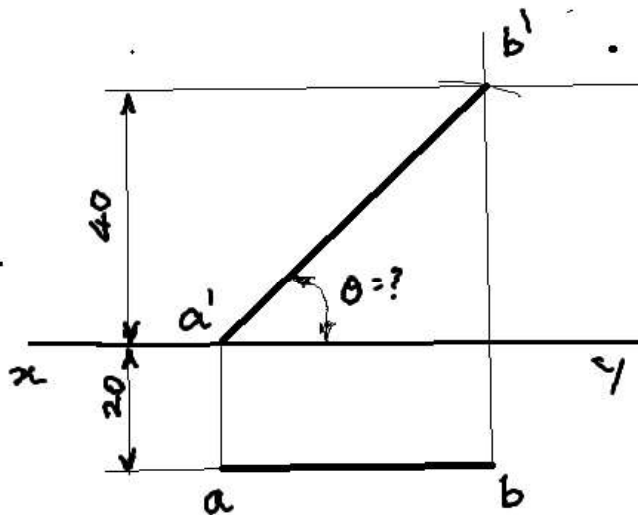




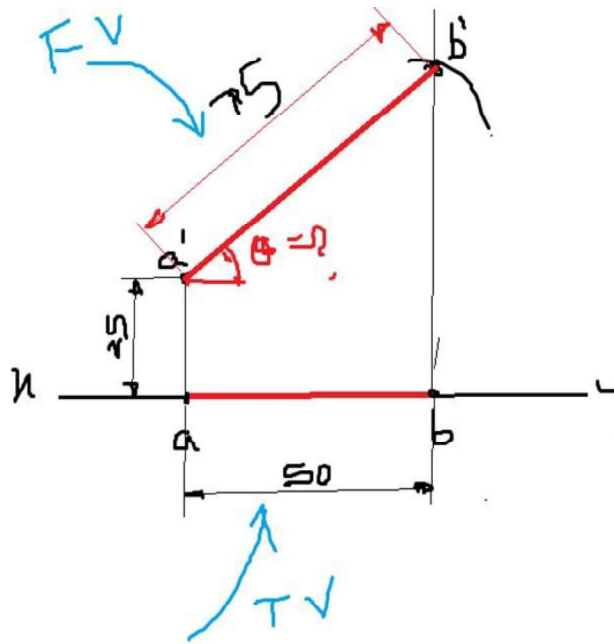
2. A line of 100 long is parallel to and 30 above HP. Its two ends are 25 and 50 in front of VP respectively. Find its inclination with VP.



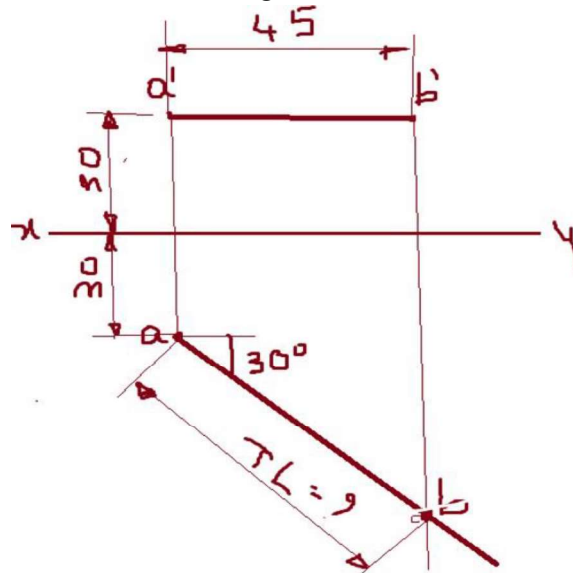
Explanation



4. The top view of a line, which is 75 long, measures 50. The line, is in VP, its one end being 25 above HP. Draw its projections.



5. The front view of a line, which is inclined at 30° to VP, is 45 long. Draw the projections of the line when it is parallel to and 30 above HP, its one end being 30 in front of VP. Determine the true length of the line.



SHEET No. 9 (UNIT-II)

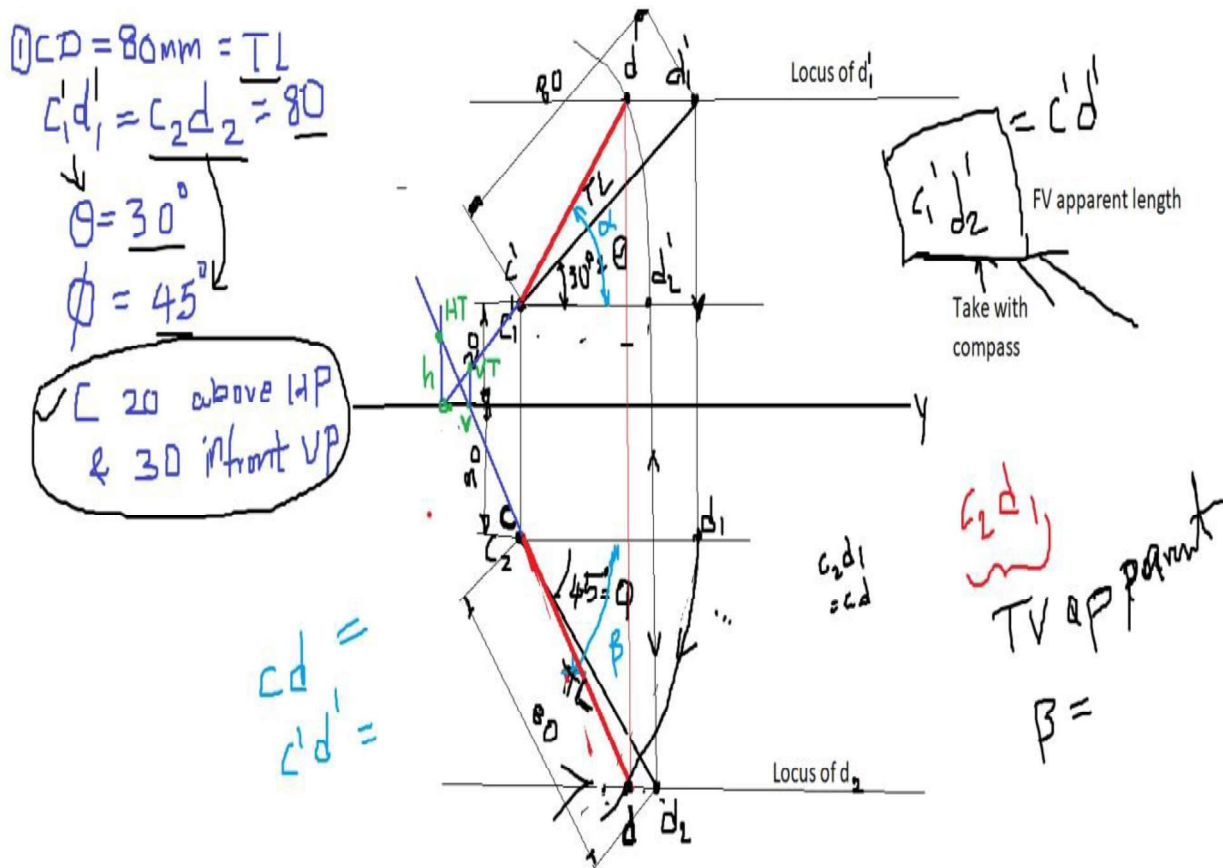
PROJECTIONS OF LINES:

Line inclined to both the planes

1. The line $CD = 80\text{mm}$ long, is inclined at 30° to H.P. and 45° to V.P. its one end C is 20mm above H.P. and 30mm in front of V.P. Draw its projections and locate its traces.

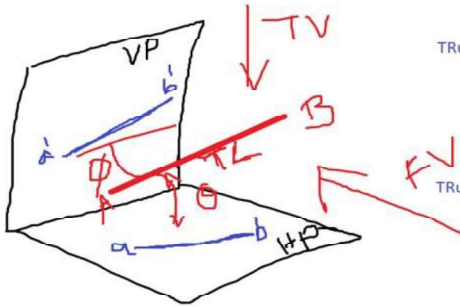
<https://www.youtube.com/watch?v=uCVMKQxnV9U>

Answer:



Line inclined to Both the planes Concept:-

Line is inclined to both the planes (HP & VP)

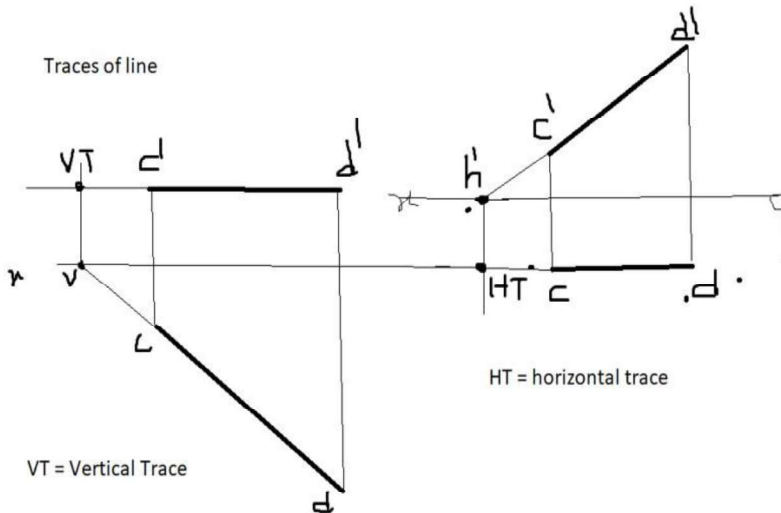


Step1
True length of line IS INCLINED TO HP $\rightarrow$
True length above XY line $\rightarrow$ FV $\rightarrow \theta$
Below XY $\rightarrow$ TV (Apparent length)
Apparent angle = α

Step2
True length of Line IS inclined to VP $\rightarrow \phi$
True length below XY line $\rightarrow$ TV
Above XY $\rightarrow$ FV (Apparent length)
Apparent angle = β

$a' b'$ = True length of a line in FV
 $a b$ = True length of a line in TV
 $a' b$ = Apparent length in FV
 Shorten length
 $a b'$ = Apparent length in TV
 Shorten length

Traces of line



HT = horizontal trace

VT = Vertical Trace

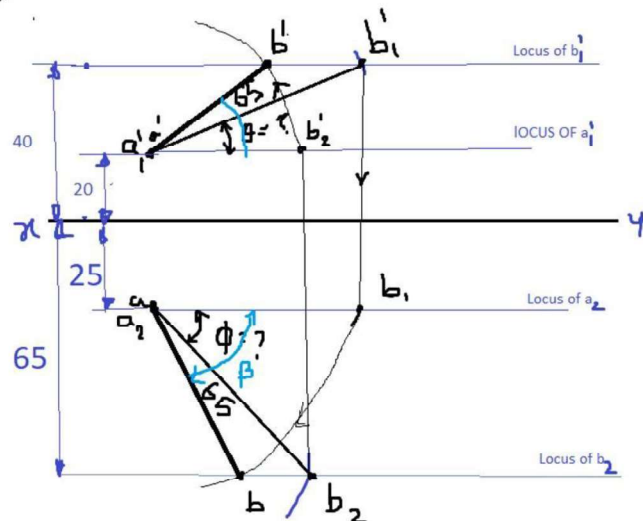
2. A line AB 65mm long has its end A 20mm above H.P and 25mm in front of V.P. the end B is 40mm above H.P. and 65mm in front of V.P. draw its projections and find its inclinations with the HP and VP. Mention the values of apparent angles also and locate its traces.

https://www.youtube.com/watch?v=8Y3_7Q1n1_w

GIVEN :
 $a_1b_1 = a_2b_2 = AB = TL = 65$
 A 20 above HP & a_1
 25 in front VP a_2
 B 40 above HP & b_1
 65 in front VP b_2

$a'b' = ?$
 $ab = ?$
 $\alpha = ?$
 $\beta = ?$
 HT, VT = ?

2



3. The mid point M of a straight line of AB is 40mm from H.P. and 50mm from V.P. the line AB is 70mm long and makes 40° with H.P and 30° with V.P. draw the projection of the line and locate its traces.

<https://www.youtube.com/watch?v=pq5669NIYdk>

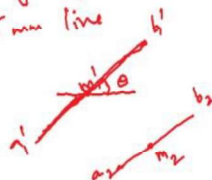
CONCEPT:-

3. The mid point M of a straight line of AB is 40mm from H.P. and 50mm from V.P. the line AB is 70mm long and makes 40° with H.P and 30° with V.P. draw the projection of the line and locate its traces.

$VT, HT = ?$

70mm = True length of line

$\frac{70}{2} = 35$ mm line



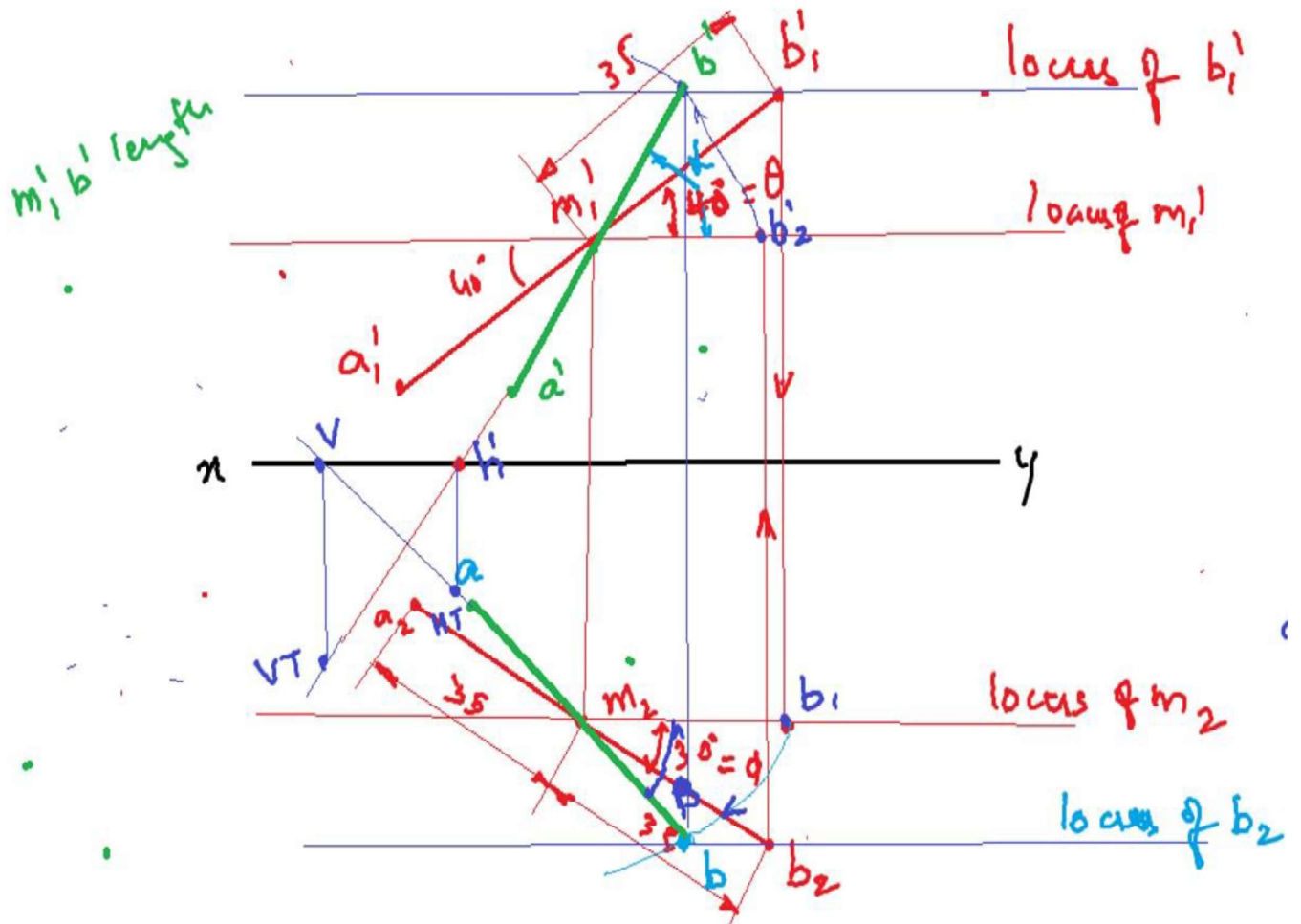
Apparent FV length $\rightarrow TV$
 $\alpha, \beta = ?$

$\frac{a'b'}{ab} = \frac{\alpha}{\beta}$
 apparent
 $m_1, b_2 =$ distance with compass
 Draw arc locus of b_1'

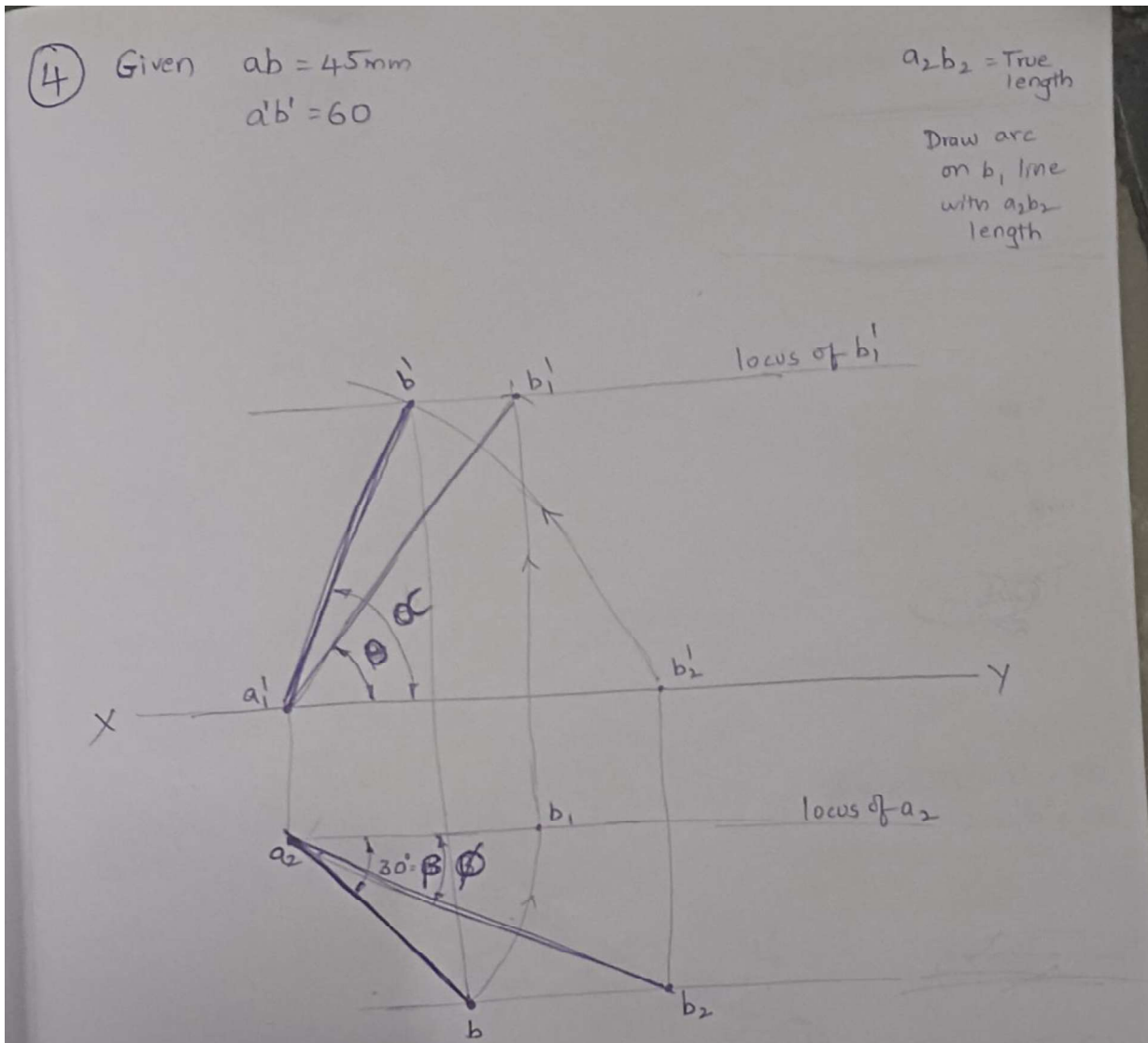
Take m_2, b_1 distance & draw arc on b_2 locus line

b_1b' should be on the line Vertical same

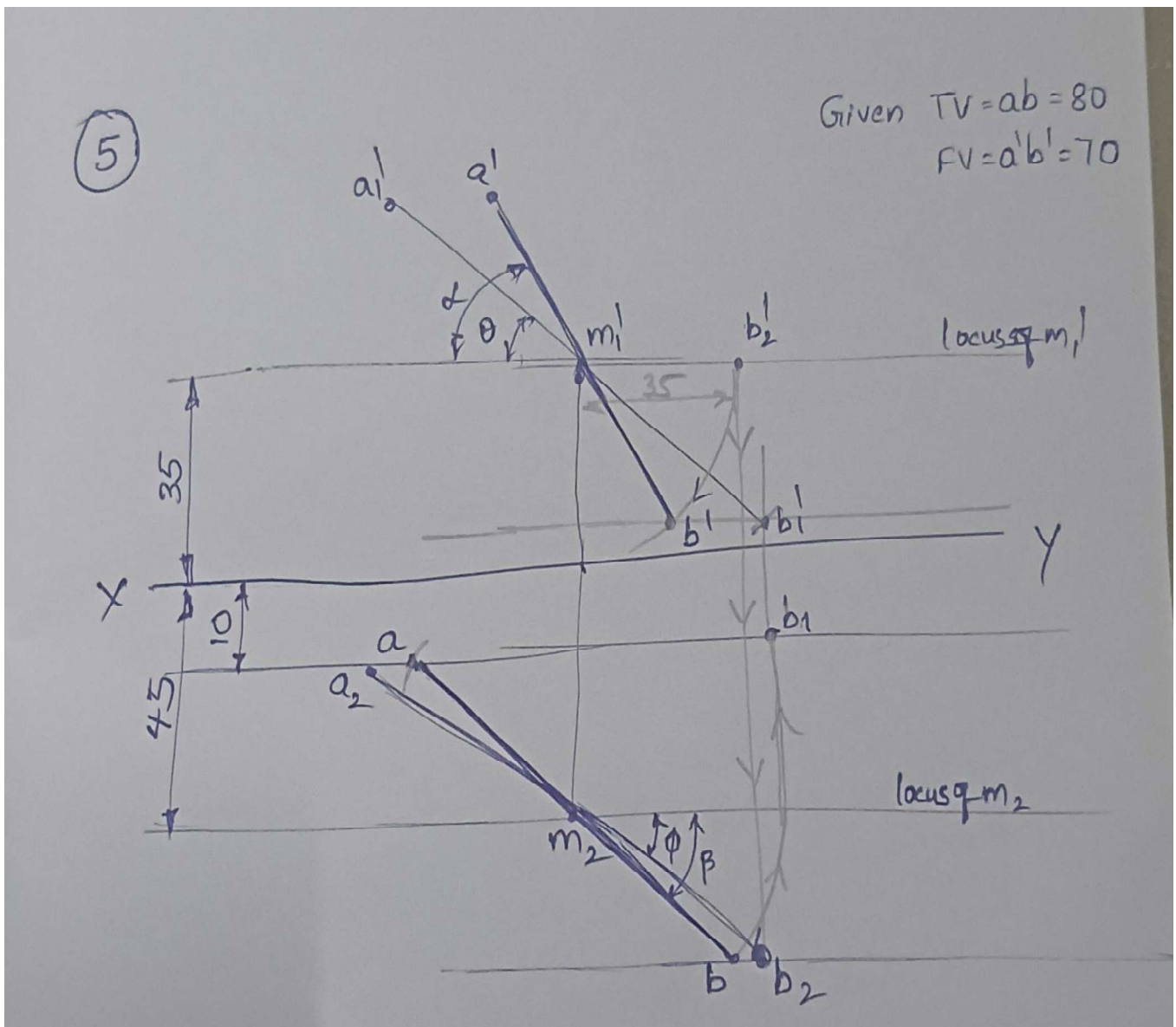
ANSWER



4. The length of the top view of a straight line is 45mm and the length of the front view is 60mm. The top view 'ab' is inclined at 30° to XY. Draw the projections of the line AB assuming the point A to be in the H.P and 30mm in front of V.P. Determine the length and inclination of the line with H.P and V.P and locate its traces.



- 5 The projections of a line measure 80mm in the top view and 70mm in the front view. The mid point of the line is 45mm in front of V.P. and 35mm above H.P. one end is 10mm in front of VP and nearer to it. The other end is nearer to H.P Draw the projections of the line; find its true length and true inclinations and locate its traces.

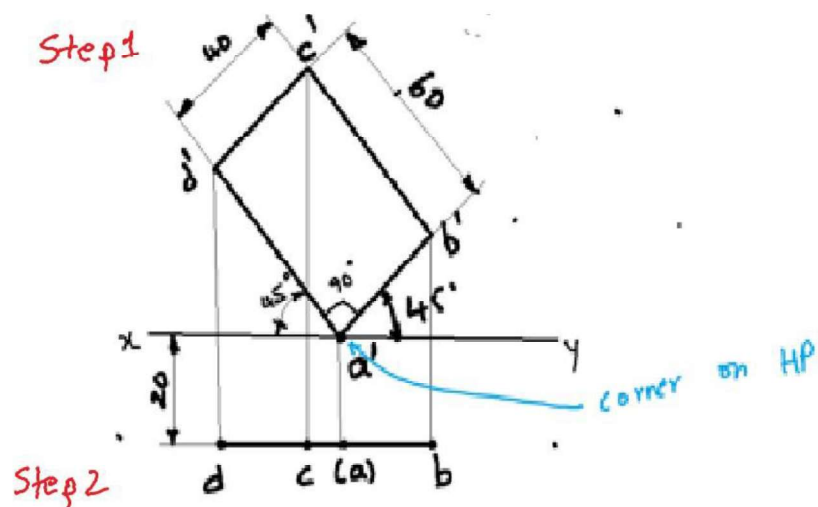


SHEET No. 10 (UNIT-II)

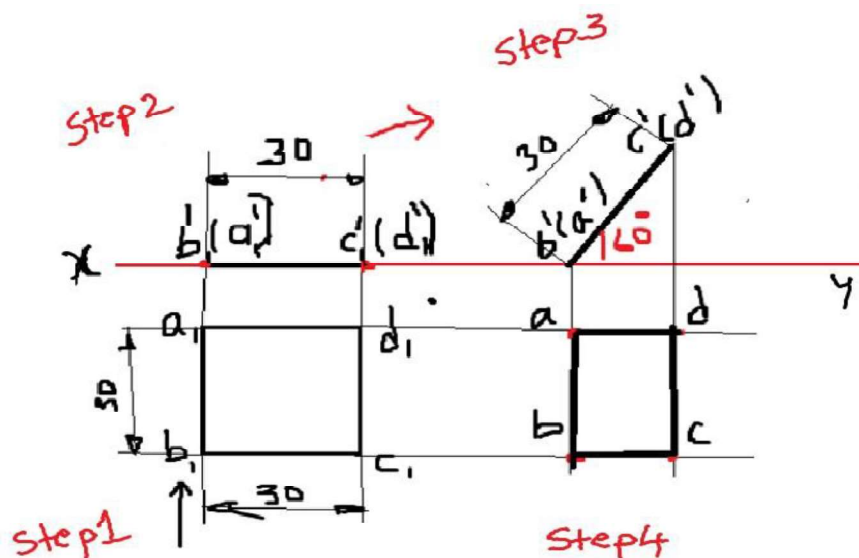
PROJECTIONS OF PLANES

Parallel/ Perpendicular to one plane and inclined to other plane

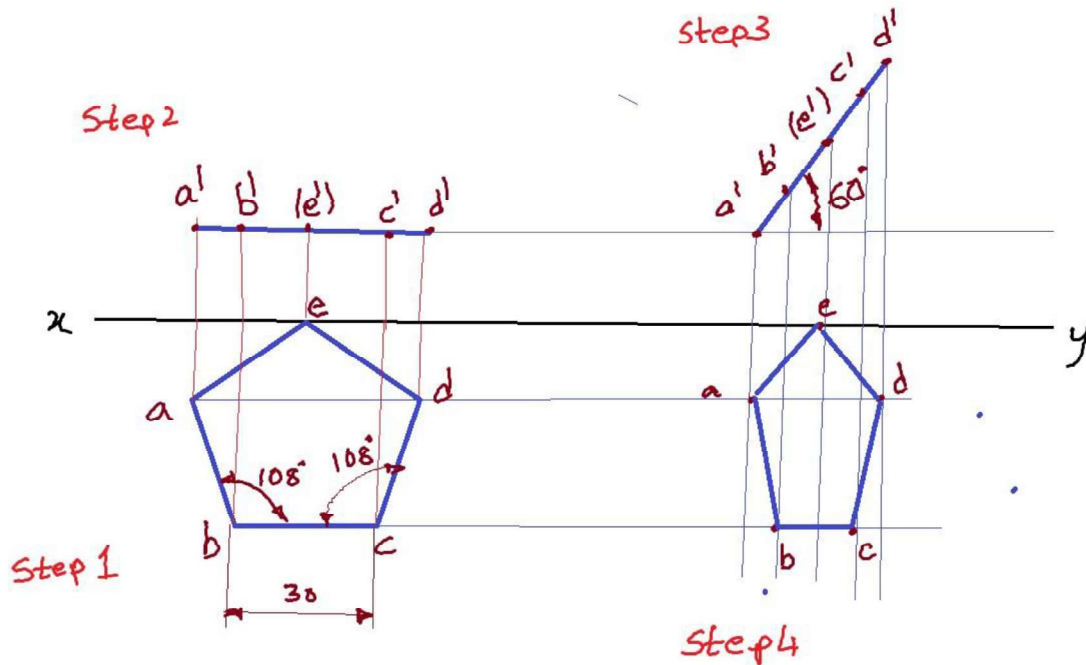
1. A rectangle ABCD of size 40x60 has a corner on H.P and 20 away from V.P. all the sides of the rectangle are equally inclined to H.P and parallel to V.P. draw its projections.



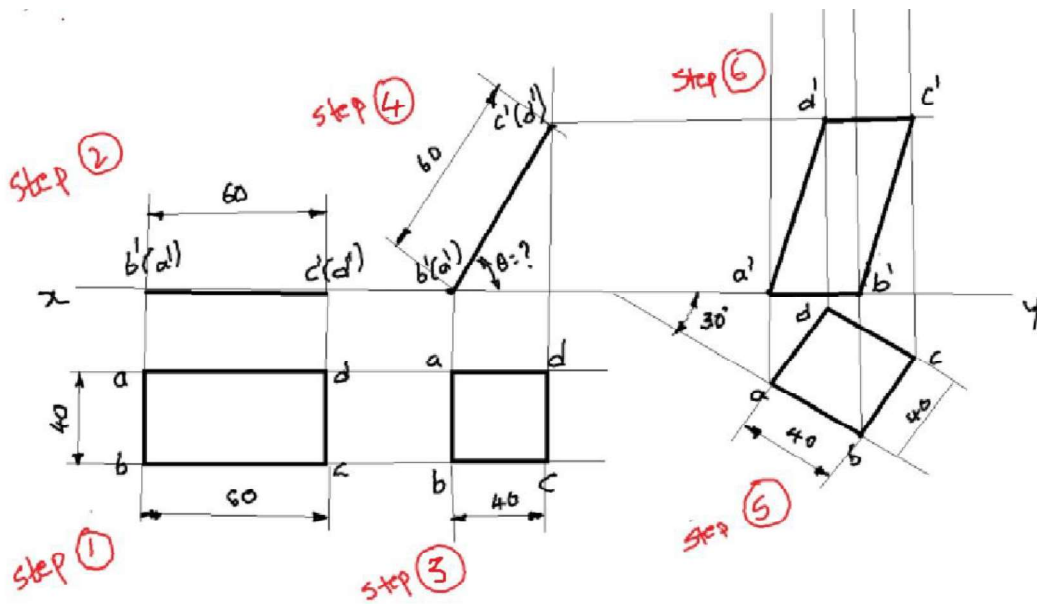
2. A square of 30 side has one side on H.P. its plane is inclined at 60° to H.P and perpendicular to V.P. draw its projections



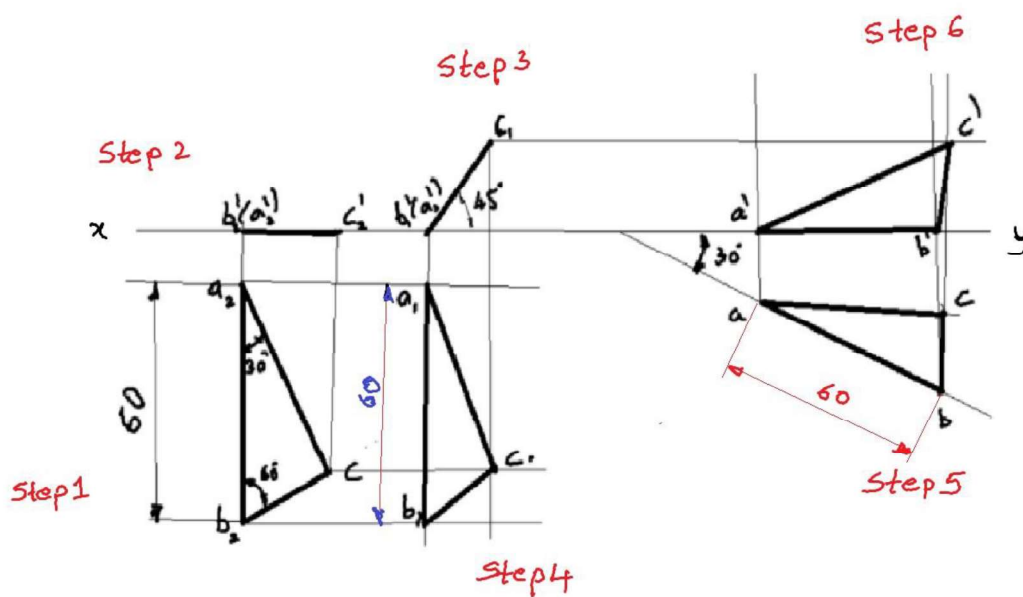
3. A pentagon of 30 side has one corner on V.P. its plane is inclined at 60° to H.P and perpendicular to V.P. draw its projections.



4. A thin rectangular plate 60x40 size has its shorter edge on H.P and inclined at 30° to V.P. draw the projections of the plate, when its top view is a square of 40 side.



5. A thin 30° - 60° set square has its longest edge (diagonal) on H.P and inclined at 30° to V.P. its surface makes an angle 45° with H.P. draw the projections, choosing suitable size of the set square.

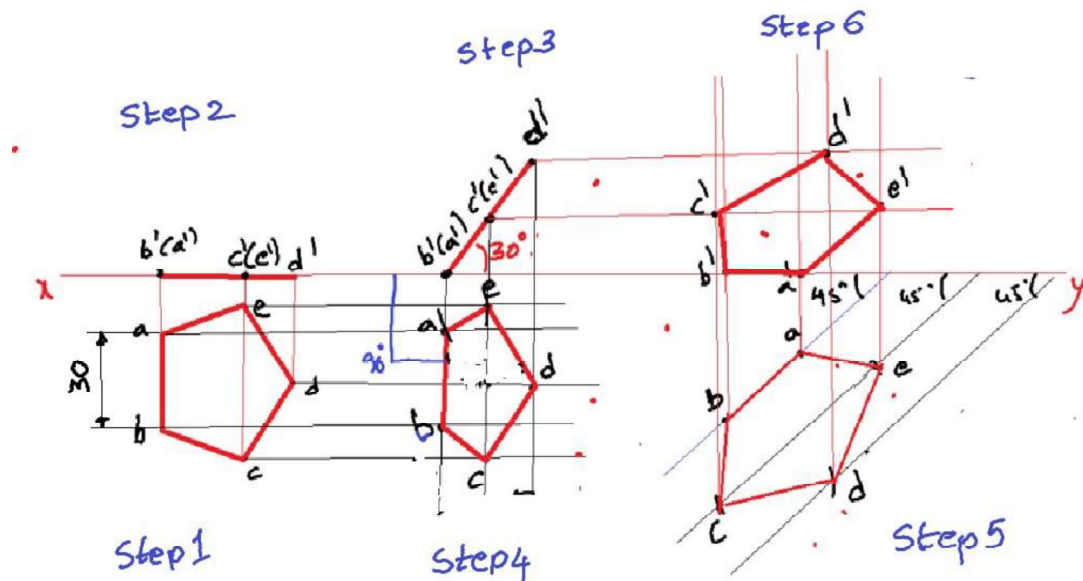


SHEET No. 11 (UNIT-II)

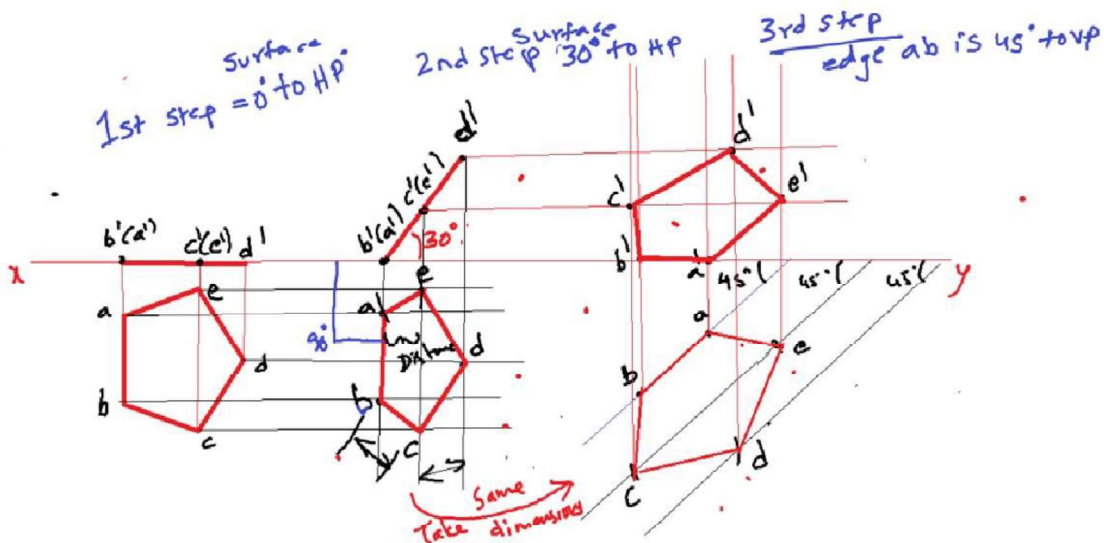
PROJECTIONS OF PLANES Inclined to both the Planes

1. A regular pentagon of 30 side is resting on one of its edges on H.P, which is inclined at 45° to V.P. its surface is inclined at 30° to H.P. draw its projections.

Answer :-

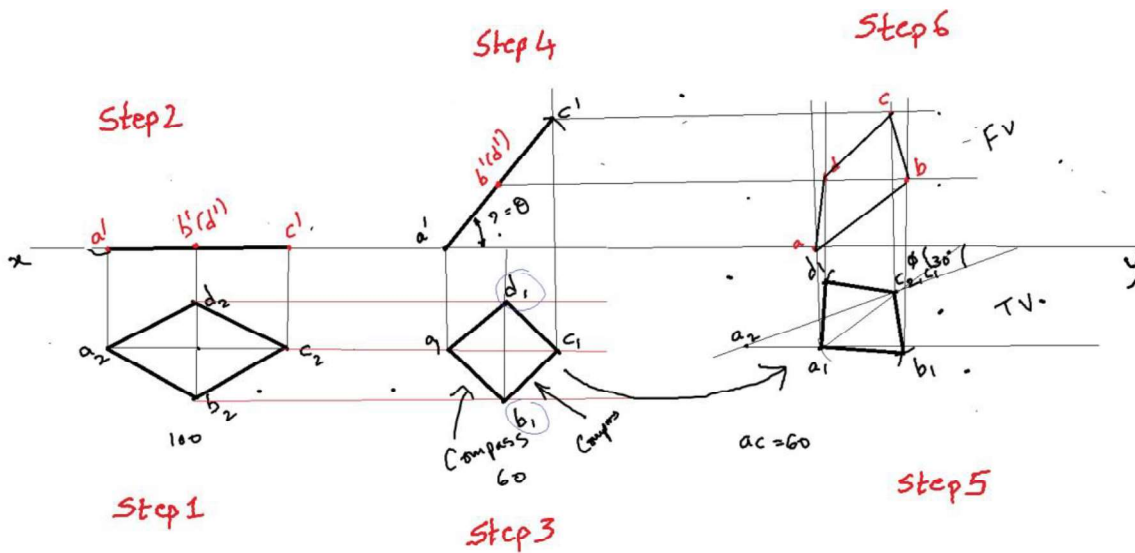


Concept :-

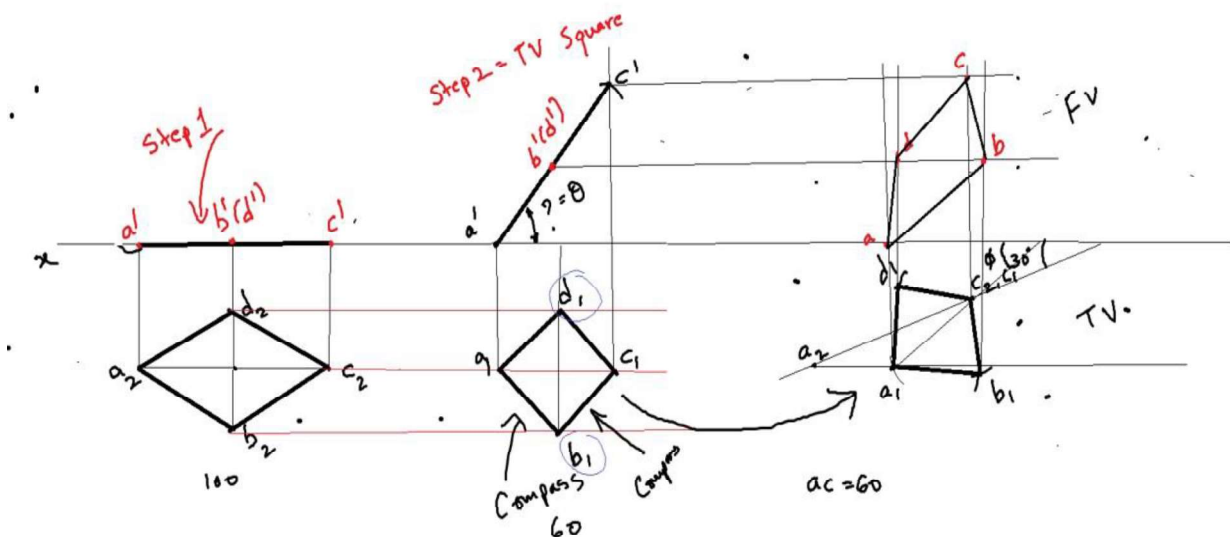


2. A rhombus has its diagonals 100 and 60 long. Draw the projections of rhombus, when it is so placed that its top view appears to be a square of diagonal 60 long and the vertical plane through the longer diagonal makes 30° with V.P.

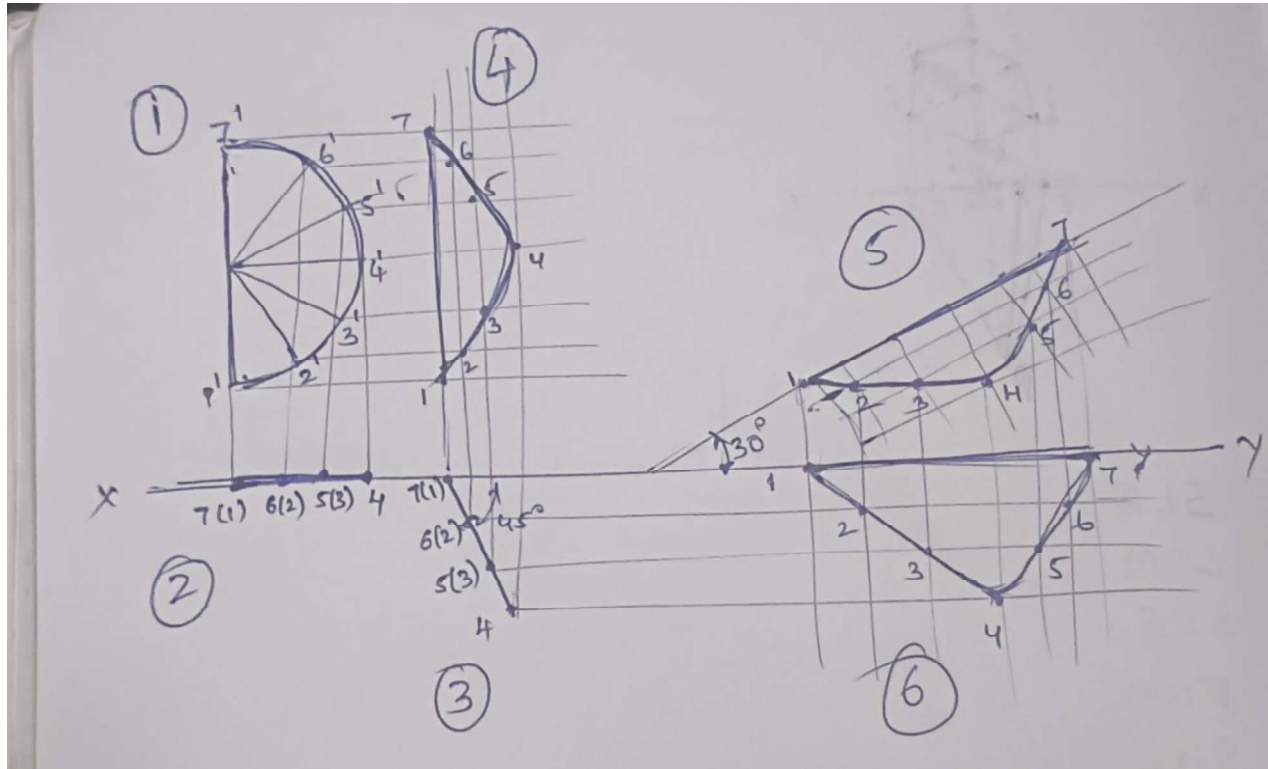
Answer:-



Concept :-

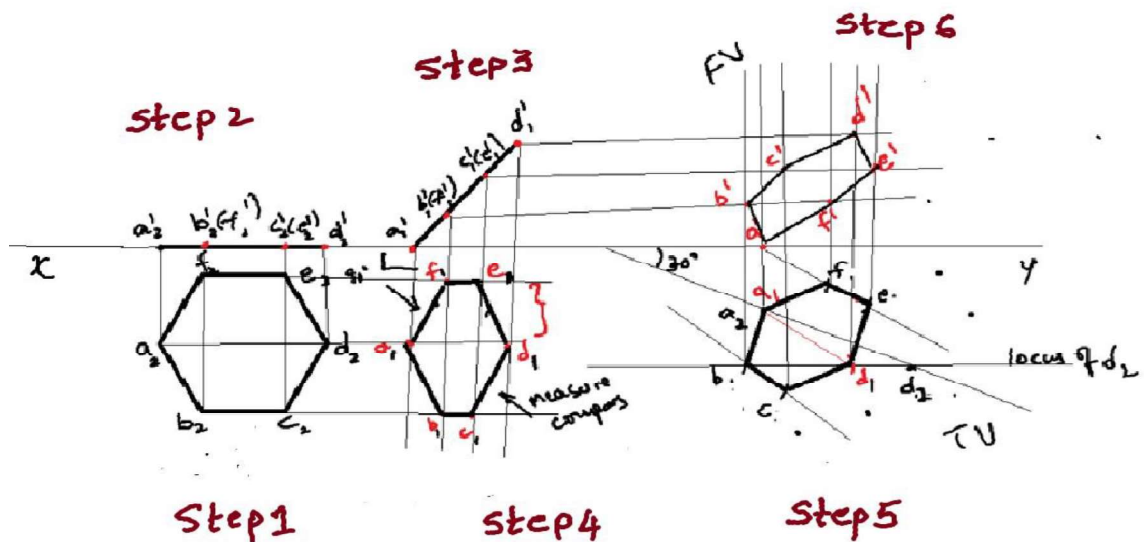


3. A semi-circular plate of 80 diameters, has its straight edge on V.P and inclined at 30° to H.P, its surface is inclined at 45° to V.P. Draw the projections of the plate.



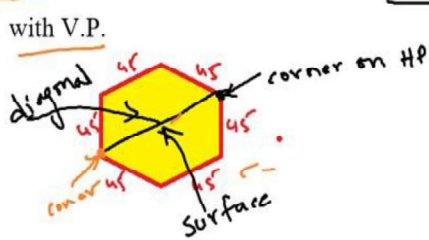
4. A regular hexagonal plane of 45 side has a corner on H.P and its surface is inclined at 45° to H.P. draw the projections when the diagonal through the corner, which is on H.P makes 30° with V.P.

Answer

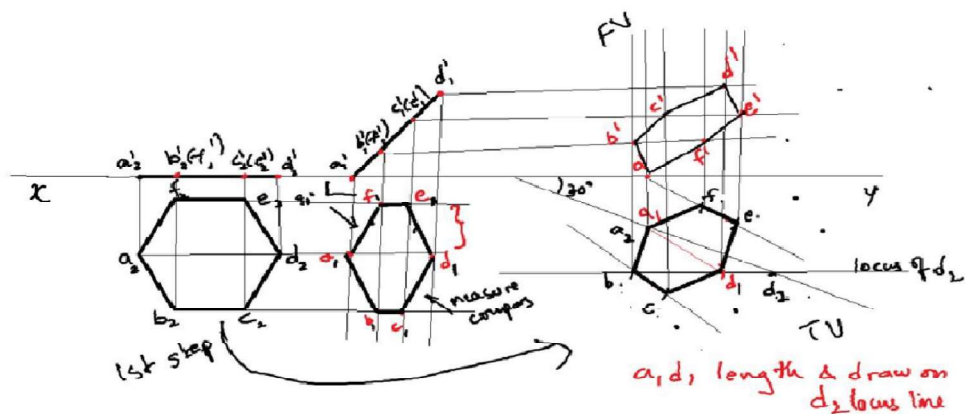


Concept :-

A regular hexagonal plane of 45 side has a corner on H.P and its surface is inclined at 45° to H.P. draw the projections when the diagonal through the corner, which is on H.P makes 30° with V.P.

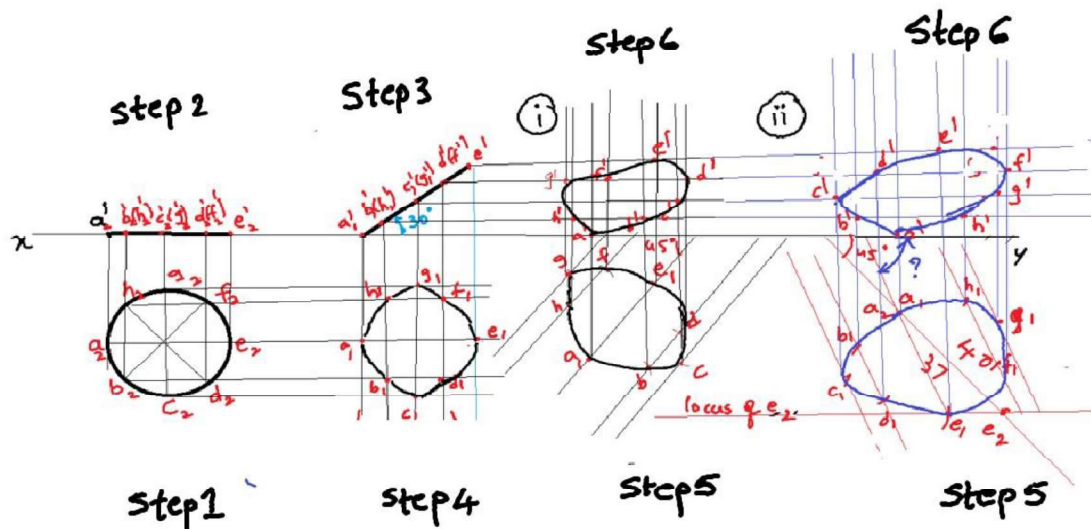


1st step is Surface 0° to HP
 2nd step Surface 45° to HP
Corner on HP

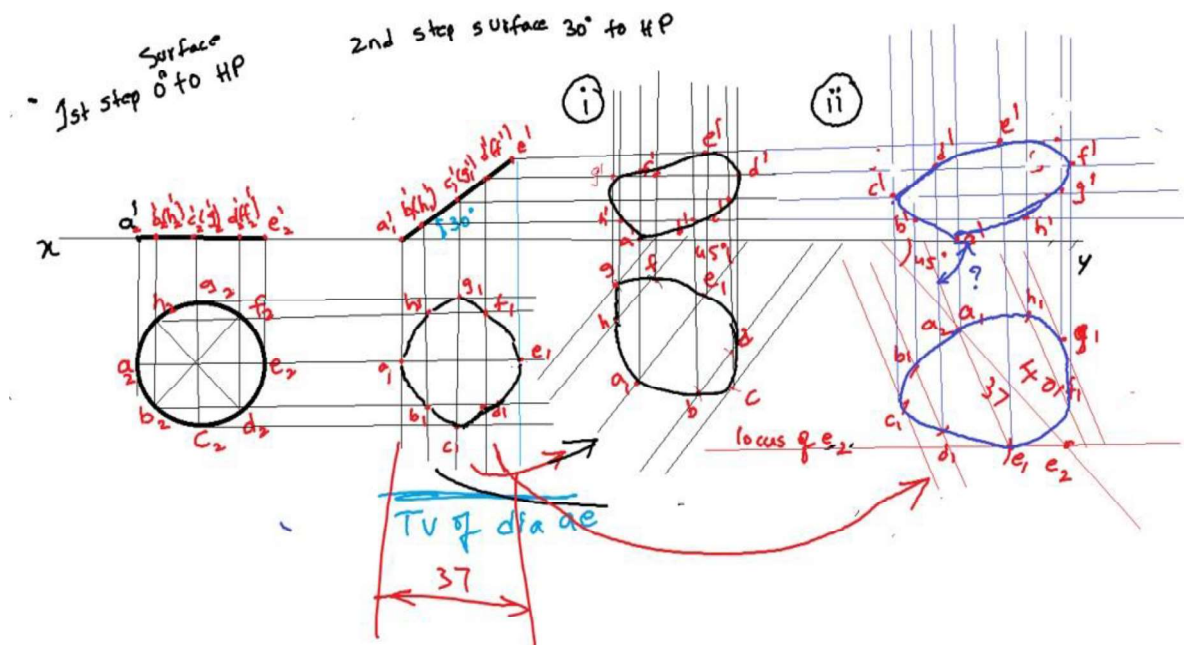


5. A circle of 40 diameter is resting on H.P on a point, with its surface inclined at 30° to H.P. Draw the projections of the circle, when (i) the top view of the diameter, through the resting point, makes an angle 45° with xy and (ii) the diameter passing through the resting point makes an angle 45° with V.P.

ANSWER



CONCEPT

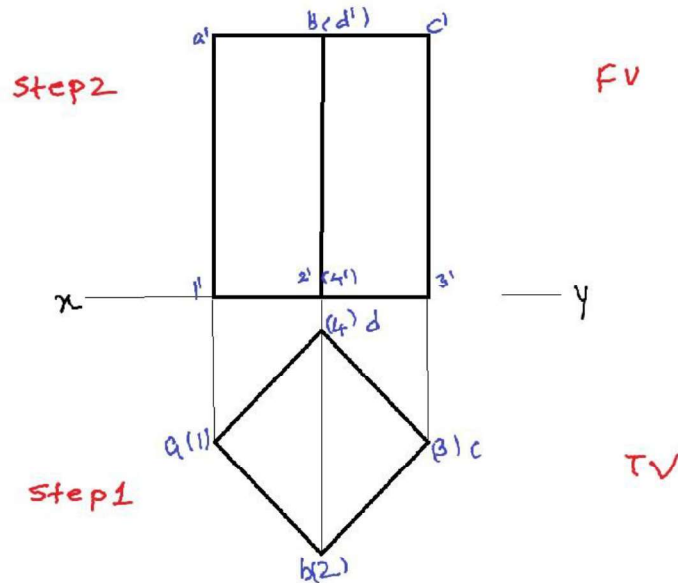


SHEET No. 12 (UNIT III)

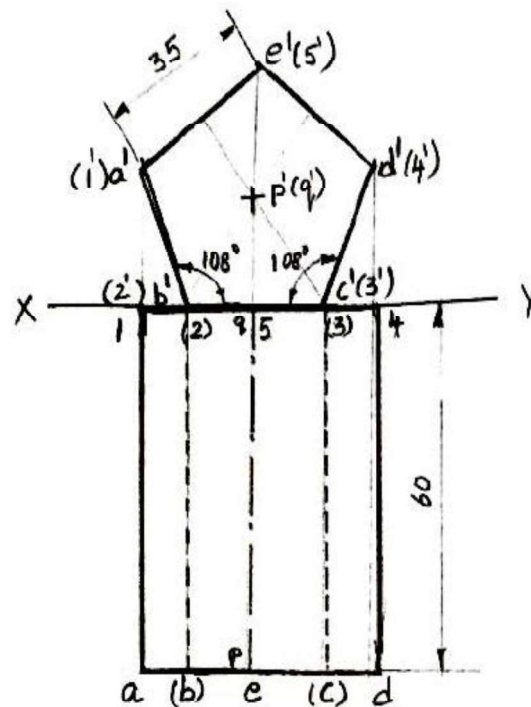
PROJECTIONS OF SOLIDS (PRISMS , CYLINDERS, PYRAMIDS & CONES)

1. Draw the projections of the following solids, situated in their respective positions, taking the side of the base equal to 40 and the length of axis as 70.

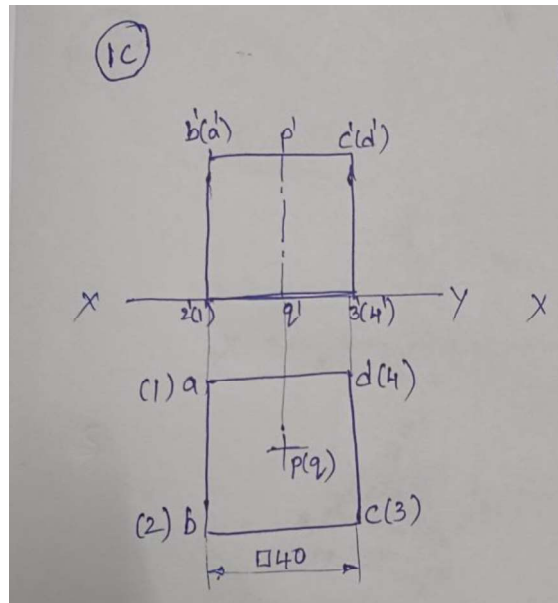
a) A square prism with base on H.P and a side of the base inclined at 45°



1 b) A pentagonal prism with a rectangular face on H.P, axis perpendicular V.P and one base on V.P.

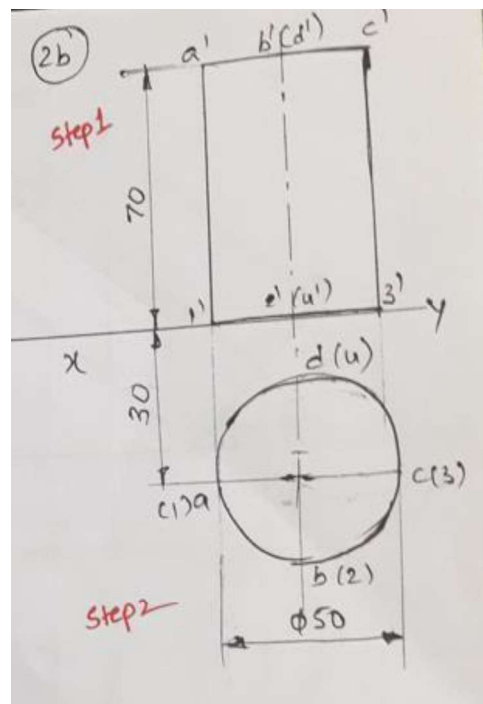
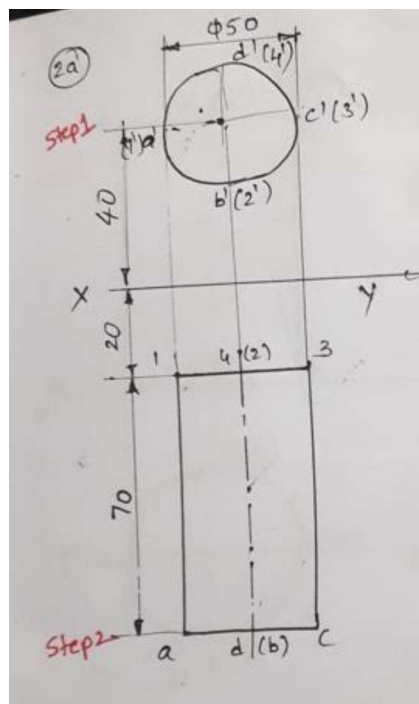


1c) A cube of edge 40, with a base on H.P and a face perpendicular to V.P



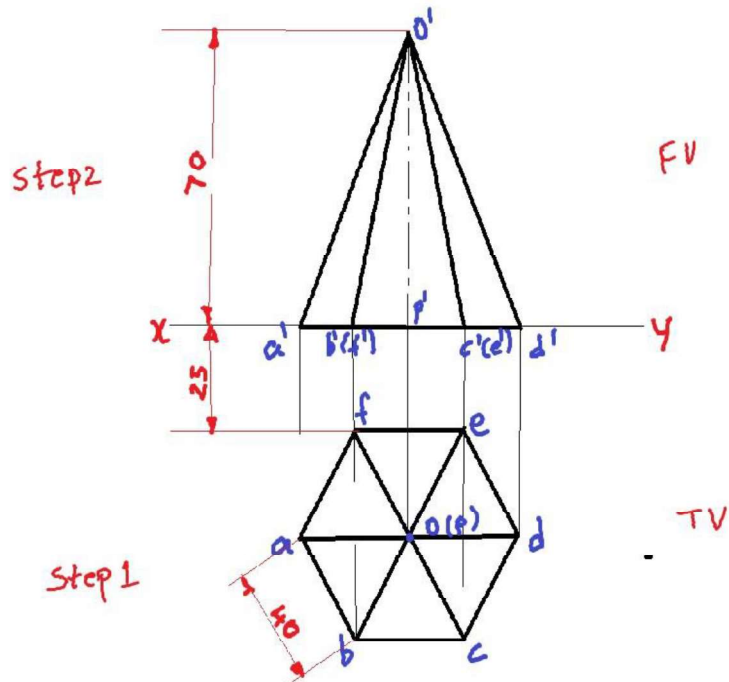
2. Draw the projections of the following solids, situated in their respective positions, taking the diameter of the base equal to 50 and the length of the axis 70.

- A cylinder, with its axis perpendicular to V.P and 40 above H.P and one end at 20 from V.P.
- A cylinder, resting with its base on H.P and axis 30 in front of V.P

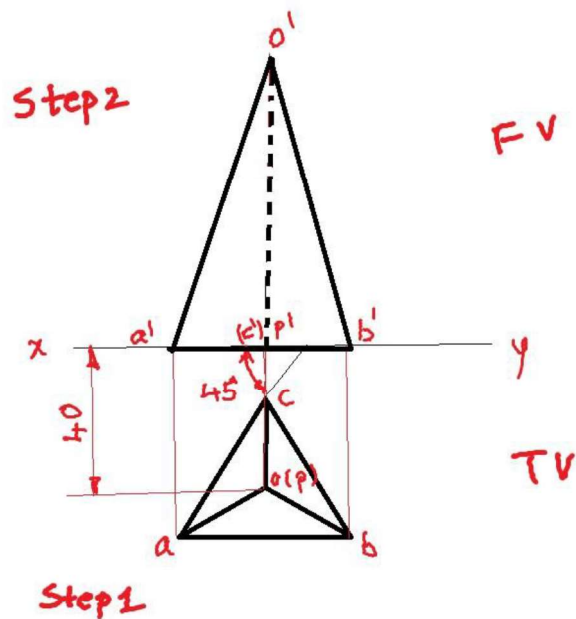


3. Draw the projections of the following solids, situated in their respective positions, taking the side of the base equal to 40 and the length of axis as 70.

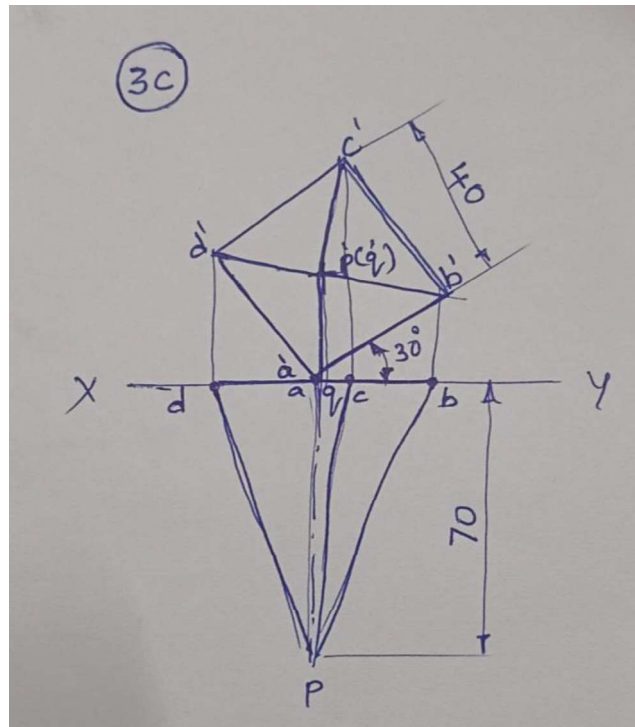
3a) A hexagonal pyramid with base on H.P and a side of the base parallel to and 25 from V.P



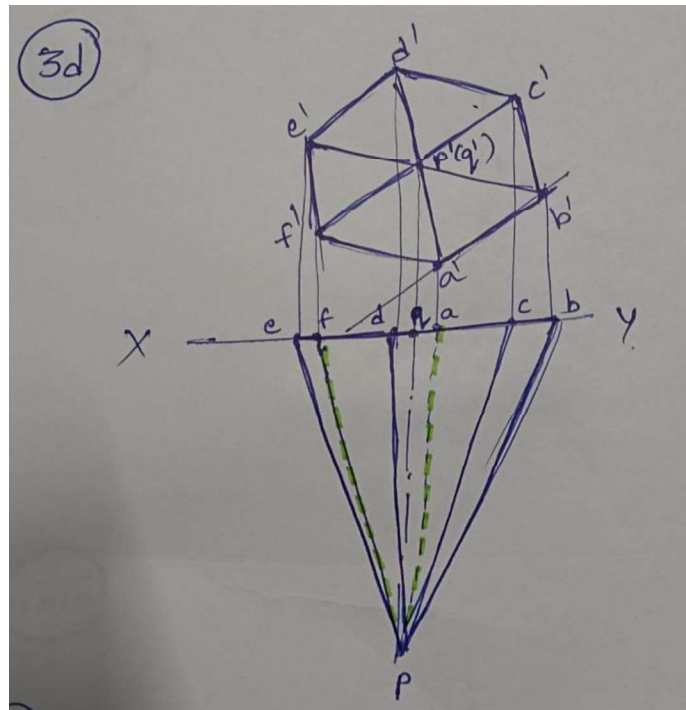
3b) A triangular pyramid with the base on H.P, an edge of the base inclined at 45° to V.P and the apex at 40 from V.P.



3c. A square pyramid with its base on V.P and one edge of the base inclined at 30° to H.P. and a corner contained by that edge is on H.P.

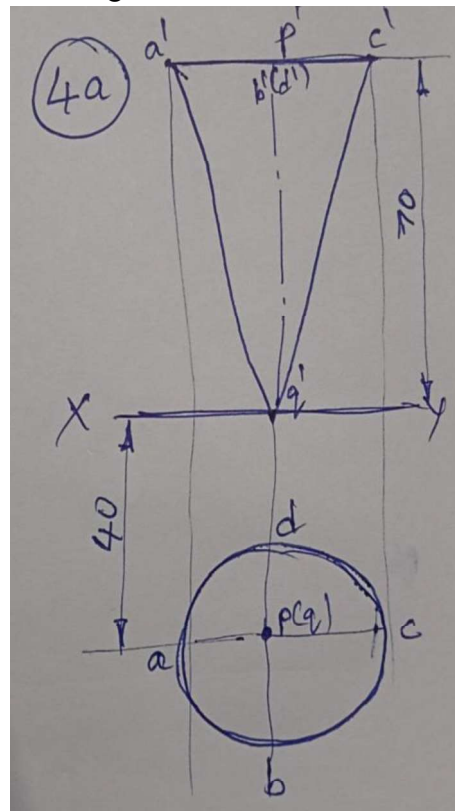


3d. A hexagonal pyramid with its base on V.P and an edge of the base inclined at 20° to H.P.



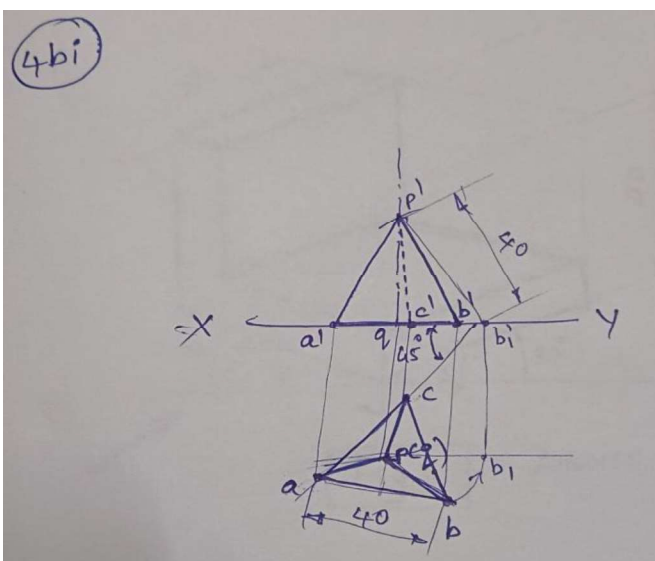
4. a. A cone with its apex on H.P and 40 from V.P., the axis being perpendicular to H.P.

Taking diameter of cone = 30mm & axis height = 60mm

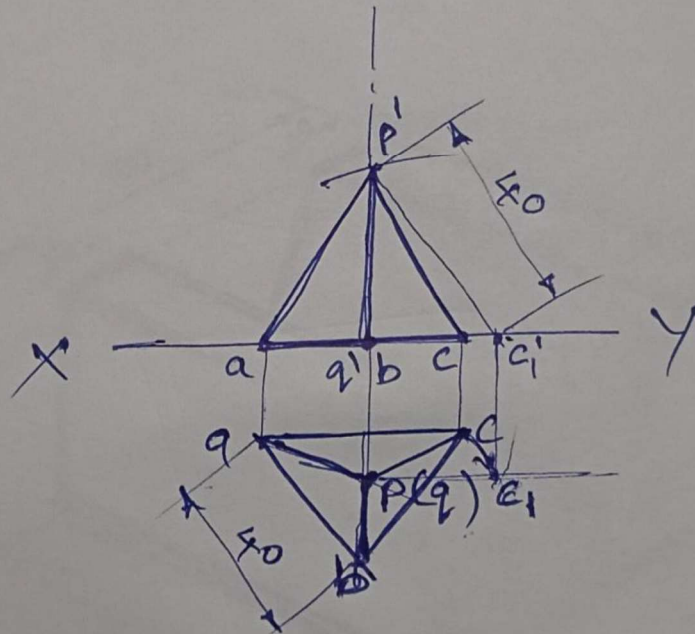


4b

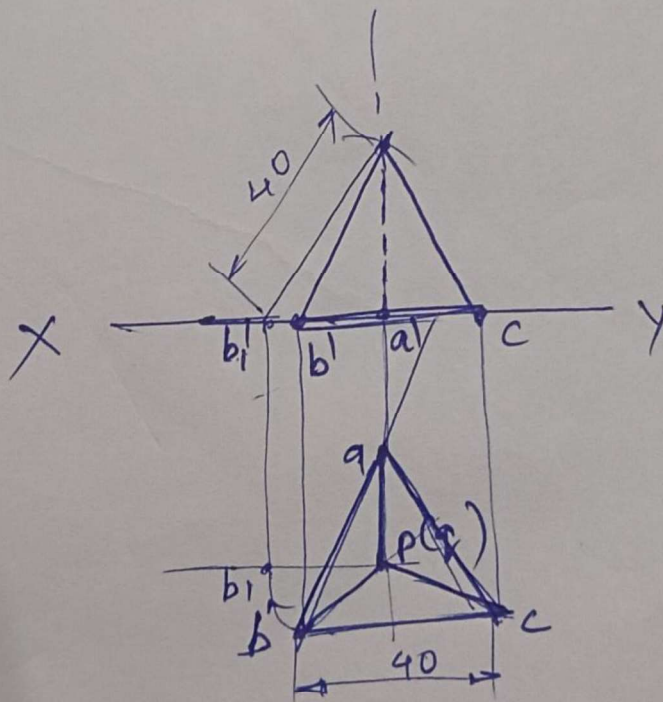
Draw the projections of a tetrahedron of side 50, when it is resting on H.P on one of its faces, with an edge of its face (i) inclined at 45° to V.P (ii) parallel to V.P and (iii) inclined at 60° to V.P.



4bii



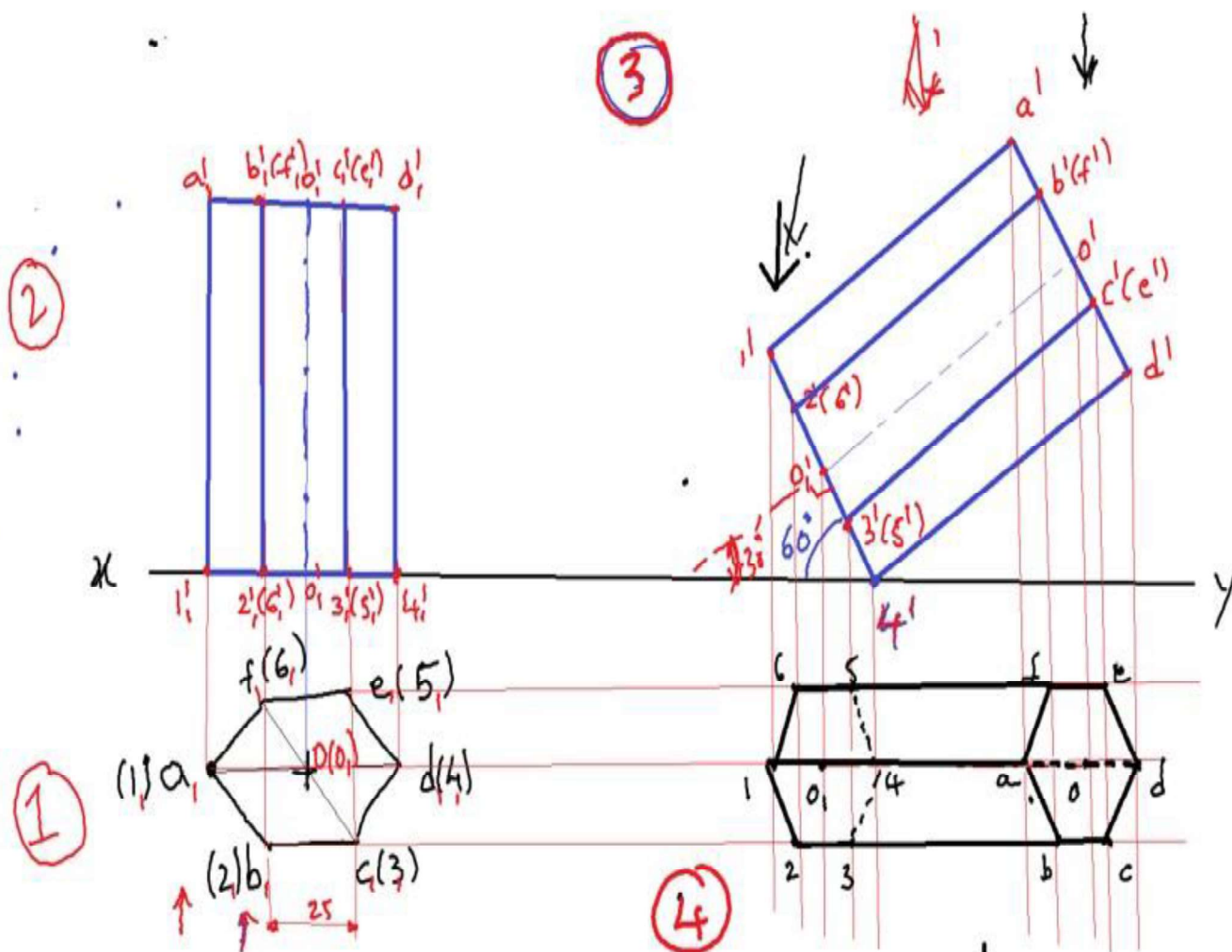
4biii



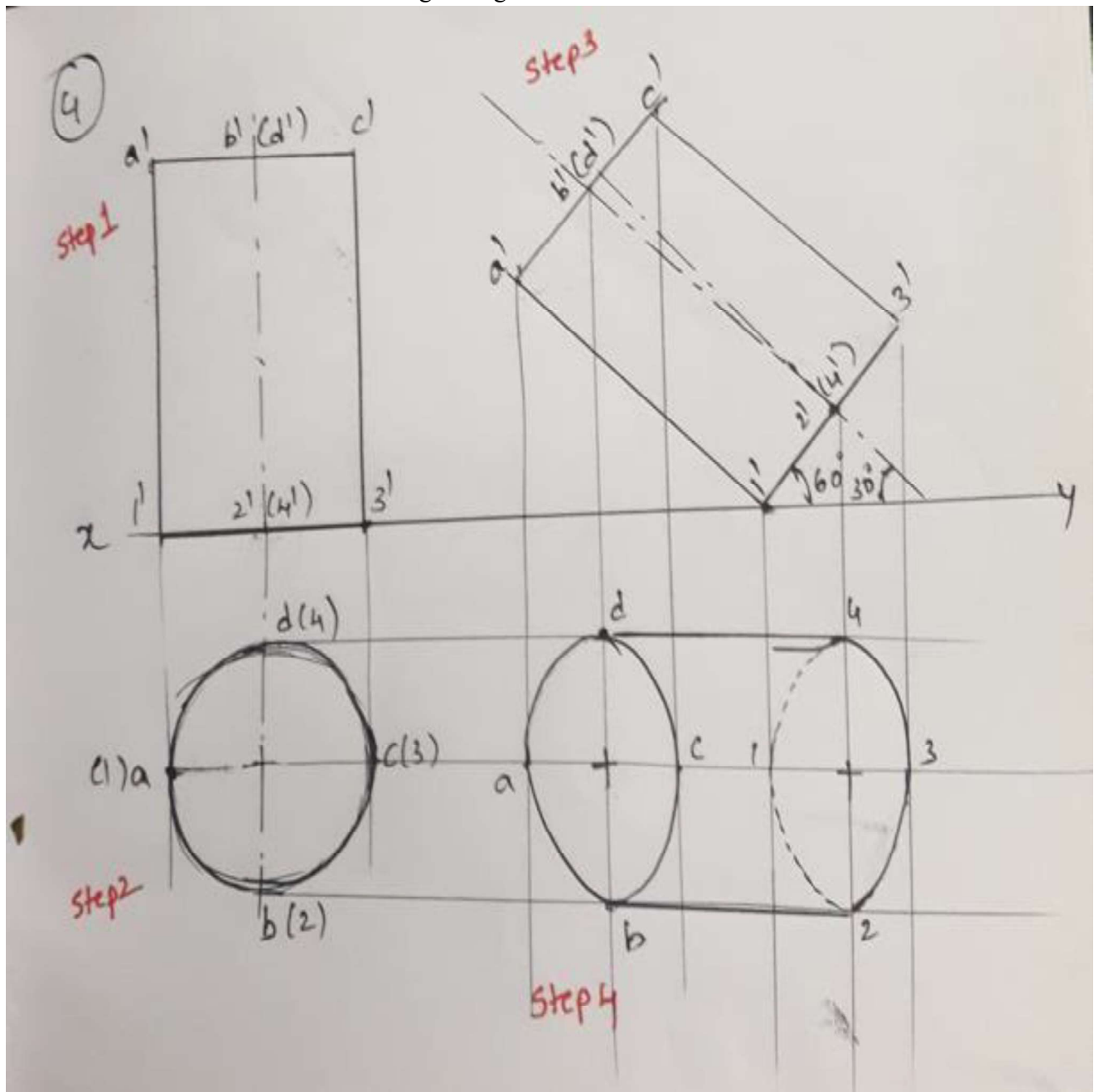
SHEET No. 13 (UNIT- III)

PROJECTIONS OF SOLIDS (PRISMS , CYLINDERS, PYRAMIDS & CONES)

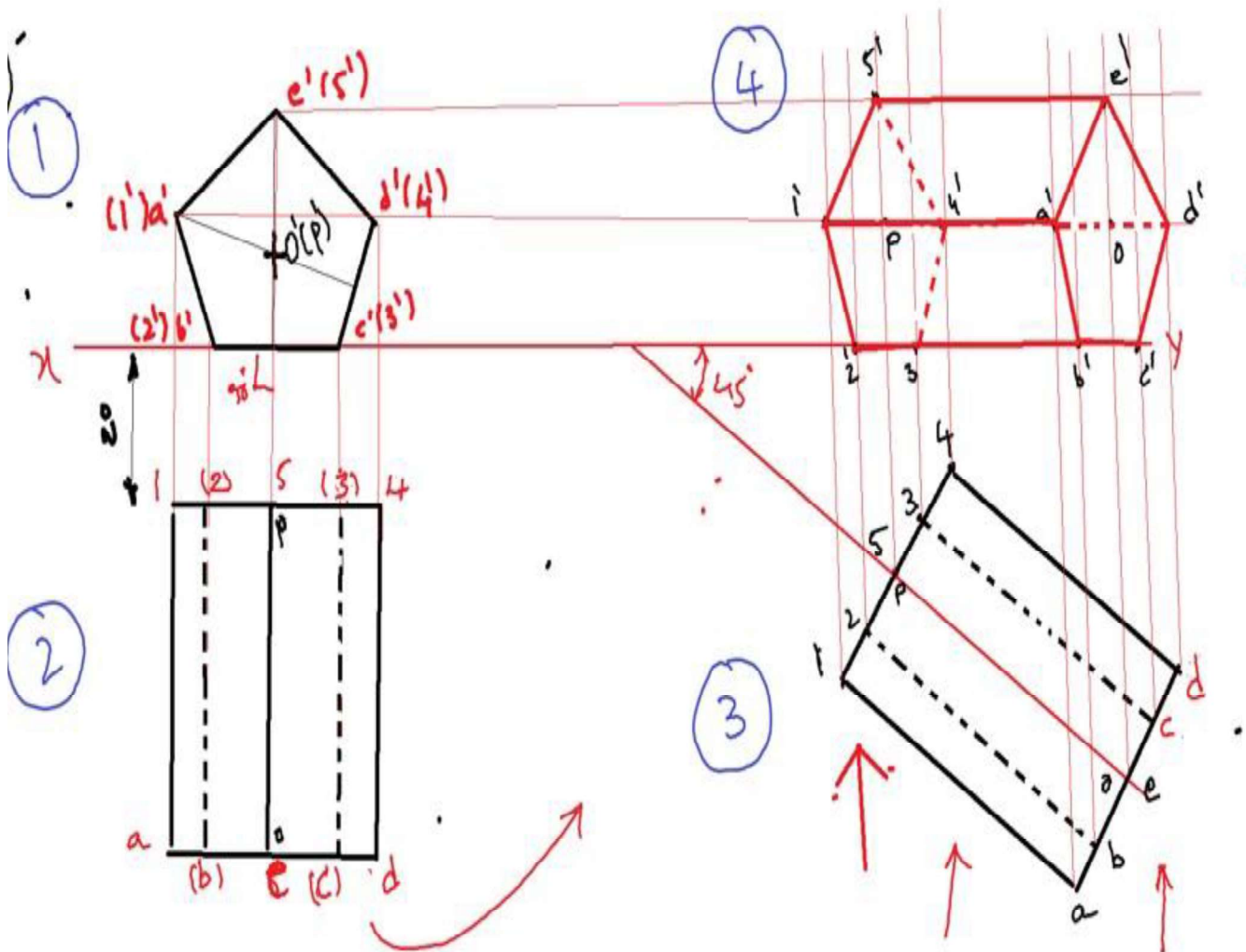
1. A hexagonal prism, side of base 25mm and axis 50mm long rests with one of its base corners on H.P. such that its base makes an angle of 60° to H.P. and its axis is parallel to V.P. Draw its Projections.



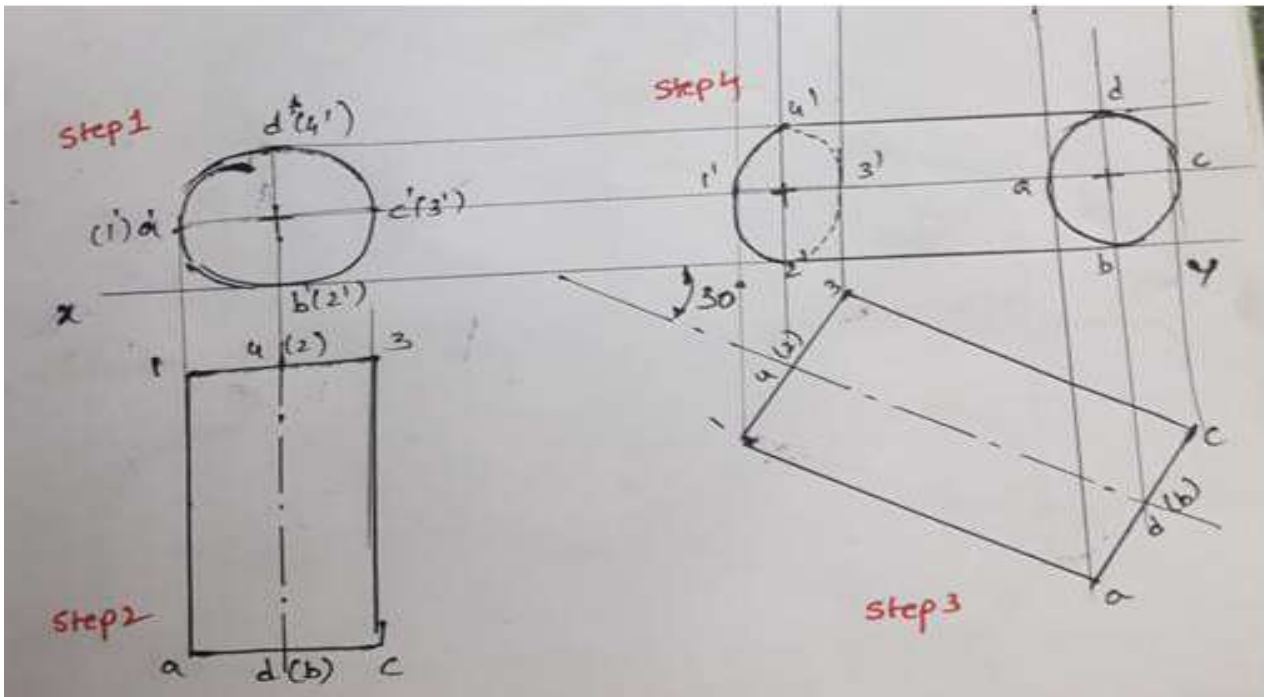
2. Draw the projections of a cylinder, base 30mm diameter and axis 40mm long, rests with a point of its base circle on H.P. such that the axis is making an angle of 30° with H.P.



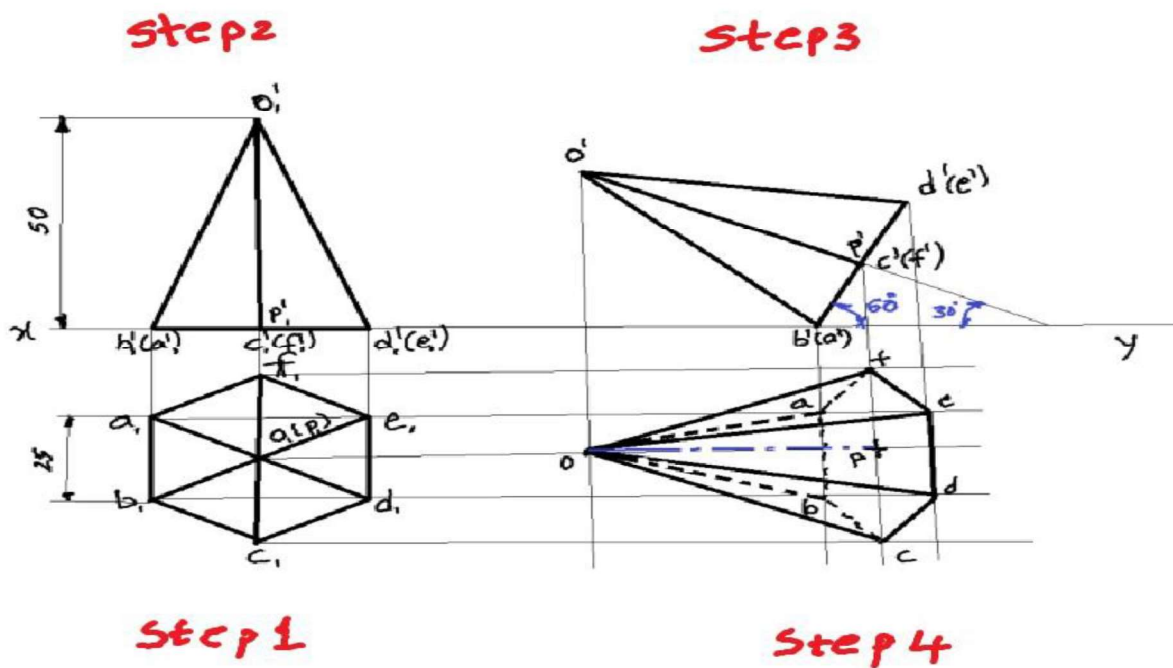
3. A regular pentagonal prism of side of base 30 and length 70 lies with one of its rectangular faces on H.P and its axis is inclined to V.P at 45° . Draw the projections of the prism.



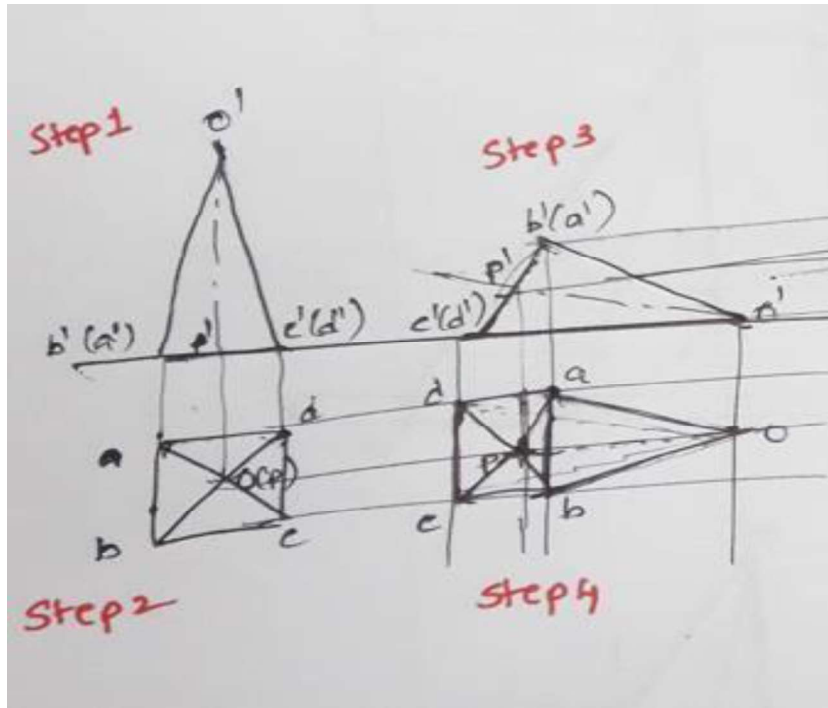
4. Draw the projections of a cylinder of diameter 50 and length 70, lying on H.P with its axis inclined at 30° to V.P



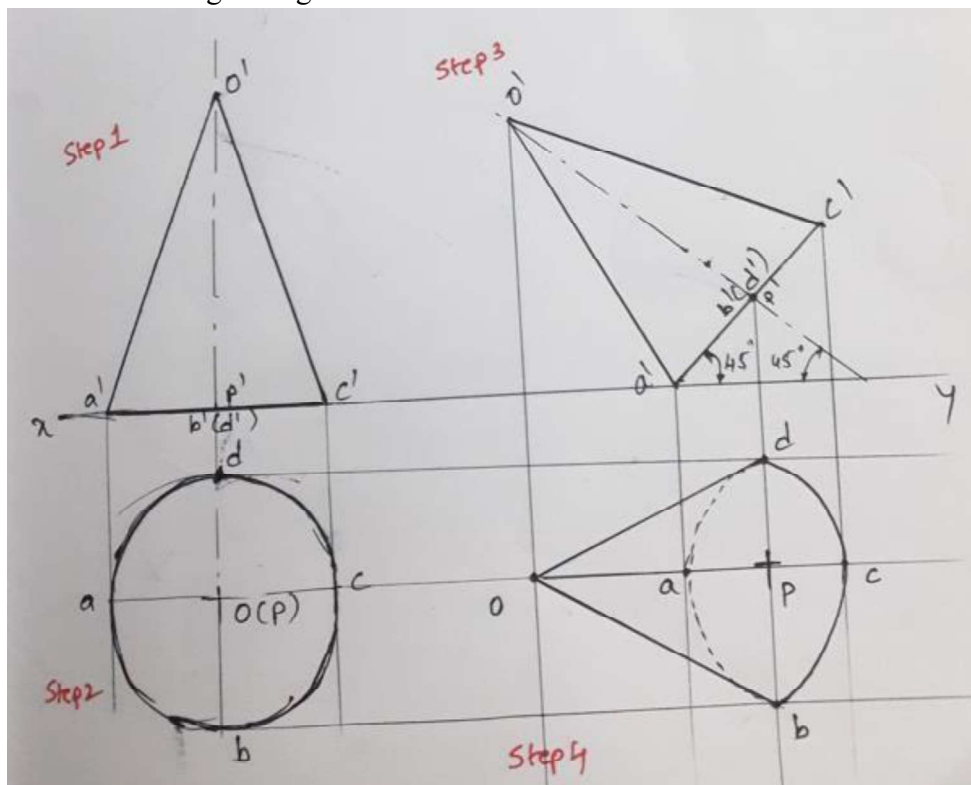
5. A hexagonal pyramid side of base 25mm and axis 50mm long rests with one of the edges of its base on H.P. and its axis is inclined at 30° to H.P. and parallel to V.P. draw its projections.



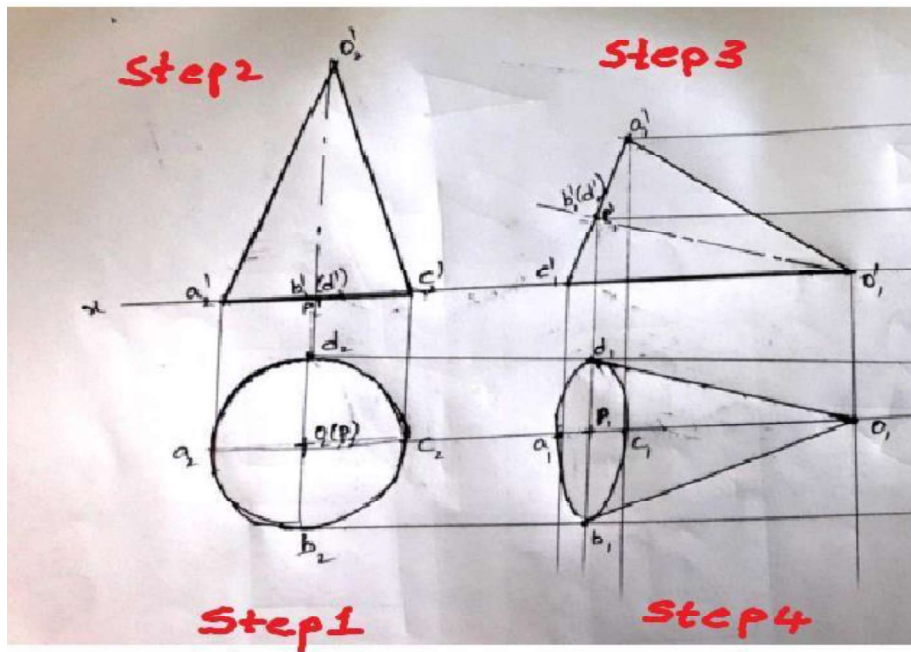
6. A square pyramid of base 40 side and axis 50 long, is resting on one of its triangular faces on H.P, draw the projections of the pyramid.



7. Draw the projection of a cone, base 30mm diameter and axis 50mm long, resting on H.P., on a point of base circle with the axis making an angle of 45° with H.P.



8. Draw the projections of a cone 70 height and 50 diameter, resting on one of its generators on H.P



SHEET No. 14 (UNIT- IV)

SECTIONS OF SOLIDS :

1

A hexagonal prism, side of base 30 mm and axis 60 mm long, rests with its base on HP such that one of its rectangular faces is parallel to VP. A section plane perpendicular to HP and parallel to VP cuts the prism at a distance of 10 mm from its axis. Draw its top and sectional front views.

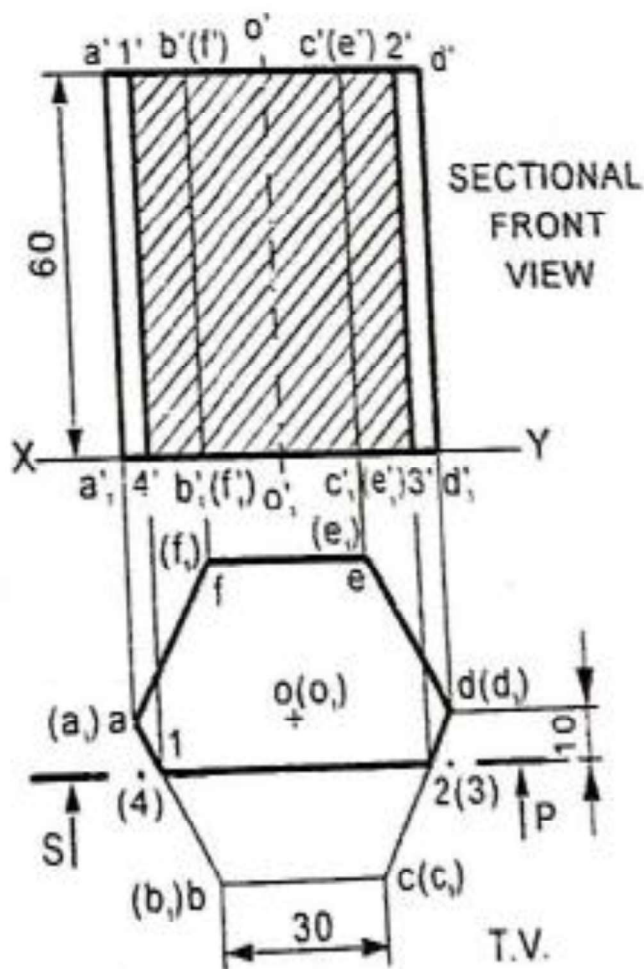
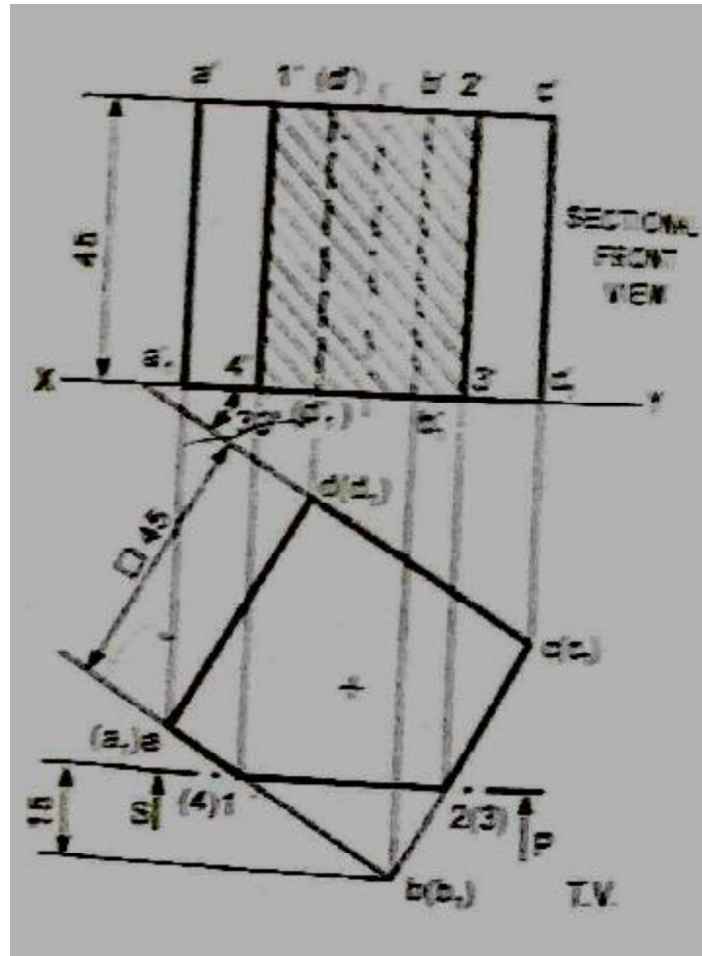


FIG. 5

2

A cube of 45 mm side rests with a face on HP such that one of its vertical faces is inclined at 30° to VP. A section plane, parallel to VP cuts the cube at a distance of 15 mm from the vertical edge nearer to the observer. Draw its top and sectional front views.



A square prism, side of base 30 mm and axis 60 mm long, rests with its base on HP and one of its rectangular faces is inclined at 30° to VP. A section plane perpendicular to VP and inclined at 60° to HP cuts the axis of the prism at a point 20 mm from its top end. Draw sectional top view and true shape of section.

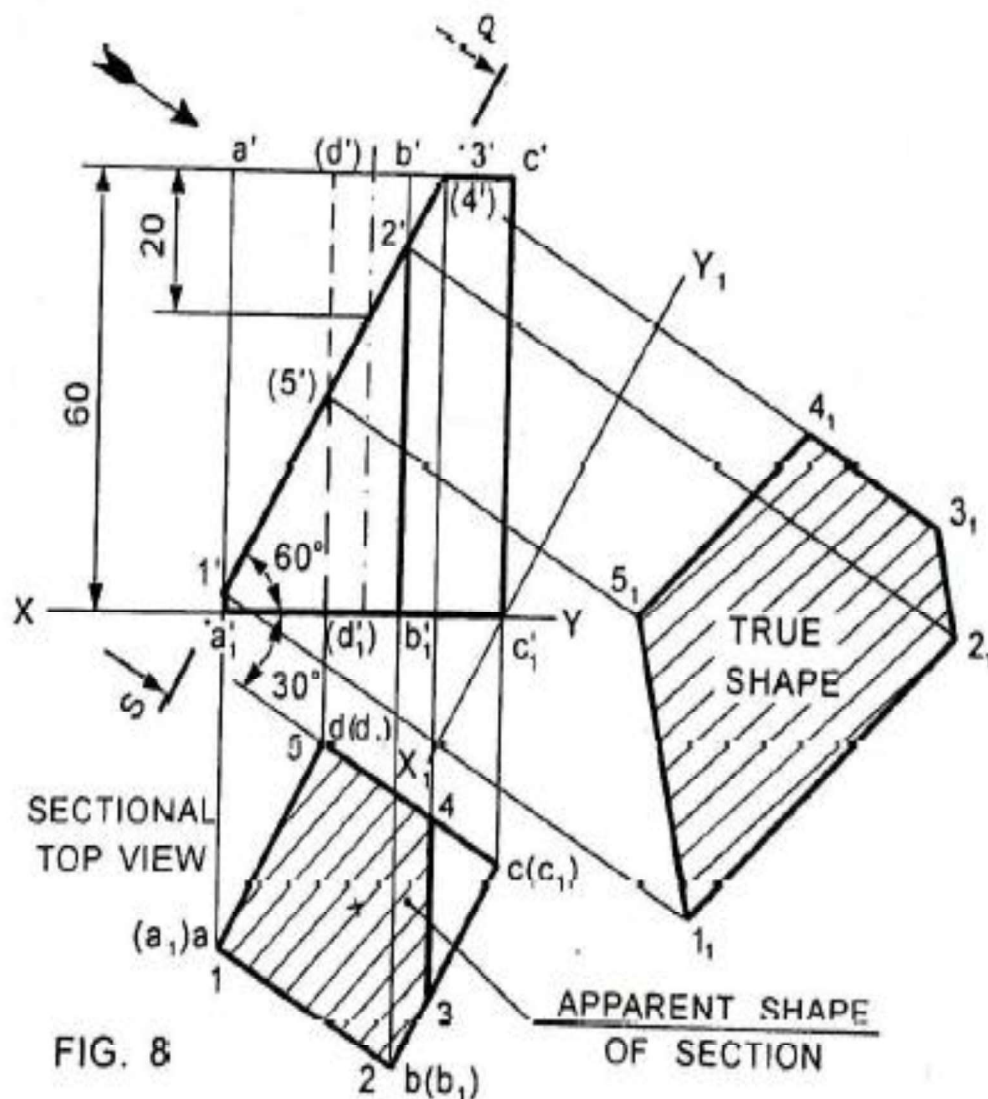
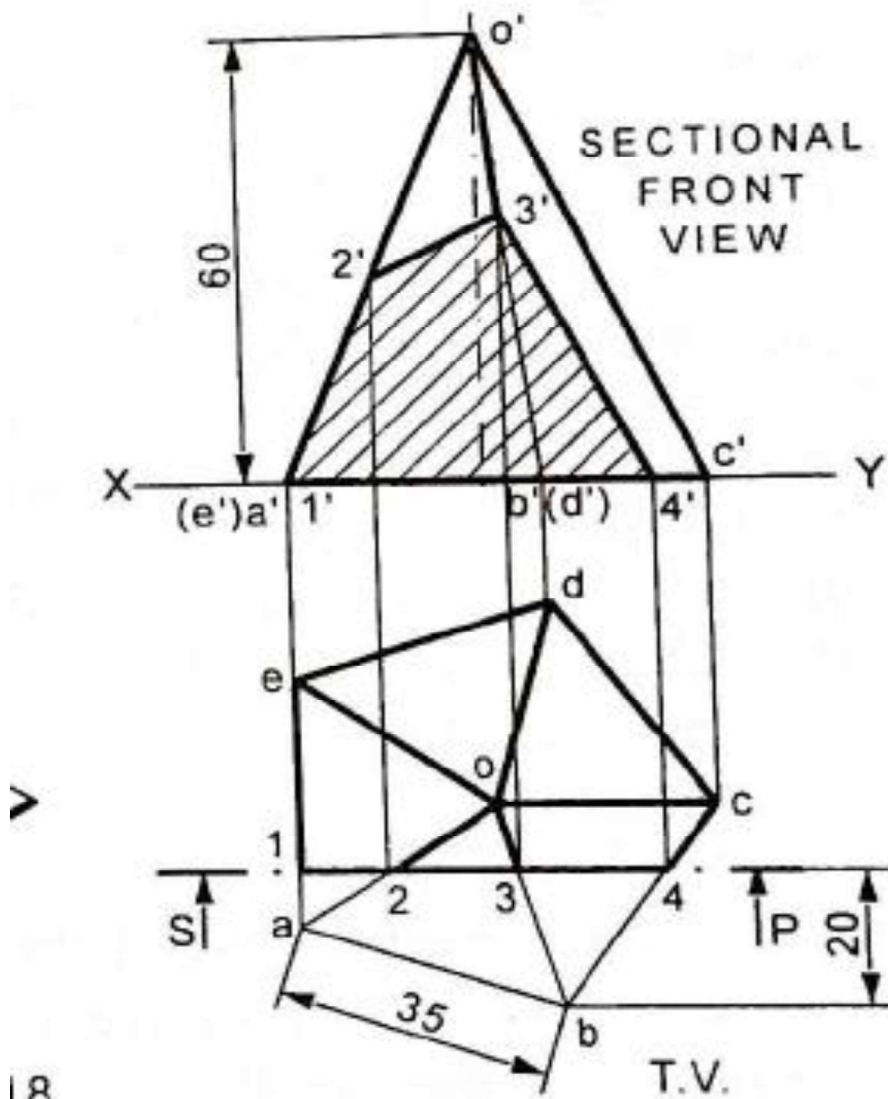
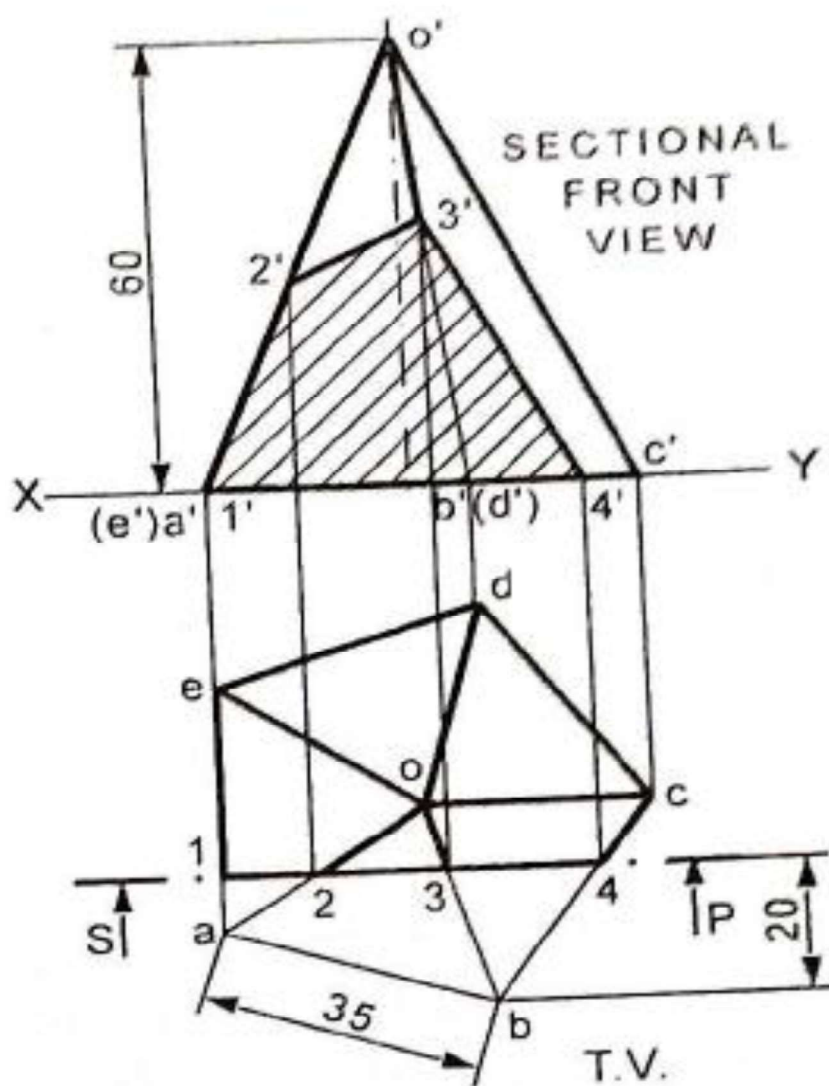


FIG. 8

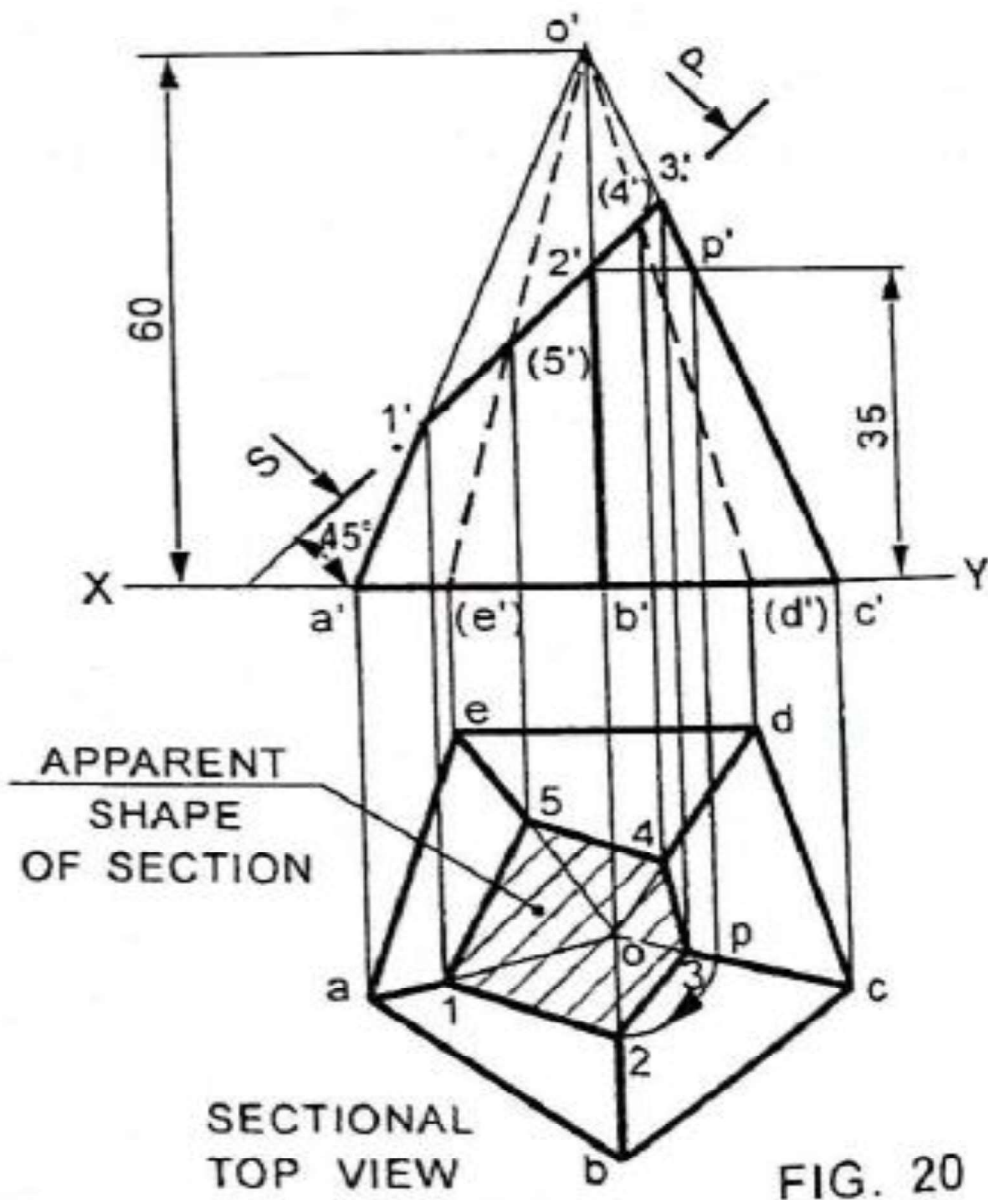
A pentagonal pyramid, side of base 35 mm and axis 60 mm long, rests with its base on HP such that one of the edges of the base is perpendicular to VP. A section plane perpendicular to HP and parallel to VP cuts the pyramid at a distance of 20 mm from the corner of the base nearer to the observer. Draw its top and sectional front views.



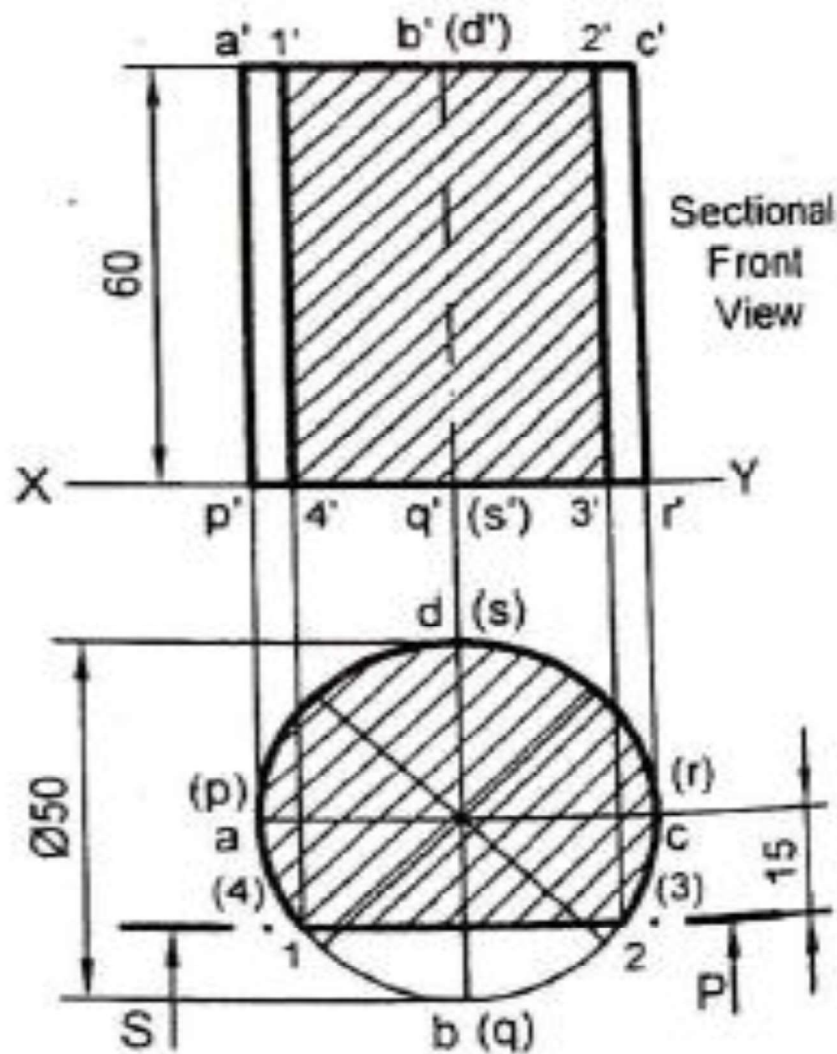
A pentagonal pyramid, side of base 35 mm and axis 60 mm long, rests with its base on HP such that one of the edges of the base is perpendicular to VP. A section plane perpendicular to HP and parallel to VP cuts the pyramid at a distance of 20 mm from the corner of the base nearer to the observer. Draw its top and sectional front views.



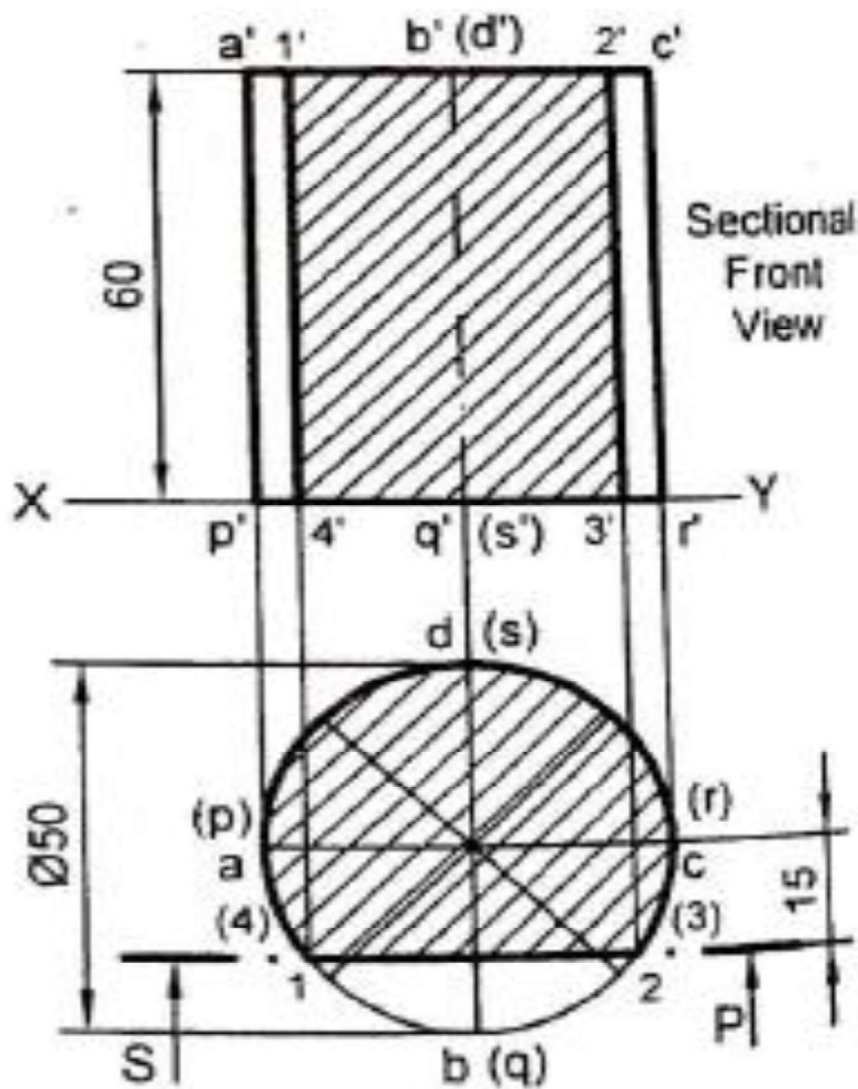
A pentagonal pyramid, side of base 30 mm and axis 60 mm long, rests with its base on HP and one of the edges of its base is perpendicular to VP. It is cut by a section plane perpendicular to VP, parallel to HP and passing through the axis at a point 35 mm above the base. Draw the front and sectional top views.



A cylinder of 50 mm diameter and 60 mm height stands vertically with its base on HP. It is cut by a section plane perpendicular to HP, parallel to VP and 15 mm from the axis. Draw its top and sectional front views.



A cylinder of 50 mm diameter and 60 mm height stands vertically with its base on HP. It is cut by a section plane perpendicular to HP, parallel to VP and 15 mm from the axis. Draw its top and sectional front views.



A cone, base 40 mm diameter and axis 60 mm long, rests with its base on HP. It is cut by a section plane perpendicular to VP, parallel to one of the end generators and passing through a point on the axis 25 mm from the apex. Draw the sectional top view and true shape of the section.

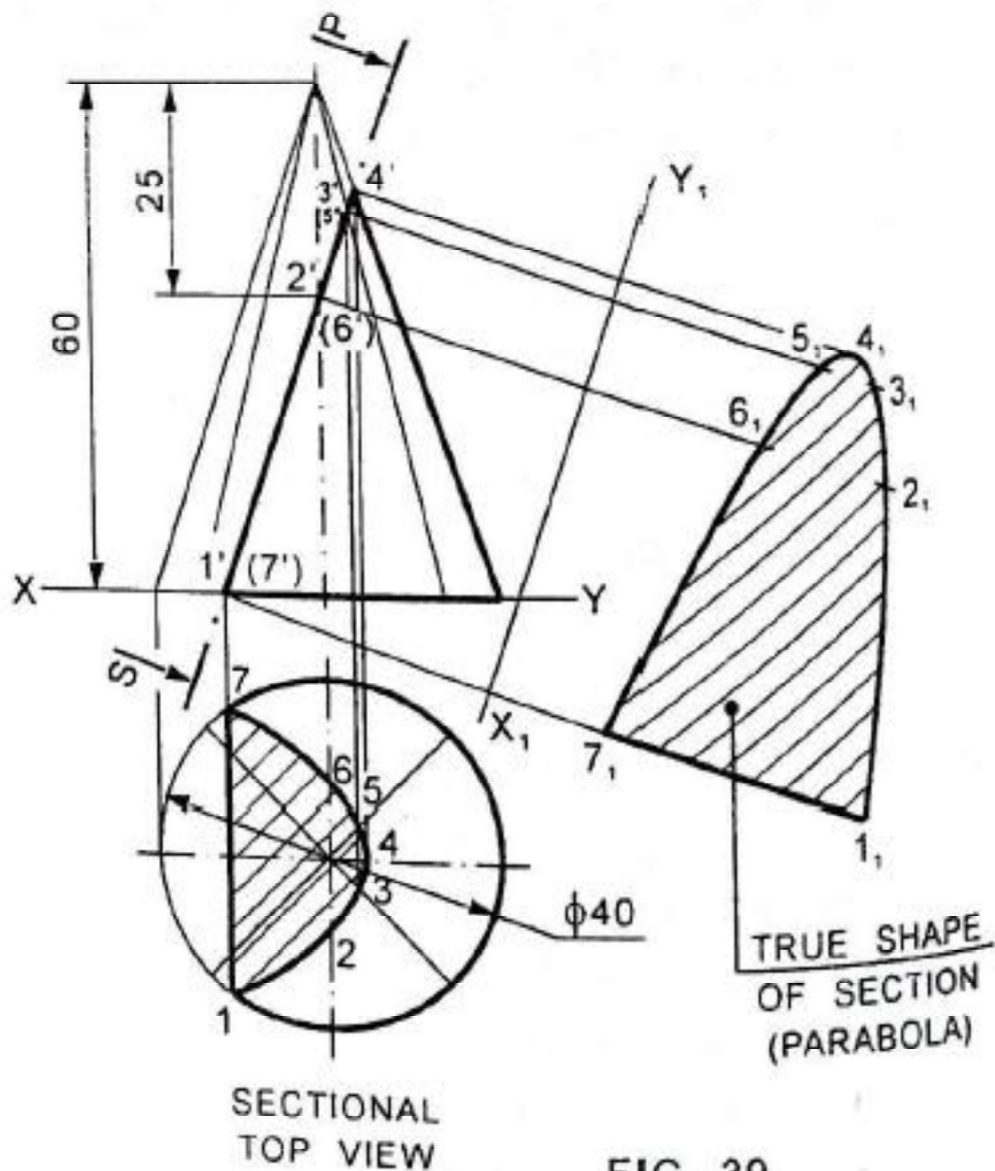
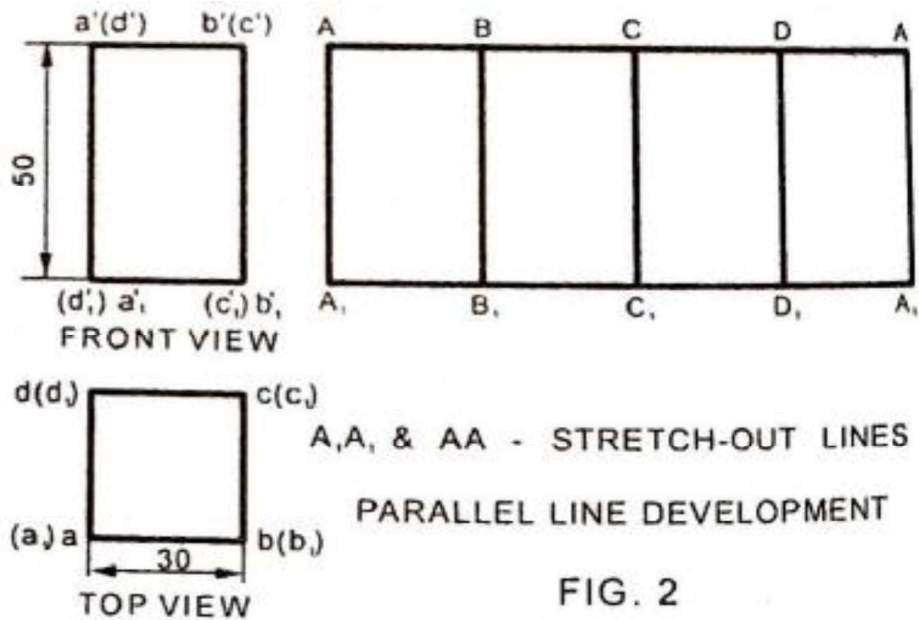


FIG. 39

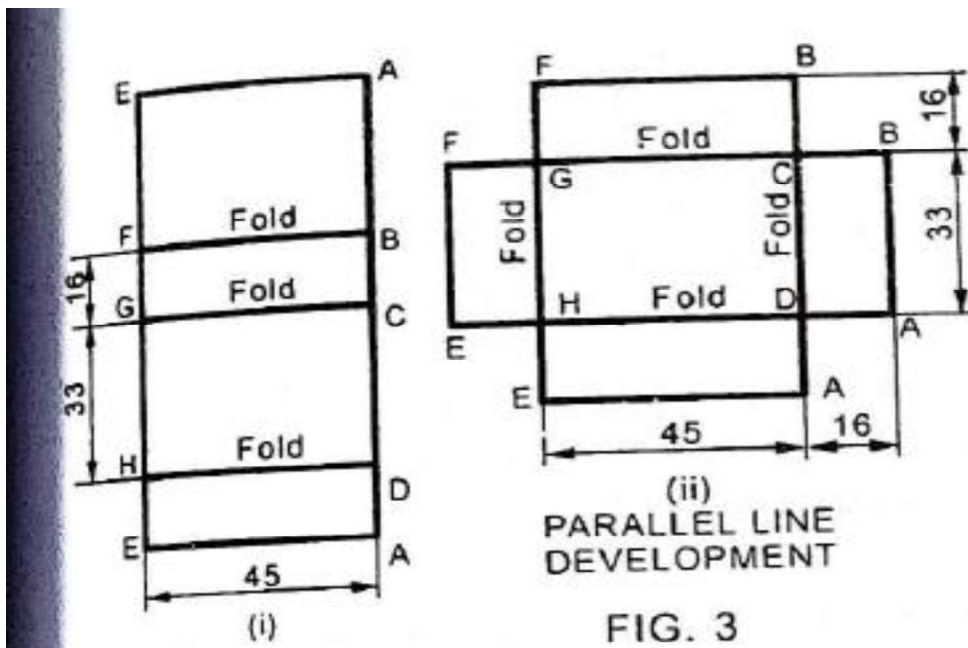
1

Draw the development of the lateral surfaces of a right square prism of edge of base 30 mm and axis 50 mm long.



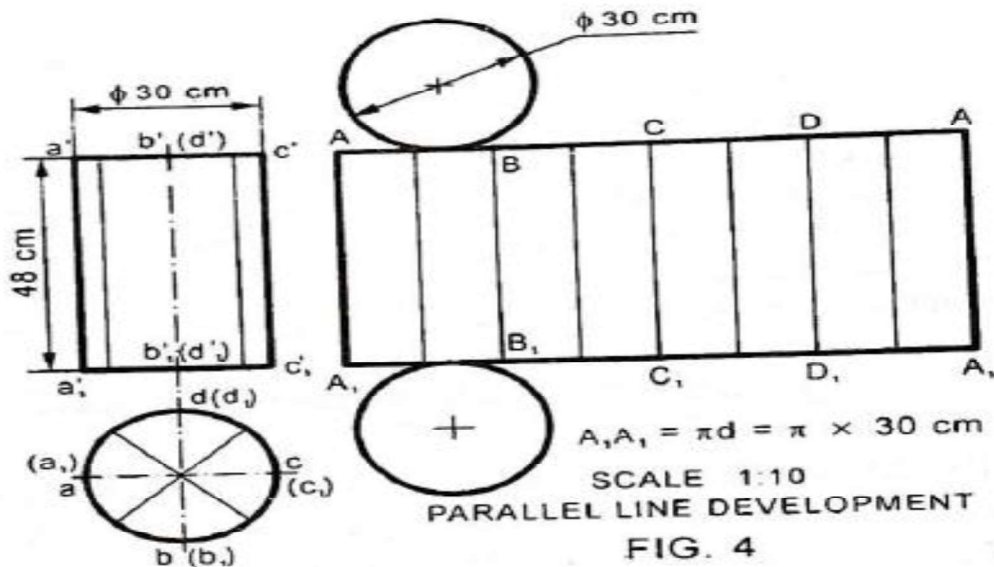
2

Draw the development of the outside case and tray of a match box of size 45 mm x 33 mm x 16 mm.



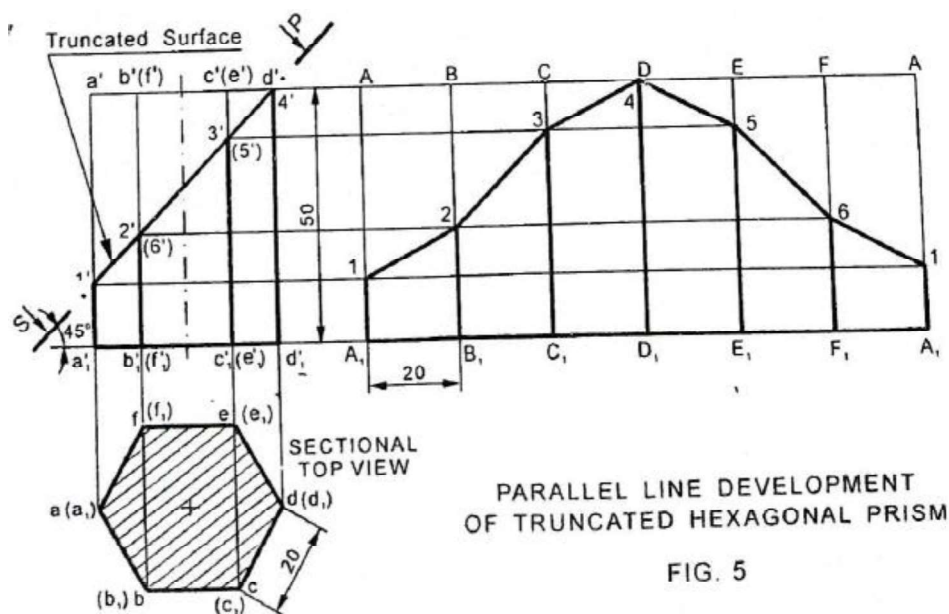
3

Draw the development of the complete surface of a *G.I.* cylindrical drum with lid. Diameter is 30 cm and height is 1.6 times the diameter.



4

A hexagonal prism, edge of base 20 mm and axis 50 mm long, rests with its base on HP such that one of its rectangular faces is parallel to VP. It is cut by a plane perpendicular to VP, inclined at 45° to HP and passing through the right corner of the top face of the prism. (i) Draw the sectional top view. (ii) Develop the lateral surfaces of the truncated prism.



5

Draw the development of the lateral surface of the lower portion of a cylinder of diameter 50 mm and axis 70 mm when sectioned by a plane inclined at 40° to HP and $\perp$ to VP and bisecting the axis.

olding and hence are accep...

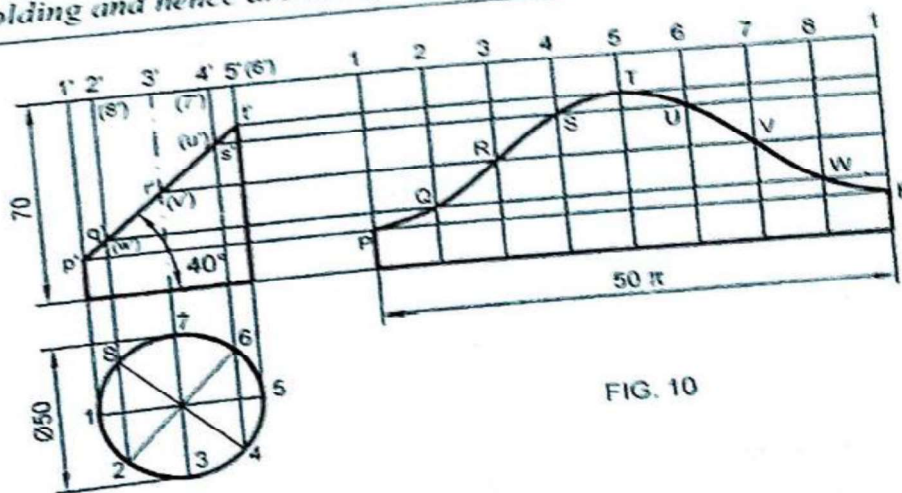
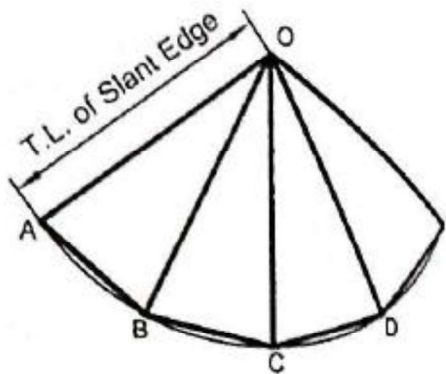
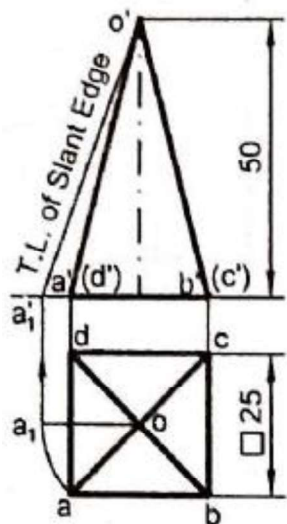


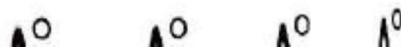
FIG. 10

6

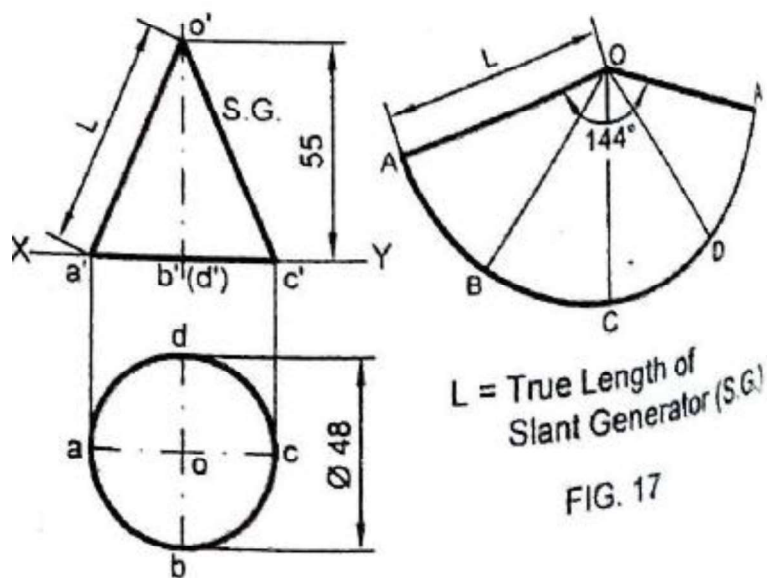
Draw the development of the lateral surfaces of a square pyramid, side of base 25 mm and height 50 mm, resting with its base on HP and an edge of the base parallel to VP.



(ii) Radial Line Development
(Economical Development)

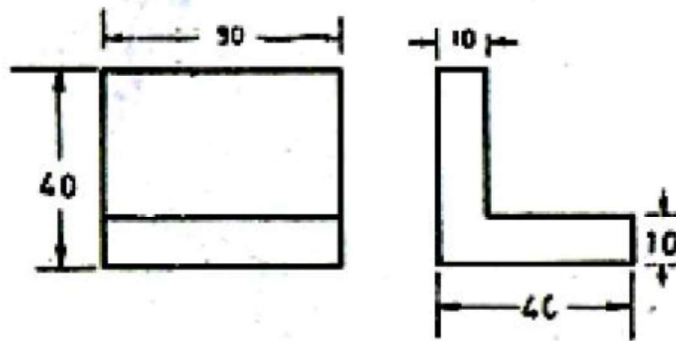


Draw the development of the lateral surface of a cone of base diameter 48 mm and altitude 55 mm.



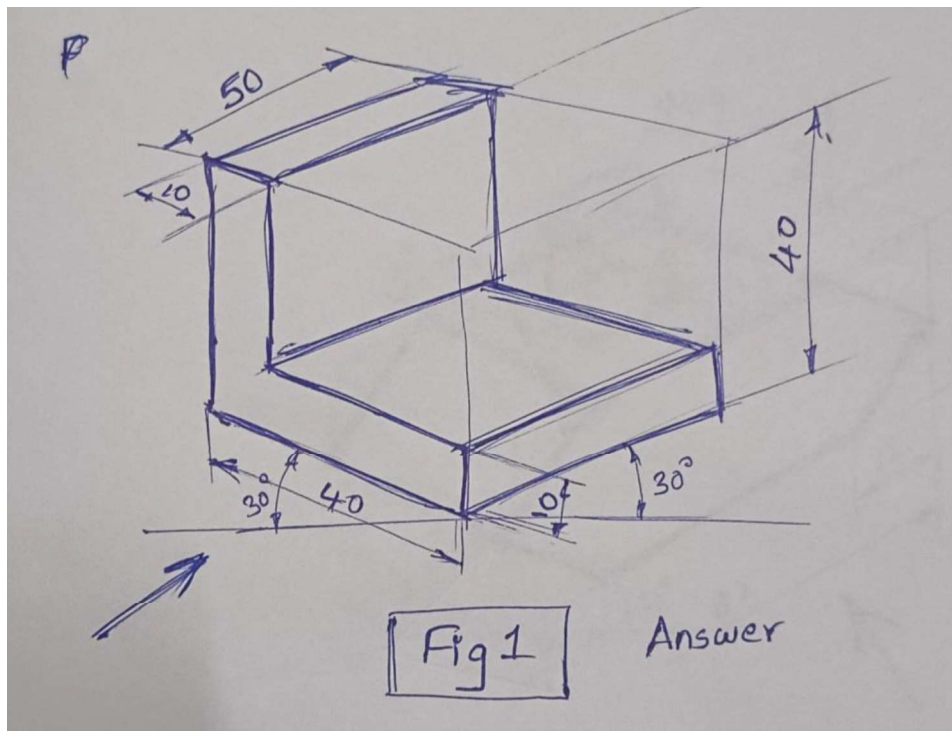
SHEET No. 16 (UNIT-V)

CONVERSION OF ORTHOGRAPHIC VIEWS TO ISOMETRIC VIEWS



(i)

FIG 1



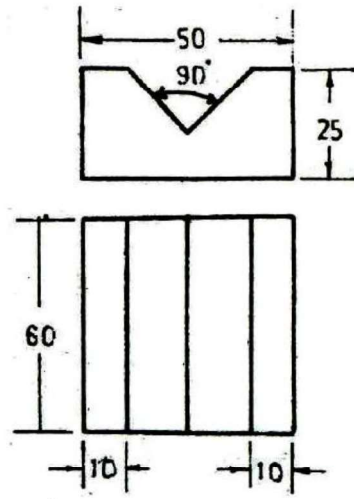
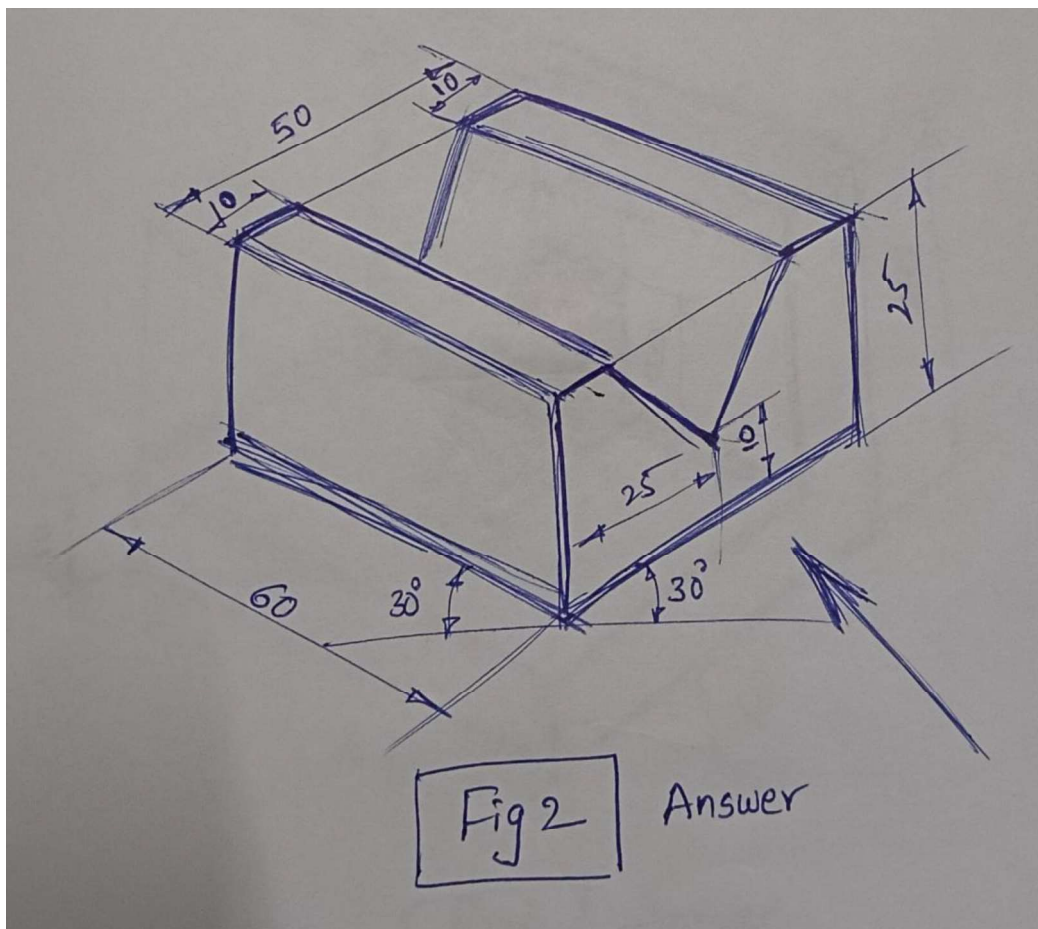


FIG 2



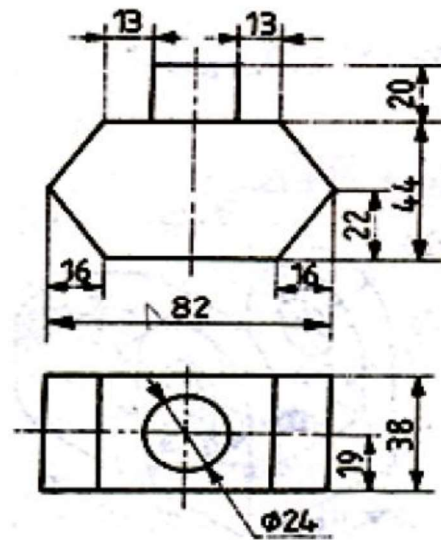
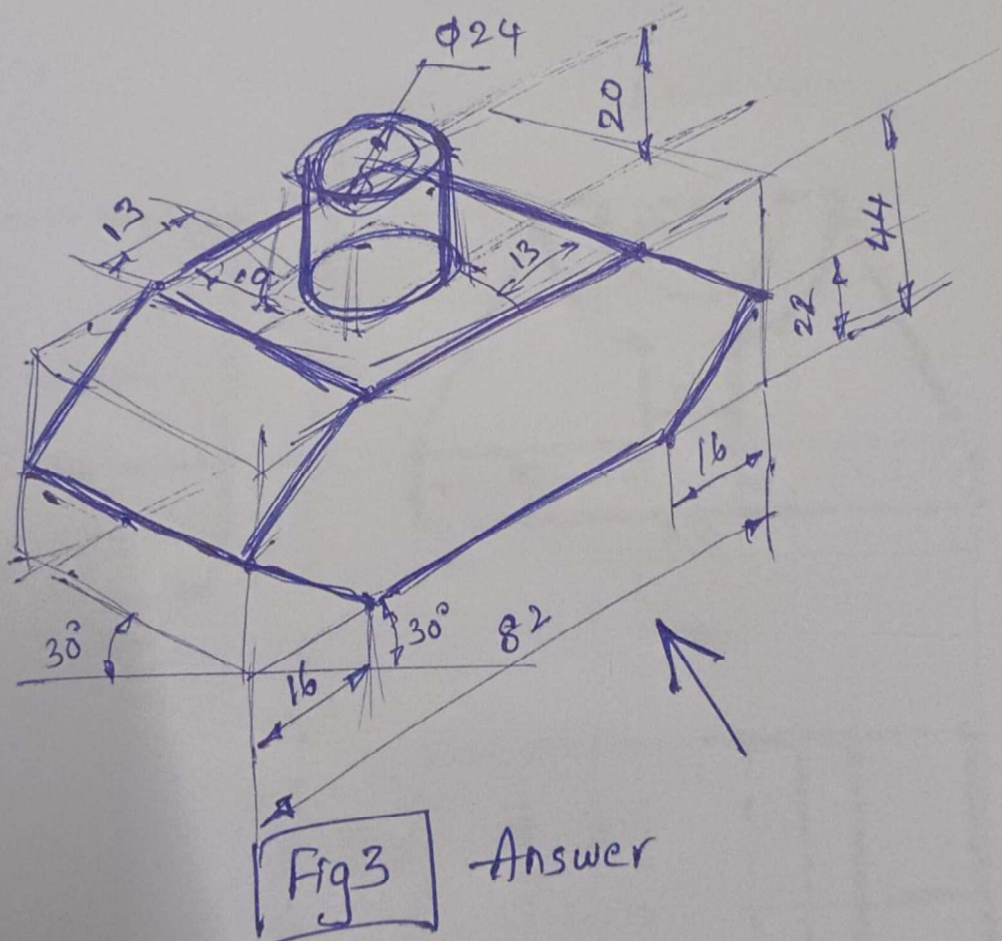


FIG 3



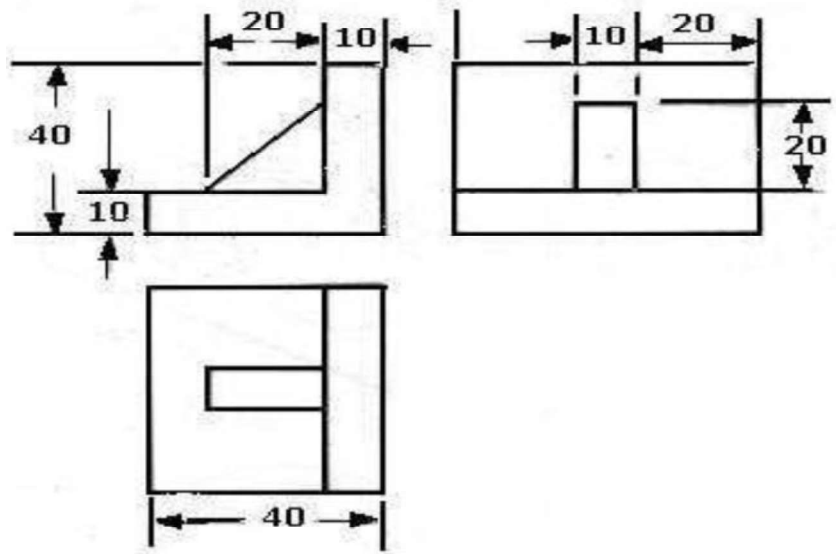


FIG 4

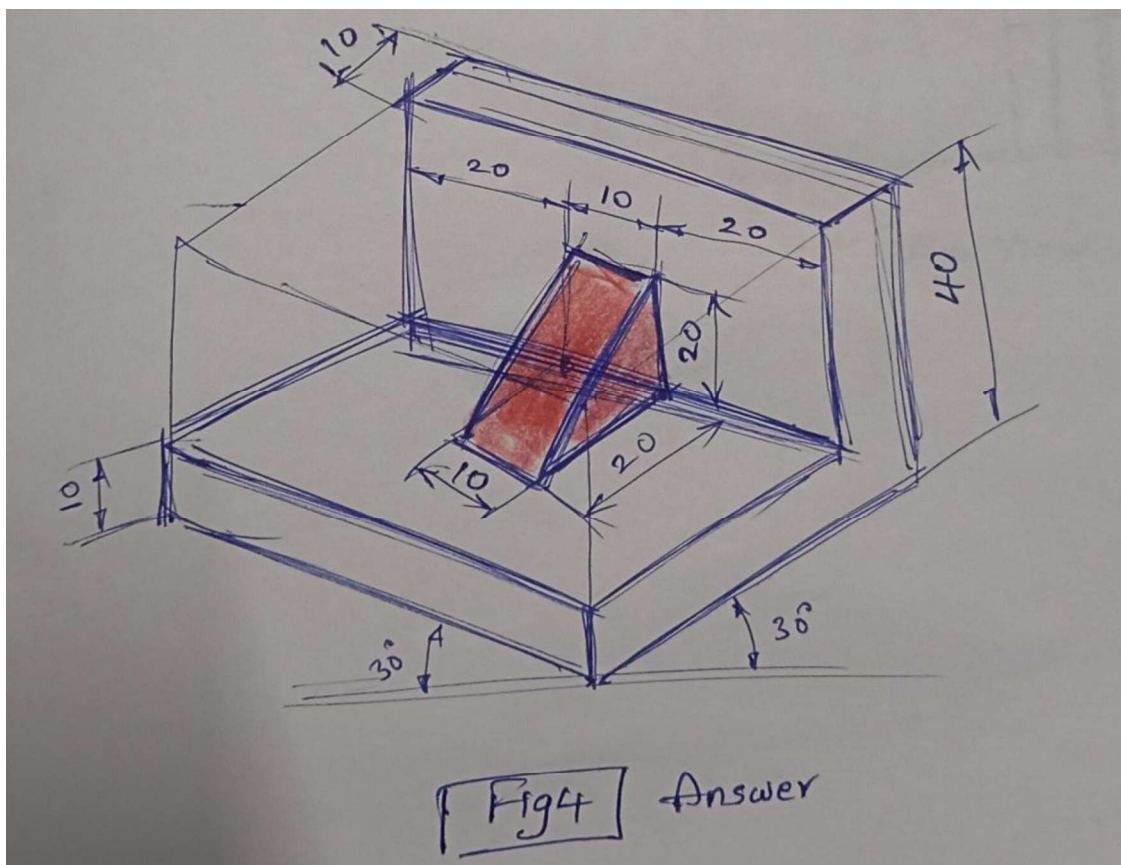


Fig4 Answer

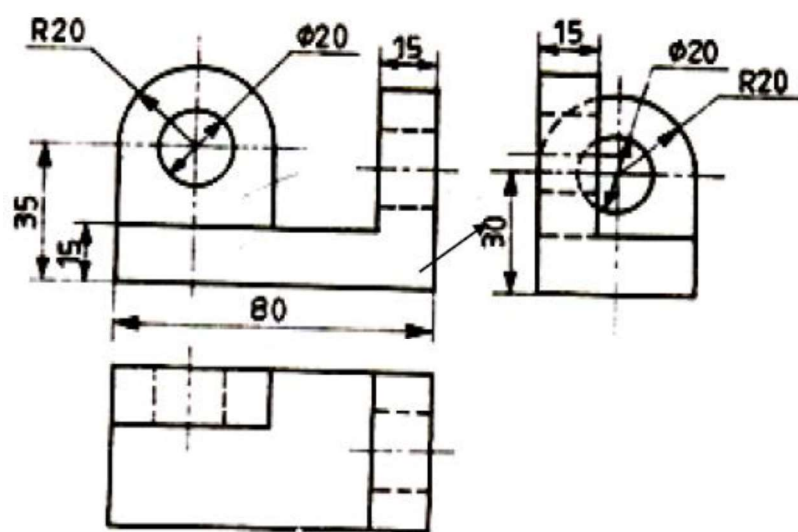


FIG 5

SHEET No.17 (UNIT-V)

CONVERSION OF ISOMETRIC VIEWS TO ORTHOGRAPHIC VIEWS

Question 1

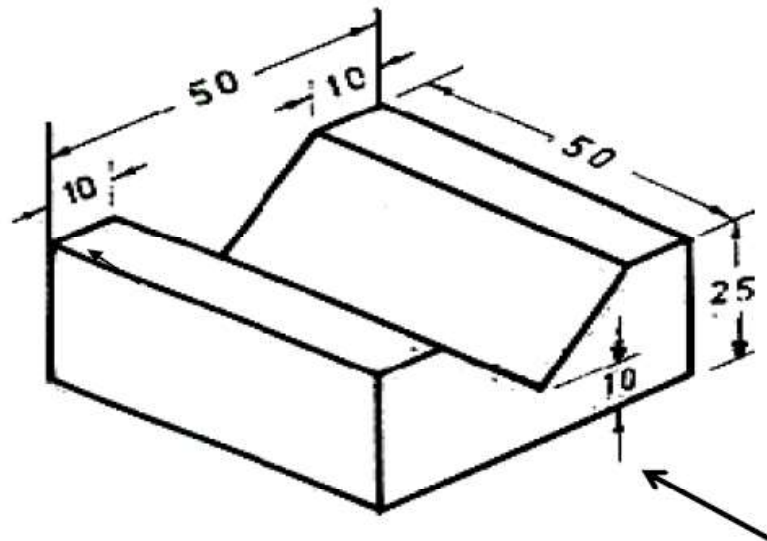
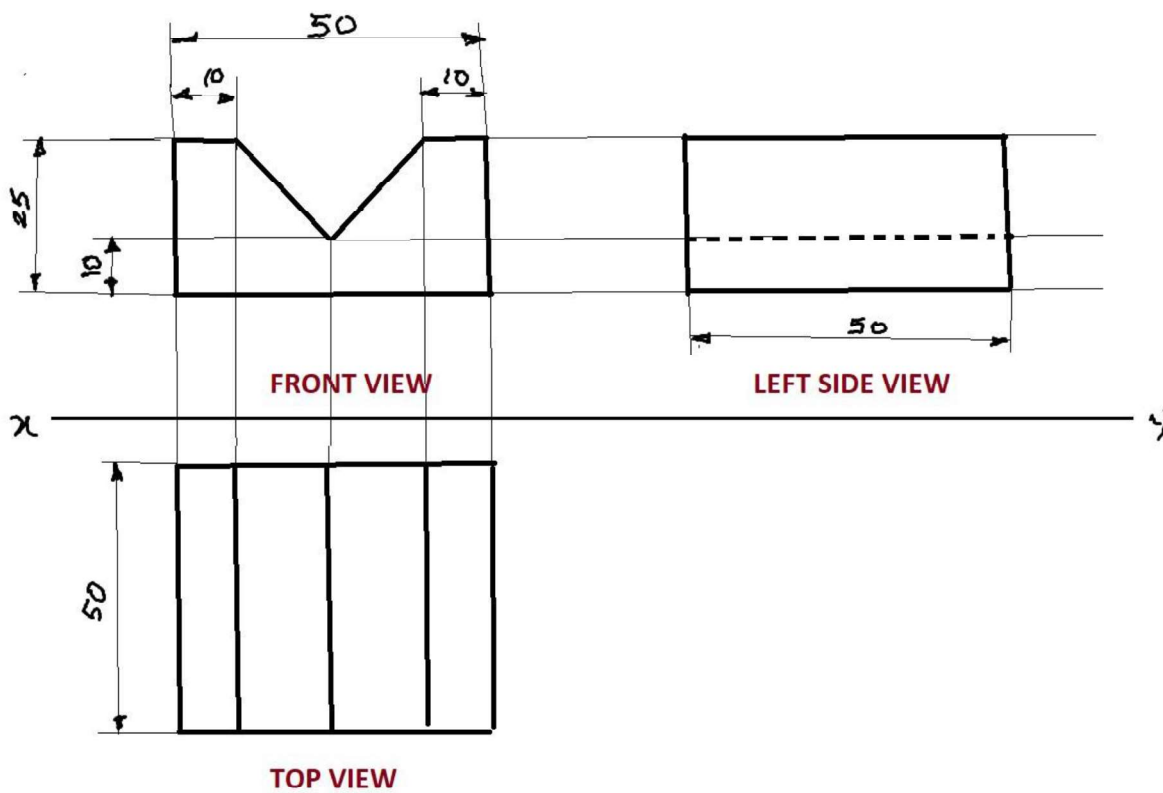


FIG 1

Answer:-



Question 2:-

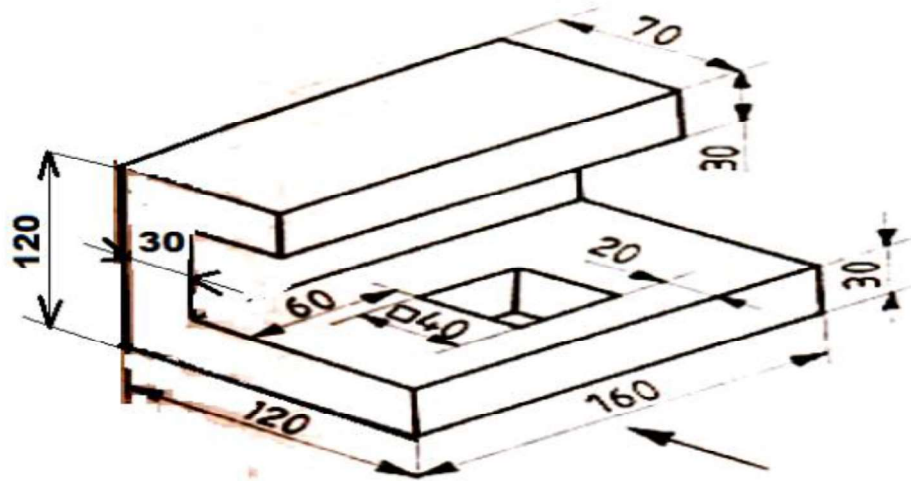
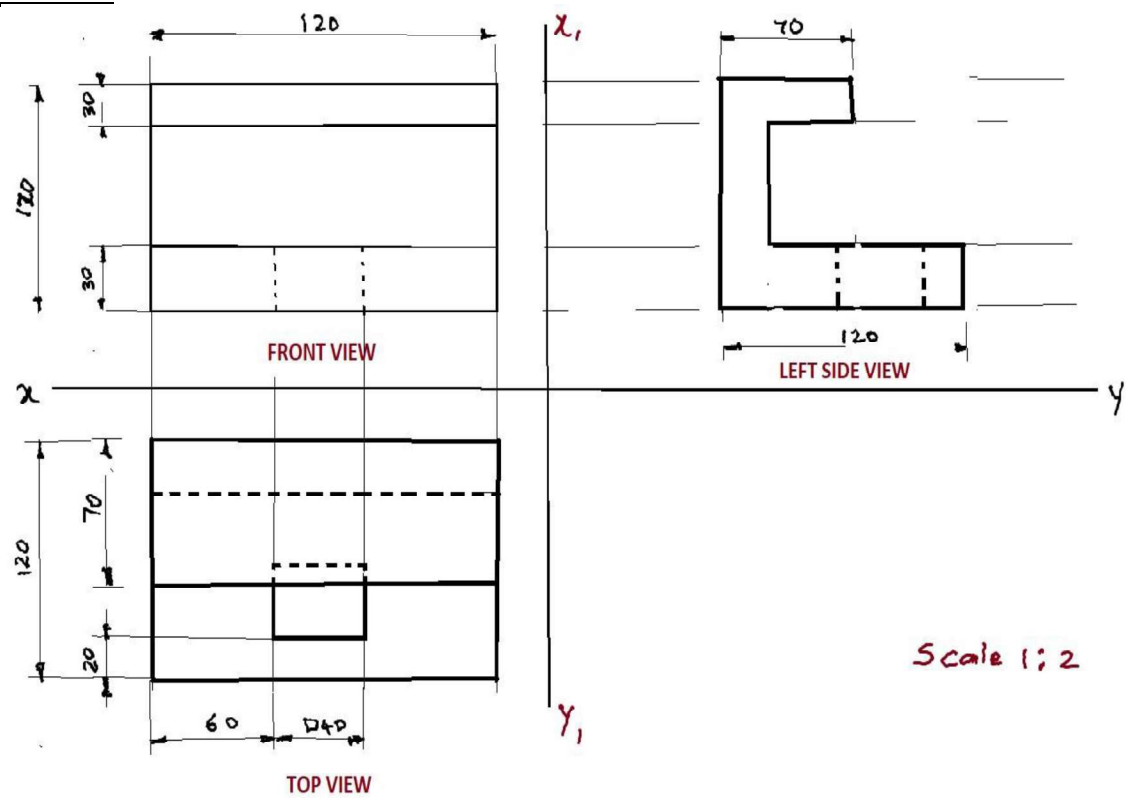


FIG 2

Answer:-



Question 3:-

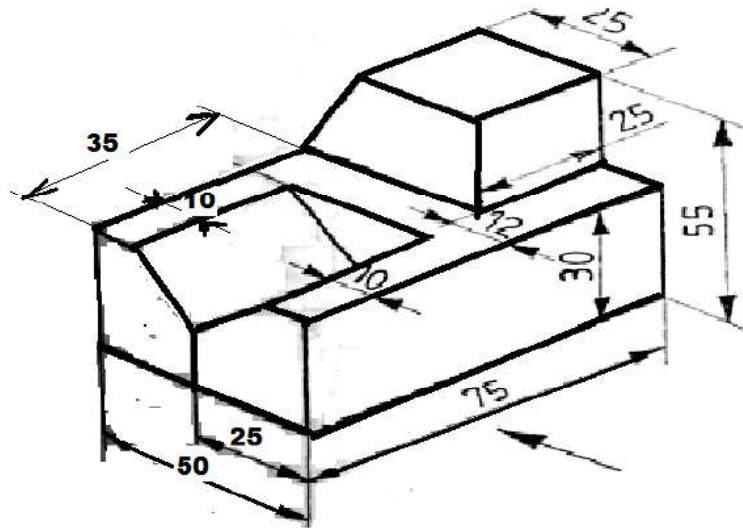
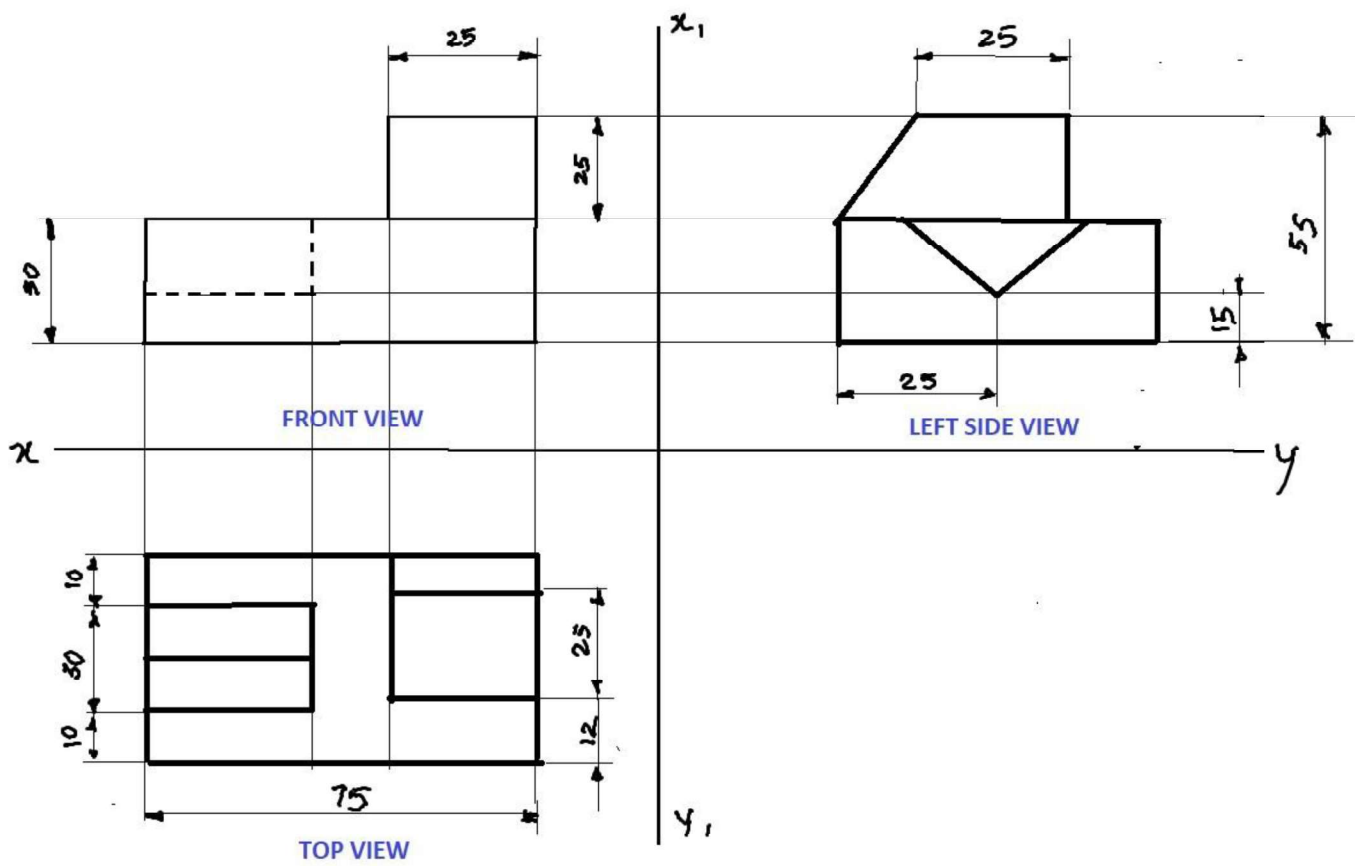
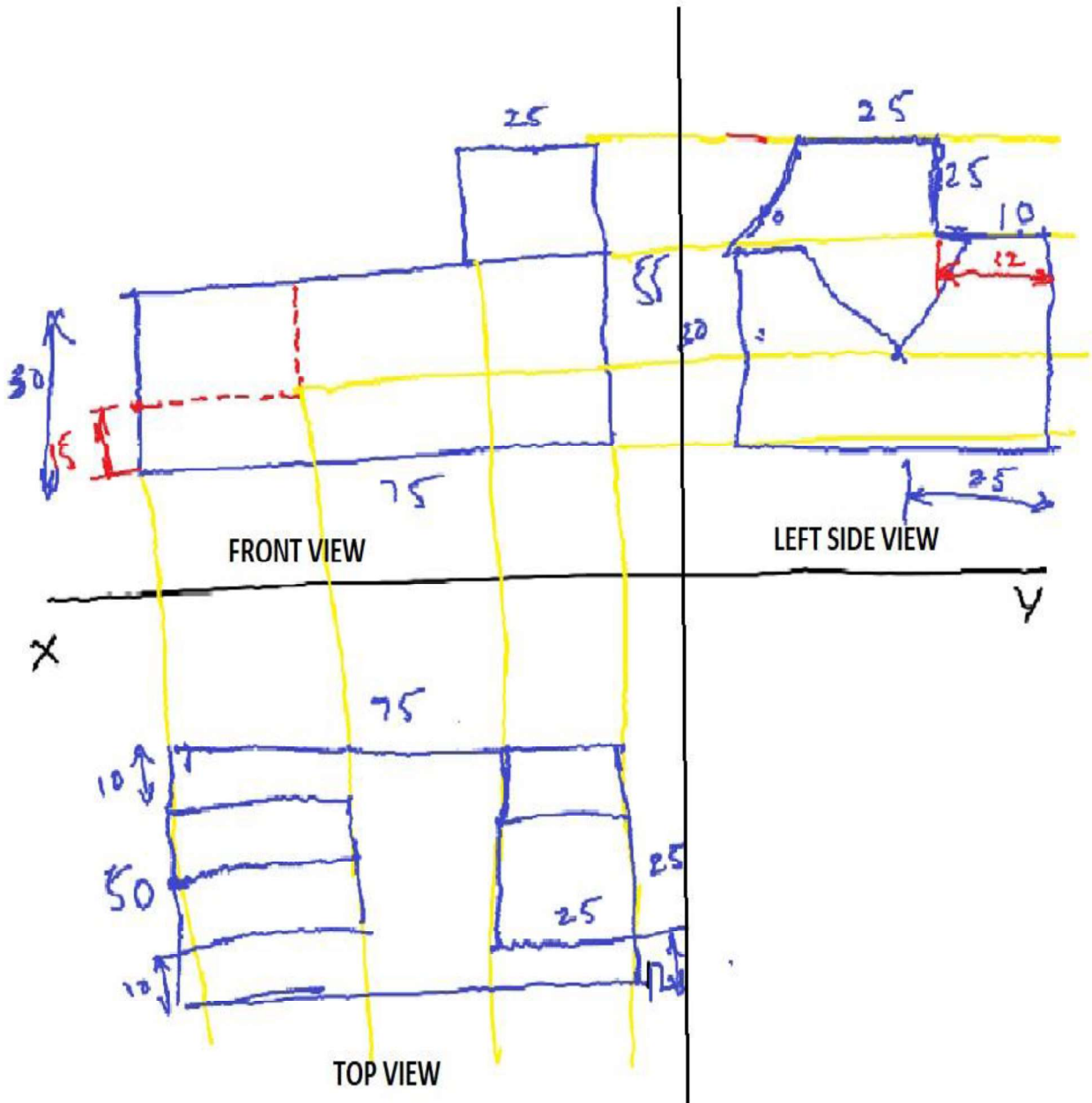


FIG 3

Answer:-



Concept for Fig3



Question 4

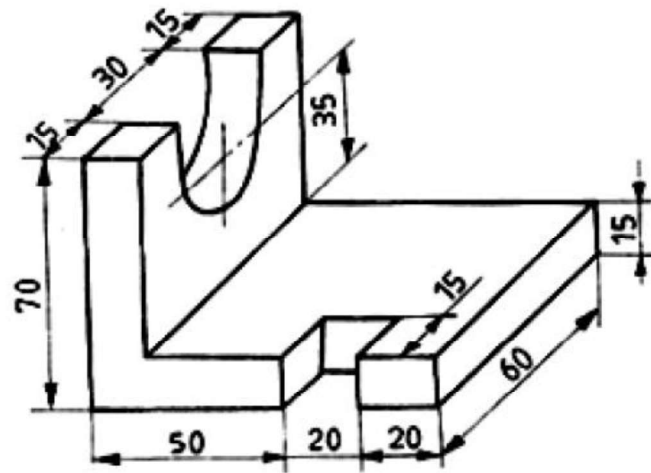
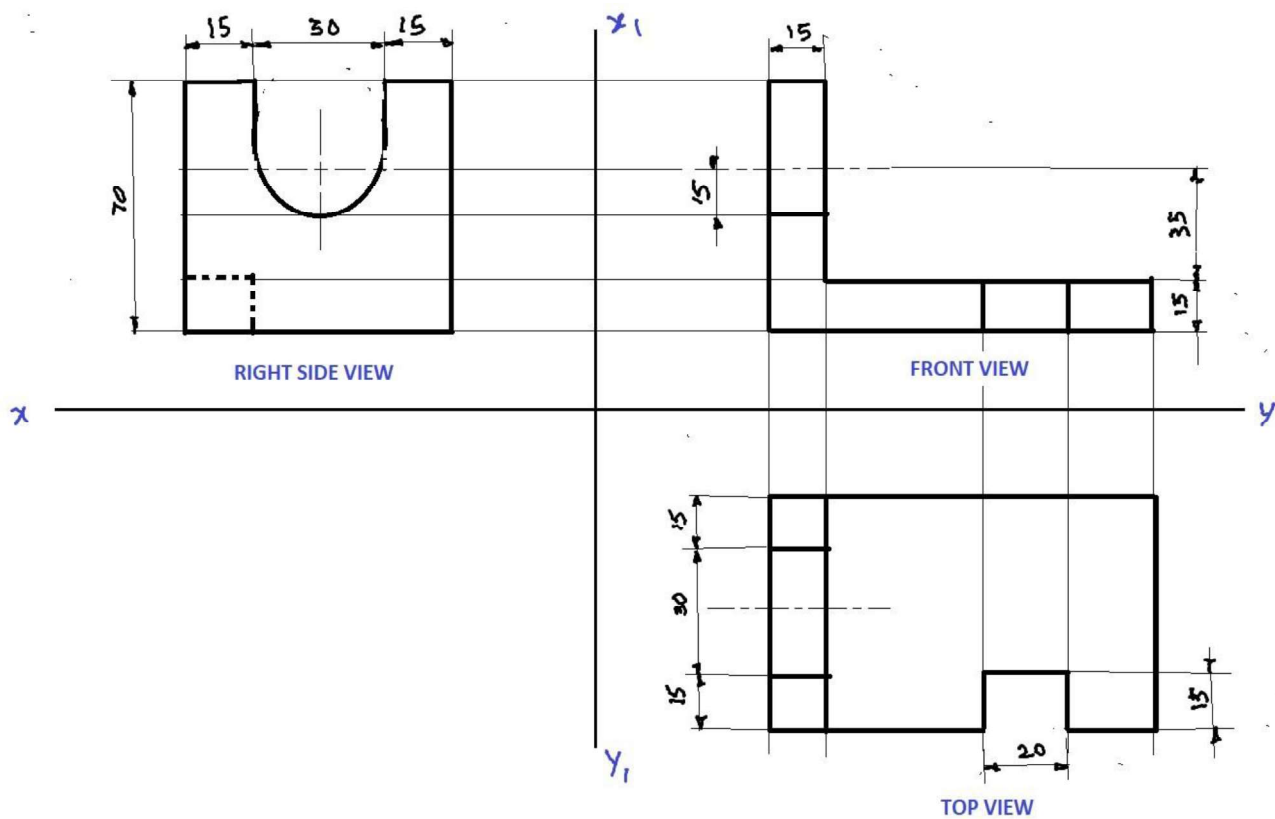
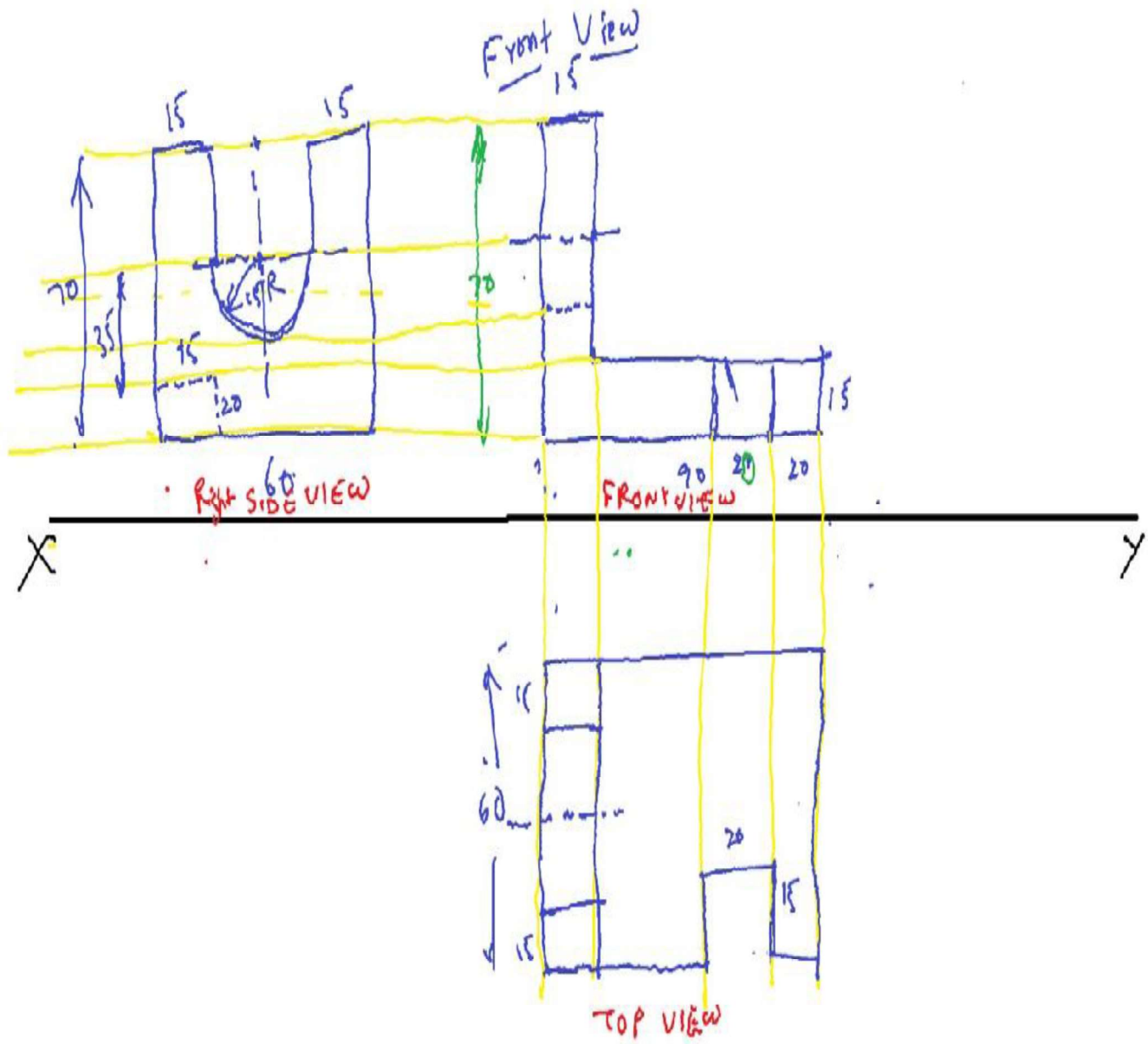


FIG 4

Answer:-



Concept for Fig4



Question 5:-

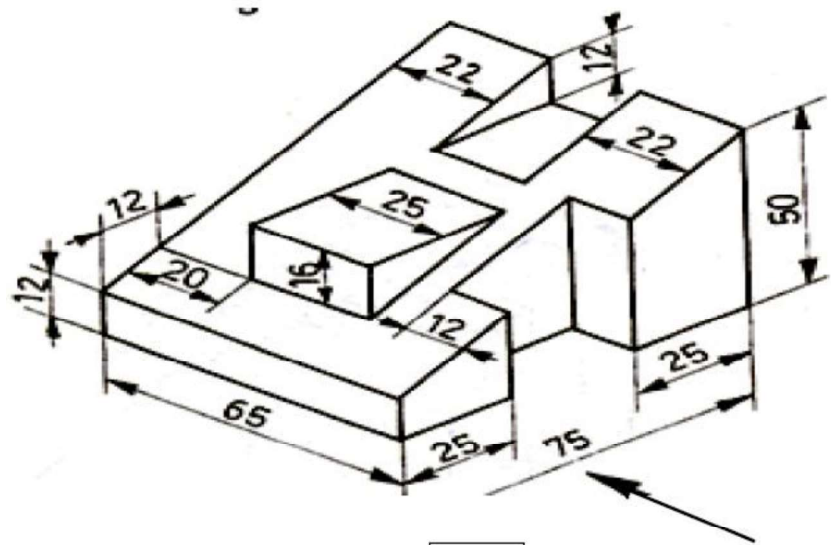
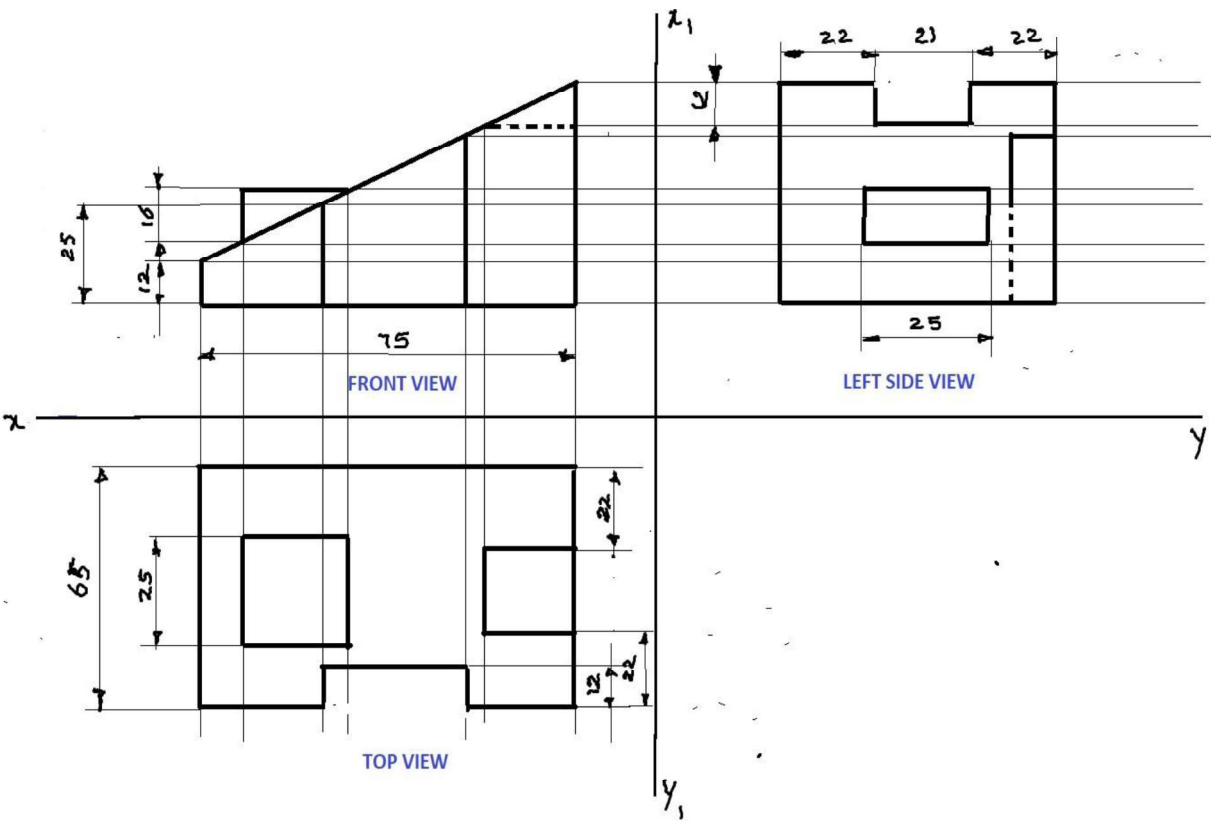


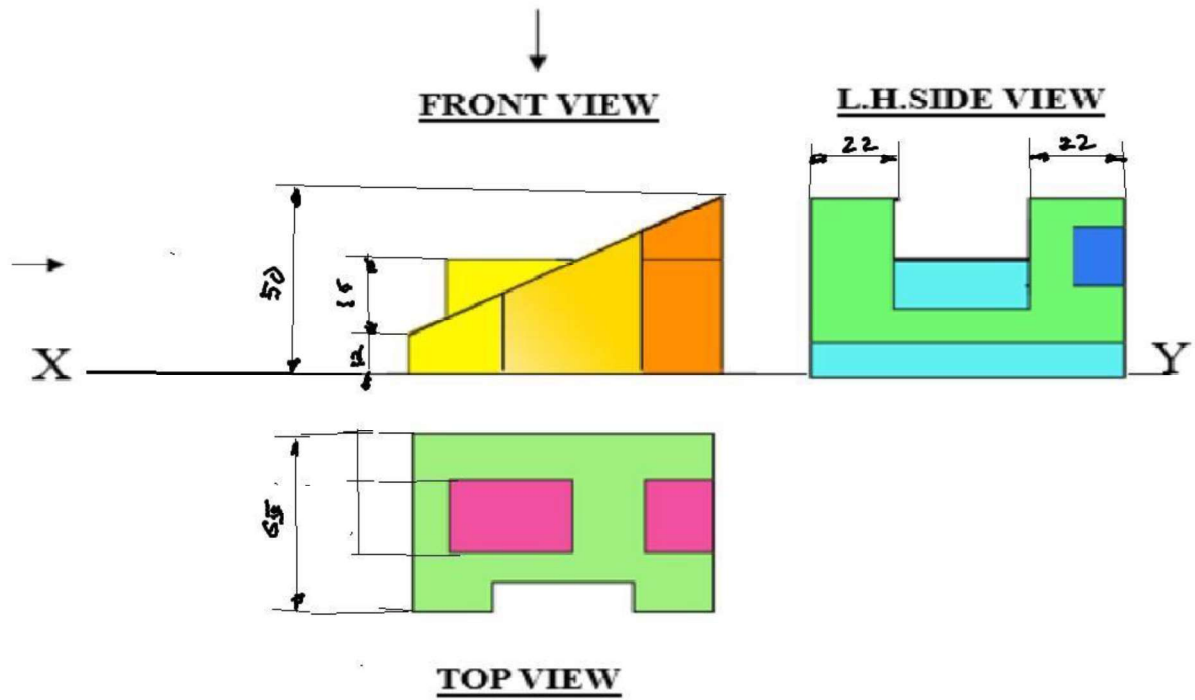
FIG 5

Answer:-



Concept for fig 5

ORTHOGRAPHIC PROJECTIONS



Question 6

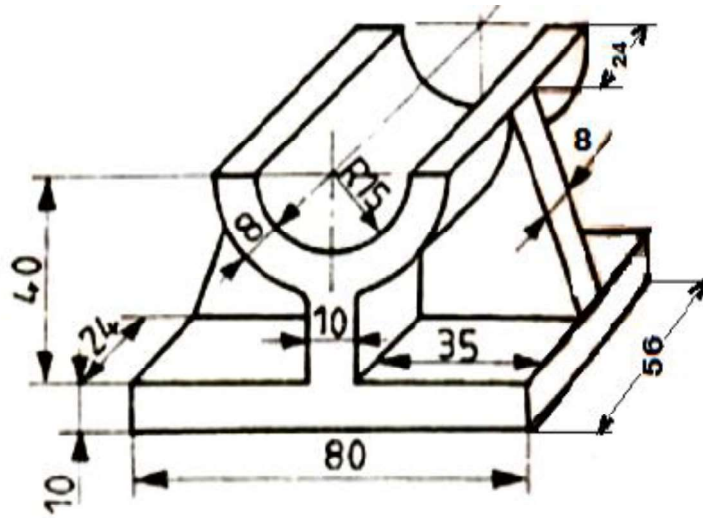


FIG 6

